

Stochastic Analysis

LectureNotes

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Chapter 1

Intro

Riemann-Stieltjes Integral: If f is continuous and g is of finite variation, then $\int_a^b f(x) \, dg(x)$ is well-defined, and is given by

$$\int_a^b f(x) \, dg(x) := \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(\xi_i) (g(x_{i+1}) - g(x_i))$$

where $a \leq x_0 < x_1 < \dots < x_n \leq b$ and $\xi_i \in [x_i, x_{i+1}]$.

Main Task: define the Itô integral

$$(H \cdot X)_t = \int_0^t H_s \, dX_s.$$

The most important case is when X is Brownian motion.

Theorem 1.0.1. $\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right)^2 \xrightarrow{a.s.} 1.$

Recall that the finite variation of a function $g : [a, b] \rightarrow \mathbb{R}$ is

$$V_g([a, b]) = \int_a^b |dg(x)| = \sup_n \sum_{i=1}^n |g(x_{i+1}) - g(x_i)|$$

Corollary 1.0.1. $V_B([0, 1]) = \infty$ a.s.

Proof for Corollary.

■.

$$\begin{aligned} (\star) \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right)^2 &\leq \sup_{1 \leq k \leq 2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \cdot \sum_{k=1}^{2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \\ &\leq V_B([0, 1]) \cdot \sup_{1 \leq k \leq 2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \end{aligned}$$

If $V_B([0, 1]) < \infty$, then by the continuity of Brownian paths,

$$\sup_{1 \leq k \leq 2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \rightarrow 0,$$

and hence $(\star) \rightarrow 0$.

This contradicts $(\star) \rightarrow 1$. Therefore, with probability one,

$$V_B([0, 1]) = \infty.$$

□

This shows that Brownian motion is too irregular to be treated by the classical Riemann–Stieltjes theory. Therefore, to define

$$(H \cdot X)_t = \int_0^t H_s \, dX_s,$$

we need a new framework, where typically H is previsible (predictable) and X is a semimartingale.

Example . Let $H_n \in \mathcal{F}_{n-1}$ and let (M_n) be a martingale.

Then $(H \cdot M)_n = \sum_{k=1}^n H_k(M_k - M_{k-1})$ is a martingale.

1.1 Set-up

Throughout, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A stochastic process $X = (X_t)_{t \geq 0}$ is a family of random variables taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$:

$$\begin{aligned} X: [0, \infty) \times \Omega &\rightarrow \mathbb{R}^d \\ (t, \omega) &\mapsto X_t(\omega). \end{aligned}$$

For each fixed ω , the map $t \mapsto X_t(\omega)$ is called the sample path of X corresponding to ω .

Definition 1.1.1. For two stochastic processes X and Y , we say that Y is a modification of X if

$$\forall t \geq 0, \quad \mathbb{P}(X_t = Y_t) = 1.$$

Definition 1.1.2. We say that X and Y are indistinguishable if

$$\mathbb{P}(X_t = Y_t, \forall t \geq 0) = 1.$$

In this case, Y is also called a version of X .

Example . Let Y be a modification of X , and suppose that both X and Y have a.s. right-continuous sample paths. Then X and Y are indistinguishable.

Definition 1.1.3. A function f is right-continuous if

$$f(x) = f(x+) = \lim_{t \downarrow x} f(t).$$

Proof. Since $\forall t \geq 0$, $P(X_t \neq Y_t) = 0$, the two processes agree almost surely at every fixed rational time. More precisely,

$$P\left(\bigcup_{q \in \mathbb{Q}^+} X_q \neq Y_q\right) \leq \sum_{q \in \mathbb{Q}^+} P(X_q \neq Y_q) = 0.$$

Therefore,

$$P\left(\bigcap_{q \in \mathbb{Q}^+} X_q = Y_q\right) = 1.$$

On the other hand, if two right-continuous functions f and g agree on $\mathbb{Q}^+$, then they must agree on all of $[0, \infty)$, since the rationals are dense.

Applying this observation pathwise to the sample paths of X and Y , we conclude that

$$P(X_t = Y_t, \forall t \geq 0) = 1.$$

□

Example . Let $T \sim \text{Exp}(1)$. Define

$$\begin{aligned} X_t &\equiv 0, \quad \forall t \geq 0 \\ Y_t &= \begin{cases} 0, & t \neq T \\ 1, & t = T. \end{cases} \end{aligned}$$

Then for every fixed $t \geq 0$,

$$P(X_t \neq Y_t) = P(Y_t = 1) = P(t = T) = 0.$$

Hence Y is a modification of X . However, X and Y are not indistinguishable, because on the event $\{T < \infty\}$ they differ at time T .

1.2 Filtration

We now introduce filtrations, which describe the information available up to each time t .

Let $(\mathcal{F}_t)_{t \geq 0}$ be a filtration of $(\Omega, \mathcal{F})$, that is,

$$\forall 0 \leq s < t < \infty, \quad \mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}.$$

Example . (Natural filtration) Let $(X_t)_{t \geq 0}$ be a stochastic process. Define

$$\mathcal{F}_t^X = \sigma(X_s, 0 \leq s \leq t).$$

Then $(\mathcal{F}_t^X)$ is a filtration.

By construction, $X_t \in \mathcal{F}_t^X$ for every $t \geq 0$.

Exercise 1.7. Let X be a process whose sample paths are all RCLL. Let

$$A = \{\omega : X(\omega) \text{ is continuous on } [0, t_0]\}$$

for some fixed $t_0 > 0$. Show that $A \in \mathcal{F}_{t_0}^X$.

RCLL: $f(x) = f(x+)$ and $f(x-)$ exists.

Proof. Since every sample path is RCLL, continuity on $[0, t_0]$ is equivalent to continuity at every point in $(0, t_0)$. Hence

$$\begin{aligned} A &= \{\omega : \forall s \in (0, t_0), X(s-, \omega) = X(s, \omega)\} \\ &= \{\omega : \forall q \in (0, t_0) \cap \mathbb{Q}, X(q-, \omega) = X(q, \omega)\} \end{aligned}$$

because for an RCLL function, possible discontinuities are jumps, and it is enough to check them on rational times.

Let $(0, t_0) \cap \mathbb{Q} = (q_m)_{m \geq 1}$. Then

$$\begin{aligned} A &= \left\{ \omega : \forall q_m \text{ with } m \geq 1, \forall k \geq 1, \exists N \geq 1, \forall n \geq N, \left| X(q_m - \frac{1}{n}, \omega) - X(q_m, \omega) \right| < \frac{1}{k} \right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{ \omega : \left| X(q_m - \frac{1}{n}, \omega) - X(q_m, \omega) \right| < \frac{1}{k} \right\} \in \mathcal{F}_{t_0}^X. \end{aligned}$$

Therefore $A \in \mathcal{F}_{t_0}^X$. □

Exercise 1.8. Let X be a process whose sample paths are RCLL a.s. Then $A \notin \mathcal{F}_{t_0}^X$ is possible.

Proof. [TODO] □

Definition 1.2.1. A filtration $\{\mathcal{F}_t\}_{t \geq 0}$ is said to satisfy the usual conditions if

$$(a) \mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s;$$

(b) $\mathcal{F}_0$ contains all the P-negligible sets in $\mathcal{F}$.

Definition 1.2.2. An event A is called P-negligible if there exists $N \in \mathcal{F}$ such that $A \subseteq N$ and $P(N) = 0$.

Definition 1.2.3. We define

$$\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s, \quad \mathcal{F}_{t-} = \sigma \left(\bigcup_{s<t} \mathcal{F}_s \right).$$

We say that the filtration is right-continuous if $\mathcal{F}_t = \mathcal{F}_{t+}$.

Definition 1.2.4. The stochastic process X is adapted to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ if $\forall t \geq 0$, X_t is $\mathcal{F}_t$ -measurable.

In particular, every process is adapted to its natural filtration.

Definition 1.2.5. The stochastic process X is called progressively measurable with respect to $(\mathcal{F}_t)$ if $\forall t \geq 0$, the map

$$\begin{aligned} (s, \omega) &\mapsto X_s(\omega) \\ ([0, t] \times \Omega, \mathcal{B}([0, t]) \otimes \mathcal{F}_t) &\rightarrow (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)) \end{aligned}$$

is measurable.

An adapted process need not be progressively measurable in general. The useful point here is that, under additional path regularity, adaptedness implies progressive measurability. In the usual hierarchy one has

$$\text{previsible} \implies \text{progressively measurable} \implies \text{adapted}.$$

Proposition 1.2.1. If X is adapted to the filtration $(\mathcal{F}_t)$ and every sample path is right-continuous or left-continuous, then X is also progressively measurable.

Proof for Proposition.

■. It is enough to treat the right-continuous case. Fix $t > 0$. For every $n \geq 1$ and $0 \leq k \leq 2^n - 1$, define

$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \quad \forall \frac{kt}{2^n} < s \leq \frac{k+1}{2^n}t$$

Then, by right-continuity of the sample paths,

$$\lim_{n \rightarrow \infty} X_s^{(n)}(\omega) = X_s(\omega)$$

for all $(s, \omega) \in [0, t] \times \Omega$.

Therefore, it suffices to show that

$$\begin{aligned} (s, \omega) &\mapsto X_s^{(n)}(\omega) \\ ([0, t] \times \Omega, \mathcal{B}([0, t]) \otimes \mathcal{F}_t) &\rightarrow (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)) \end{aligned}$$

is measurable.

Indeed, for every $A \in \mathcal{B}(\mathbb{R}^d)$,

$$\begin{aligned} & \left\{ (s, \omega) : X_s^{(n)}(\omega) \in A \right\} \\ &= \bigcup_{k=0}^{2^n-1} \left(\frac{k}{2^n}t, \frac{k+1}{2^n}t \right] \times \left\{ X_{\frac{k+1}{2^n}t} \in A \right\} \in \mathcal{B}([0, t]) \otimes \mathcal{F}_t \end{aligned}$$

because X is adapted and therefore $X_{\frac{k+1}{2^n}t}$ is $\mathcal{F}_t$ -measurable.

Thus $X_s^{(n)}(\omega)$ is measurable for every n . Passing to the pointwise limit, we conclude that X is progressively measurable. $\square$

1.3 Stopping Times

We now study random times that are compatible with the filtration $(\mathcal{F}_t)_{t \geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 1.3.1. A random time T is called a stopping time with respect to $(\mathcal{F}_t)$ if

$$\forall t \geq 0, \quad \{T \leq t\} \in \mathcal{F}_t.$$

Definition 1.3.2. A random time T is called an optional time with respect to $(\mathcal{F}_t)$ if

$$\forall t \geq 0, \quad \{T < t\} \in \mathcal{F}_t.$$

Proposition 1.3.1. Every stopping time is an optional time. Moreover, if the filtration is right-continuous, then the two notions are equivalent.

Proof for Proposition.

■. “ $\Rightarrow$ ” If T is a stopping time, then

$$\{T < t\} = \bigcup_{n=1}^{\infty} \left\{ T \leq t - \frac{1}{n} \right\} \in \mathcal{F}_t.$$

“ $\Leftarrow$ ” Now assume that T is an optional time and that the filtration is right-continuous. Then for every $m \geq 1$,

$$\{T \leq t\} = \bigcap_{n=m}^{\infty} \left\{ T < t + \frac{1}{n} \right\}.$$

Since $\{T < t + \frac{1}{n}\} \in \mathcal{F}_{t+\frac{1}{n}} \subseteq \mathcal{F}_{t+\frac{1}{m}}$, we obtain

$$\{T \leq t\} \in \mathcal{F}_{t+\frac{1}{m}}.$$

Therefore

$$\{T \leq t\} \in \bigcap_{m=1}^{\infty} \mathcal{F}_{t+\frac{1}{m}} = \mathcal{F}_{t^+} = \mathcal{F}_t,$$

so T is a stopping time. $\square$

Lemma 1.3.1. If T and S are stopping times, then $T \wedge S$, $T \vee S$, $T + S$ are stopping times.

Proof. $\forall t \geq 0$,

1. $\{T \wedge S \leq t\} = \{T \leq t\} \cup \{S \leq t\} \in \mathcal{F}_t$.
2. $\{T \vee S \leq t\} = \{T \leq t\} \cap \{S \leq t\} \in \mathcal{F}_t$.
3. To prove $\{T + S \leq t\} \in \mathcal{F}_t$, it is enough to show that $\{T + S > t\} \in \mathcal{F}_t$.

$$\begin{aligned} & \{T + S > t\} \\ &= \{T = 0, S > t\} \cup \{S = 0, T > t\} \cup \{0 < T < t, T + S > t\} \cup \{S > 0, T \geq t\} \\ &\in \mathcal{F}_t \cup \mathcal{F}_t \cup \{0 < T < t, T + S > t\} \cup \mathcal{F}_t. \end{aligned}$$

Thus it remains to show that

$$\{0 < T < t, T + S > t\} = \bigcup_{\substack{r \in \mathbb{Q}^+ \\ 0 < r < t}} \{r < T < t, S > t - r\}.$$

The right-hand side belongs to $\mathcal{F}_t$.

“ $\Leftarrow$ ” $S > t - r > t - T$, so $S + T > t$.

“ $\Rightarrow$ ” Since $S + T > t$, we can choose $r \in \mathbb{Q}^+$ with $0 < r < T$ and $S + r > t$. $\square$

Lemma 1.3.2. If $\{T_n\}_{n \geq 1}$ are stopping times, then $\sup_{n \geq 1} T_n$ is a stopping time.

Proof. $\forall t \geq 0$, $\{\sup_{n \geq 1} T_n \leq t\} = \bigcap_{n=1}^{\infty} \{T_n \leq t\} \in \mathcal{F}_t$. $\square$

Definition 1.3.3. The stopping σ -algebra associated with a stopping time T is

$$\mathcal{F}_T = \{A \in \mathcal{F} \mid \forall t \geq 0, A \cap \{T \leq t\} \in \mathcal{F}_t\}.$$

Proposition 1.3.2. If $T(\omega) = t_0$ for some fixed $t_0 \geq 0$, $\forall \omega \in \Omega$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.

Proof. Note that $\forall t \geq 0$,

$$\{T \leq t\} = \begin{cases} \emptyset, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

“ $\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$ ”

$\forall A \in \mathcal{F}_{t_0}, \forall t \geq 0$,

① $t \geq t_0, A \cap \{T \leq t\} = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$;

$$\textcircled{2} \quad t < t_0, A \cap \{T \leq t\} = \emptyset \in \mathcal{F}_t.$$

Hence $\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$.

“ $\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$ ”

$\forall A \in \mathcal{F}_T$, we have $A \cap \{T \leq t\} \in \mathcal{F}_t$ by definition. Let $t = t_0$, then $\{T \leq t_0\} = \Omega$, so

$$A \cap \{T \leq t_0\} \in \mathcal{F}_{t_0} \Rightarrow A \cap \Omega \in \mathcal{F}_{t_0} \Rightarrow A \in \mathcal{F}_{t_0}.$$

Hence $\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$. □

Lemma 1.3.3. $\forall$ stopping time T and S , $\forall A \in \mathcal{F}_S$, we have $A \cap \{S \leq T\} \in \mathcal{F}_T$. In particular, if $S \leq T$ for all $\omega \in \Omega$, then $\mathcal{F}_S \subseteq \mathcal{F}_T$.

Proof. First note that $T \wedge t$ is also a stopping time. Hence $T \wedge t$ is $\mathcal{F}_t$ -measurable.

To prove $A \cap \{S \leq T\} \in \mathcal{F}_T$, it is enough to show that for every $t \geq 0$,

$$A \cap \{S \leq T\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

We have

$$A \cap \{S \leq T\} \cap \{T \leq t\} = \underbrace{A \cap \{S \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{T \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{S \wedge t \leq T \wedge t\}}_{\mathcal{F}_t}.$$

Indeed, on $\{S \leq t\} \cap \{T \leq t\}$, the event $\{S \leq T\}$ is exactly the same as $\{S \wedge t \leq T \wedge t\}$. Therefore the whole intersection belongs to $\mathcal{F}_t$, and hence $A \cap \{S \leq T\} \in \mathcal{F}_T$. □

Lemma 1.3.4. Let S, T be two stopping times. Then $\mathcal{F}_{T \wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$ and $\{T < S\}$, $\{S < T\}$, $\{T \leq S\}$, $\{S \leq T\}$, $\{T = S\}$ belong to $\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}$.

Proof. $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_S$ and $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \Rightarrow \mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \cap \mathcal{F}_S$.

To prove $\mathcal{F}_T \cap \mathcal{F}_S \subseteq \mathcal{F}_{T \wedge S}$, $\forall A \in \mathcal{F}_T \cap \mathcal{F}_S$, we will prove $\forall t \geq 0$, $A \cap \{T \wedge S \leq t\} \in \mathcal{F}_t$.

$$\begin{aligned} A \cap \{T \wedge S \leq t\} &= A \cap (\{T \leq t\} \cup \{S \leq t\}) \\ &= (A \cap \{T \leq t\}) \cup (A \cap \{S \leq t\}) \in \mathcal{F}_t. \end{aligned}$$

Next, $\{S \leq T\} \in \mathcal{F}_T$. It remains to prove $\{S < T\} \in \mathcal{F}_T$.

Assuming this, by interchanging T and S , we get $\{T \leq S\}$, $\{T > S\}$, $\{T < S\} \in \mathcal{F}_S$.

Therefore $\{S \leq T\} \in \mathcal{F}_S \cap \mathcal{F}_T$.

$$\{T = S\} = \{S \leq T\} \cap \{T \leq S\} \in \mathcal{F}_S \cap \mathcal{F}_T.$$

To prove $\{S < T\} \in \mathcal{F}_T$, let $R = S \wedge T$. Then $\{S < T\} = \{R < T\}$. Since R is a stopping time, we have $R \in \mathcal{F}_R \subseteq \mathcal{F}_T$, and therefore $\{R < T\} \in \mathcal{F}_T$. □

Summary:

(i) $T \in \mathcal{F}_T$;

- (ii) If $S \leq T$, $\mathcal{F}_S \subseteq \mathcal{F}_T$;
- (iii) $T \wedge t \in \mathcal{F}_{T \wedge t} \subseteq \mathcal{F}_t$;
- (iv) $\mathcal{F}_{T \wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$.

For each $\omega \in \Omega$, if $T(\omega) \in [0, \infty)$ and $X_t(\omega) \in \mathbb{R}^d$ for all $t \geq 0$, we define

$$X_T(\omega) = X_{T(\omega)}(\omega).$$

Proposition 1.3.3. Let $\{X_t\}_{t \geq 0}$ be a progressively measurable process, and let T be a stopping time. Then X_T , defined on $\{T < \infty\}$, is $\mathcal{F}_T$ -measurable, and the stopped process $\{X_{T \wedge t}\}_{t \geq 0}$ is progressively measurable.

Proof. Let $B \in \mathcal{B}(\mathbb{R}^d)$. By definition of $\mathcal{F}_T$,

$$\{X_T \in B\} \in \mathcal{F}_T \iff \forall t \geq 0, \{X_T \in B\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

On the event $\{T \leq t\}$ we have $X_T = X_{T \wedge t}$, so

$$\{X_T \in B\} \cap \{T \leq t\} = \{X_{T \wedge t} \in B\} \cap \{T \leq t\}.$$

Therefore, it is enough to prove that $X_{T \wedge t}$ is $\mathcal{F}_t$ -measurable for every $t \geq 0$.

Observe that the map

$$(s, \omega) \mapsto (T(\omega) \wedge s, \omega)$$

is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ -measurable.

By assumption, $(s, \omega) \mapsto X_s(\omega)$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ -measurable.

Hence the composition

$$(s, \omega) \mapsto X_{T(\omega) \wedge s}(\omega)$$

is also $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ -measurable. In particular, evaluating at $s = t$ shows that $X_{T \wedge t}$ is $\mathcal{F}_t$ -measurable.

This also proves that the stopped process $(X_{T \wedge t})_{t \geq 0}$ is progressively measurable. $\square$

1.4 Cts-time martingale

We now turn to continuous-time martingales and some of their basic properties.

Definition 1.4.1. On $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, a martingale is an adapted stochastic process $(X_t)_{t \geq 0}$ such that

$$\begin{cases} (i) \forall t \geq 0, \mathbb{E}|X_t| < \infty \\ (ii) \forall t \geq s \geq 0, \mathbb{E}(X_t | \mathcal{F}_s) = X_s. \end{cases}$$

If $\mathbb{E}(X_t | \mathcal{F}_s) \geq X_s$, we call (X_t) a submartingale.

If $\mathbb{E}(X_t | \mathcal{F}_s) \leq X_s$, we call (X_t) a supermartingale.

Proposition 1.4.1. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex (concave up) function. If $(X_t)_{t \geq 0}$ is a martingale, then $(\varphi(X_t))_{t \geq 0}$ is a submartingale.

Examples. $|X_t|$, X_t^2 , $|X_t|^p$ for all $p \geq 1$, and $(X_t)^+$ are all submartingales.

We assume $\mathbb{E}|\varphi(X_t)| < \infty$, $\forall t \geq 0$.

Proof. Jensen's inequality. □

Define

$$\begin{aligned} T_b^{(0)} &= -\infty, \\ T_a^{(1)} &= \inf\{t \geq (T_b^{(0)})^+ : X_t \leq a\}, \\ T_b^{(1)} &= \inf\{t \geq T_a^{(1)} : X_t \geq b\}. \end{aligned}$$

$\forall l \geq 2$,

$$\begin{aligned} T_a^{(l)} &= \inf\{t \geq T_b^{(l-1)} : X_t \leq a\}, \\ T_b^l &= \inf\{t \geq T_a^l : X_t \geq b\} \end{aligned}$$

Define

$$U_{ab}^X[0, n] = \max\{k \geq 0 : T_b^{(k)} \leq n\}$$

to be the number of upcrossings of the interval $[a, b]$ on $[0, n]$.

Theorem 1.4.1 (Continuous-time Doob's upcrossing inequality). Let $(X_t)_{t \geq 0}$ be a right-continuous submartingale. For all $T > 0$,

$$\mathbb{E}(U_{ab}^X[0, T]) \leq \frac{\mathbb{E}(X_T - a)^+}{b - a}$$

Proof. For each $n \geq 1$, let $D_n = \mathbb{Z}/2^n = \{0, \frac{\pm 1}{2^n}, \frac{\pm 2}{2^n}, \dots\}$.

Consider the upcrossings of X on $[0, T] \cap D_n$. The sampled process $\left\{\frac{X_j}{2^n}\right\}_{j=0}^\infty$ is a discrete-time submartingale, so

$$U_{ab}^X([0, T] \cap D_n) \leq U_{ab}^X[0, T], \quad U_{ab}^X([0, T] \cap D_n) \uparrow U_{ab}^X([0, T]).$$

Applying the discrete-time upcrossing inequality, we obtain

$$\mathbb{E}(U_{ab}^X([0, T] \cap D_n)) \leq \frac{1}{b - a} \mathbb{E}(X_{T_n} - a)^+,$$

where $T_n = \frac{[T \cdot 2^n]}{2^n}$. Letting $n \rightarrow \infty$, we get

$$\mathbb{E}(U_{ab}^X([0, T])) \leq \frac{1}{b - a} \mathbb{E}(X_T - a)^+.$$

□

Proposition 1.4.2. If $(X_t)_{t \geq 0}$ is a right-continuous submartingale and $\sup_{t \geq 0} \mathbb{E}X_t^+ < \infty$, then there exists $X \in L^1$ such that $X_t \xrightarrow{a.s.} X$ as $t \rightarrow \infty$.

Proof.

$$\begin{aligned} \mathbb{E}(U_{ab}^X([0, \infty))) &= \lim_{n \rightarrow \infty} \mathbb{E}(U_{ab}^X([0, n])) \\ &\leq \limsup_{n \rightarrow \infty} \frac{1}{b-a} \mathbb{E}(X_n - a)^+ \\ &\leq \limsup_{n \rightarrow \infty} \frac{1}{b-a} (\mathbb{E}(X_n^+) + |a|) < \infty. \end{aligned}$$

Hence for every $a < b$, we have $U_{ab}^X[0, \infty) < \infty$ a.s.

Therefore, with probability one, $U_{ab}^X[0, \infty) < \infty$ for all rational $a < b$.

Hence the event

$$\left\{ \liminf_{t \rightarrow \infty} X_t \leq a \leq b < \limsup_{t \rightarrow \infty} X_t \right\}$$

cannot occur.

Therefore,

$$\liminf_{t \rightarrow \infty} X_t = \limsup_{t \rightarrow \infty} X_t$$

almost surely. □

Definition 1.4.2. A family of random variables $(X_t)_{t \geq 0}$ is uniformly integrable (U.I.) if

$$\lim_{M \rightarrow \infty} \sup_{t \geq 0} \mathbb{E}(|X_t| \mathbf{1}_{|X_t| > M}) = 0.$$

Remark. If $\sup_{t \geq 0} \mathbb{E}|X_t|^p < \infty$ for some $p > 1$, then U.I.

Theorem 1.4.2. If $X_n \xrightarrow{P} X$, then the following are equivalent:

$$\begin{cases} X_n \xrightarrow{L^1} X, \\ \mathbb{E}|X_n| \rightarrow \mathbb{E}|X|, \\ \{X_n\} \text{ is U.I..} \end{cases}$$

Proposition 1.4.3. Let $Z \in L^1$. Then

$$\{\mathbb{E}(Z \mid \mathcal{G}) : \mathcal{G} \text{ is a sub-}\sigma\text{-field of } \mathcal{F}\}$$

is uniformly integrable.

In particular, $\{\mathbb{E}(Z \mid \mathcal{F}_t)\}_{t \geq 0}$ is uniformly integrable, and it is a martingale by the tower property.

Proposition 1.4.4. A right-continuous martingale $(X_t)_{t \geq 0}$ is uniformly integrable if and only if there exists $X_\infty \in L^1$ such that

$$X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t).$$

Proof. “ $\Rightarrow$ ” Uniform integrability implies $\sup_{t \geq 0} \mathbb{E}|X_t| < \infty$.

Hence the martingale convergence theorem yields some X_∞ such that $X_t \xrightarrow{a.s.} X_\infty$ and

$$\mathbb{E} |X_\infty| = \mathbb{E} \left(\lim_{t \rightarrow \infty} |X_t| \right) \leq \liminf_{t \rightarrow \infty} \mathbb{E} |X_t| < \infty.$$

To prove that $X_t = \mathbb{E}(X_\infty | \mathcal{F}_t)$ for every $t \geq 0$, it is enough to show that

$$\mathbb{E}(X_t 1_A) = \mathbb{E}(X_\infty 1_A), \quad \forall A \in \mathcal{F}_t.$$

For any $s \geq 0$, the martingale property gives

$$\mathbb{E}(X_{t+s} | \mathcal{F}_t) = X_t,$$

and hence

$$\mathbb{E}(X_{t+s} 1_A) = \mathbb{E}(X_t 1_A).$$

On the other hand,

$$|\mathbb{E}(X_{t+s} 1_A) - \mathbb{E}(X_\infty 1_A)| \leq \mathbb{E} |X_{t+s} - X_\infty| \rightarrow 0.$$

Letting $s \rightarrow \infty$, we obtain

$$\mathbb{E}(X_t 1_A) = \mathbb{E}(X_\infty 1_A).$$

“ $\Leftarrow$ ” If there exists $X_\infty \in L^1$ such that $X_t = \mathbb{E}(X_\infty | \mathcal{F}_t)$, then $\{X_t\}_{t \geq 0}$ is uniformly integrable by the previous proposition. $\square$

Theorem 1.4.3 (Optional sampling Theorem). Let $(X_t)_{t \geq 0}$ be a right-continuous martingale. Let $S \leq T$ be two stopping times. Suppose that either

- ① $\{X_t\}$ is uniformly integrable;
- ② there exists a constant $K > 0$ such that $S, T \leq K$.

Then

$$\mathbb{E}(X_T | \mathcal{F}_S) = X_S.$$

In particular,

$$\mathbb{E}(X_T) = \mathbb{E}(X_S) = \mathbb{E}(X_0).$$

Proof. In case ①, let $Z = X_\infty$. In case ②, let $Z = X_K$.

In either case, we have $X_t = \mathbb{E}(Z | \mathcal{F}_t)$: for all $t \geq 0$ in case ①, and for all $t \leq K$ in case ②.

It therefore suffices to prove that $X_T = \mathbb{E}(Z | \mathcal{F}_T)$. Then similarly $X_S = \mathbb{E}(Z | \mathcal{F}_S)$.

Hence

$$\begin{aligned} \mathbb{E}(X_T | \mathcal{F}_S) &= \mathbb{E}(\mathbb{E}(Z | \mathcal{F}_T) | \mathcal{F}_S) \\ &= \mathbb{E}(Z | \mathcal{F}_S) \\ &= X_S. \end{aligned}$$

First suppose that T takes values in some countable set $\{t_1, t_2, \dots\}$. Then for every $A \in \mathcal{F}_T$,

$$\begin{aligned} \mathbb{E}(X_T 1_A) &= \sum_{n=1}^{\infty} \mathbb{E}(X_{t_n} 1_{\{T=t_n\}} 1_A) \\ &= \sum_{n=1}^{\infty} \mathbb{E}(X_{t_n} 1_{A \cap \{T=t_n\}}) \\ &= \sum_{n=1}^{\infty} \mathbb{E}(\mathbb{E}(Z \mid \mathcal{F}_{t_n}) 1_{A \cap \{T=t_n\}}) \\ &= \sum_{n=1}^{\infty} \mathbb{E}(Z 1_{A \cap \{T=t_n\}}) \\ &= \mathbb{E}(Z 1_A). \end{aligned}$$

because $A \cap \{T = t_n\} \in \mathcal{F}_{t_n}$ and $\mathbb{E}(Z \mid \mathcal{F}_{t_n}) = X_{t_n}$.

Now consider a general stopping time T . Define $T_k = \frac{[2^k T] + 1}{2^k}$.

For every $A \in \mathcal{F}_T \subseteq \mathcal{F}_{T_k}$, the first step gives

$$\mathbb{E}(Z 1_A) = \mathbb{E}(X_{T_k} 1_A).$$

Letting $k \rightarrow \infty$, we conclude that

$$\mathbb{E}(Z 1_A) = \mathbb{E}(X_T 1_A).$$

Indeed, $X_{T_k} \xrightarrow{a.s.} X_T$ by right-continuity, and $\{X_{T_k}\}$ is uniformly integrable because $X_{T_k} = \mathbb{E}(Z \mid \mathcal{F}_{T_k})$. Hence $X_{T_k} \xrightarrow{L^1} X_T$. $\square$

Theorem 1.4.4 (Doob's Maximal Inequality). Let $(X_t)_{t \geq 0}$ be a right-continuous submartingale. Then for every $\lambda > 0$,

$$\begin{aligned} \mathbb{P}(\sup_{0 \leq s \leq t} X_s > \lambda) &\leq \frac{1}{\lambda} \mathbb{E}X_t^+, \\ \mathbb{P}(\inf_{0 \leq s \leq t} X_s < -\lambda) &\leq \frac{1}{\lambda} (\mathbb{E}X_t^+ - \mathbb{E}X_0). \end{aligned}$$

Proof. For $\lambda > 0$, let

$$T = \inf\{t \geq 0 : X_t > \lambda\}.$$

Then

$$\mathbb{P}(\sup_{0 \leq s \leq t} X_s > \lambda) = \mathbb{P}(T \leq t).$$

Consider the stopped process $\{X_{t \wedge T}, t \geq 0\}$. Then

$$\begin{aligned} \lambda \mathbb{P}(T \leq t) &= \lambda \mathbb{E}(\mathbf{1}_{T \leq t}) \leq \mathbb{E}(X_T \mathbf{1}_{T \leq t}) \\ &= \mathbb{E}(X_{T \wedge t} \mathbf{1}_{T \leq t}) \\ &= \mathbb{E}(X_{T \wedge t}^+ \mathbf{1}_{T \leq t}) \\ &= \mathbb{E}(X_{T \wedge t}^+) \leq \mathbb{E}(X_t^+). \end{aligned}$$

Next let

$$S = \inf\{t \geq 0 : X_t < -\lambda\}.$$

Then

$$\mathbb{P}(\inf_{0 \leq s \leq t} X_s < -\lambda) = \mathbb{P}(S \leq t).$$

Therefore

$$\lambda \mathbb{P}(S \leq t) = \lambda \mathbb{E}(\mathbf{1}_{S \leq t}).$$

Because $X_S \leq -\lambda$, $\lambda \leq -X_S$, so

$$\lambda \mathbb{P}(S \leq t) \leq \mathbb{E}(-X_S \mathbf{1}_{S \leq t}) = -\mathbb{E}(X_{S \wedge t} \mathbf{1}_{S \leq t}).$$

Applying optional stopping to the submartingale, we obtain

$$\begin{aligned} \mathbb{E}X_0 &\leq \mathbb{E}X_{t \wedge S} = \mathbb{E}(X_{t \wedge S} \mathbf{1}_{S \leq t}) + \mathbb{E}(X_{t \wedge S} \mathbf{1}_{S > t}) \\ &\leq -\lambda \mathbb{P}(S \leq t) + \mathbb{E}(X_{t \wedge S}^+) \\ &\leq -\lambda \mathbb{P}(S \leq t) + \mathbb{E}(X_t^+). \end{aligned}$$

□

Theorem 1.4.5. $\forall p > 1, \mathbb{E} \left(\sup_{s \leq t} |X_s| \right)^p \leq \left(\frac{p}{p-1} \right)^p \mathbb{E} |X_t|^p.$

Proof. Let $Y = \sup_{0 \leq s \leq t} |X_s|$.

$$\begin{aligned} \lambda \mathbb{P}(Y \geq \lambda) &\leq \mathbb{E}(|X_{T \wedge t}| \mathbf{1}_{T \leq t}) \\ &= \mathbb{E}(|X_{T \wedge t}|) - \mathbb{E}(|X_{T \wedge t}| \mathbf{1}_{T > t}) \\ &\leq \mathbb{E}(|X_t|) - \mathbb{E}(|X_t| \mathbf{1}_{T > t}) \\ &= \mathbb{E}|X_t| - \mathbb{E}(|X_t| \mathbf{1}_{Y < \lambda}) \\ &= \mathbb{E}(|X_t| \mathbf{1}_{Y \geq \lambda}). \end{aligned}$$

Hence $\mathbb{P}(Y \geq \lambda) \leq \frac{1}{\lambda} \mathbb{E}(|X_t| \mathbf{1}_{Y \geq \lambda})$.

$$\begin{aligned}
 \mathbb{E}Y^p &= \int_0^\infty p\lambda^{p-1} \mathbb{P}(Y \geq \lambda) d\lambda \\
 &\leq \int_0^\infty p\lambda^{p-2} \mathbb{E}(|X_t| \mathbf{1}_{Y \geq \lambda}) d\lambda \\
 &= \mathbb{E} \left(|X_t| \int_0^\infty p\lambda^{p-2} \mathbf{1}_{Y \geq \lambda} d\lambda \right) \\
 &= \mathbb{E} \left(|X_t| \int_0^Y p\lambda^{p-2} d\lambda \right) \\
 &= \mathbb{E} \left(|X_t| \frac{p}{p-1} Y^{p-1} \right) \\
 &\leq \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{\frac{1}{p}} (\mathbb{E}(Y^{p-1})^q)^{\frac{1}{q}} \\
 &= \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{\frac{1}{p}} (\mathbb{E}Y^p)^{\frac{1}{q}}.
 \end{aligned}$$

Therefore

$$(\mathbb{E}Y^p)^{\frac{1}{p}} \leq \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{\frac{1}{p}}.$$

To justify the argument rigorously, consider $Y \wedge k$ for $k > 0$. Then

$$\begin{aligned}
 \mathbb{E}(Y \wedge k)^p &= \int_0^\infty p\lambda^{p-1} \mathbb{P}(Y \wedge k \geq \lambda) d\lambda \\
 &= \int_0^k p\lambda^{p-1} \mathbb{P}(Y \wedge k \geq \lambda) d\lambda + \underbrace{\int_k^\infty p\lambda^{p-1} \mathbb{P}(Y \wedge k \geq \lambda) d\lambda}_{=0} \\
 &= \int_0^k p\lambda^{p-1} \mathbb{P}(Y \geq \lambda) d\lambda \\
 &= \int_0^k p\lambda^{p-1} \mathbb{P}(Y \geq \lambda) d\lambda \\
 &\leq \mathbb{E} \left(|X_t| \int_0^k p\lambda^{p-1} \mathbf{1}_{Y \geq \lambda} d\lambda \right) \\
 &= \mathbb{E} \left(|X_t| \int_0^{k \wedge Y} p\lambda^{p-2} d\lambda \right) \\
 &= \mathbb{E} \left[|X_t| \frac{p}{p-1} (Y \wedge k)^{p-1} \right].
 \end{aligned}$$

$$\implies \mathbb{E}(Y \wedge k)^p \leq \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{\frac{1}{p}} (\mathbb{E}(Y \wedge k)^p)^{\frac{1}{q}}.$$

$$\implies (\mathbb{E}(Y \wedge k)^p)^{\frac{1}{p}} \leq \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{\frac{1}{p}}.$$

Letting $k \rightarrow \infty$, we obtain

$$(\mathbb{E}Y^p)^{\frac{1}{p}} \leq \frac{p}{p-1} (\mathbb{E}|X_t|^p)^{\frac{1}{p}}.$$

□

1.4.1 The Doob-Meyer Decomposition

We now turn to the continuous-time analogue of the discrete Doob decomposition.

Recall the discrete-time Doob decomposition

$$X_n = M_n + A_n,$$

where X_n is a submartingale, M_n is a martingale, and A_n is a predictable increasing process.

$$\begin{aligned} \mathbb{E}(M_{n+1} \mid \mathcal{F}_n) &= M_n \\ \implies \mathbb{E}(X_{n+1} - A_{n+1} \mid \mathcal{F}_n) &= X_n - A_n \\ \implies \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - A_{n+1} &= X_n - A_n \\ \implies A_{n+1} - A_n &= \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - X_n \geq 0. \end{aligned}$$

$$\begin{cases} A_n = \sum_{k=0}^{n-1} (\mathbb{E}(X_{k+1} \mid \mathcal{F}_k) - X_k) \\ A_0 = 0. \end{cases}$$

For a right-continuous submartingale $(X_t)_{t \geq 0}$, we seek a decomposition of the form

$$X_t = M_t + A_t,$$

where M_t is a martingale and A_t is an increasing and “predictable” process.

So we first clarify what properties the increasing part should satisfy.

Definition 1.4.3. An adapted process A is called increasing if, almost surely,

- (a) $A_0 = 0$;
- (b) $t \mapsto A_t$ is nondecreasing and right-continuous;
- (c) $\mathbb{E}(A_t) < \infty, \forall t \geq 0$.

We say that A is integrable if

$$\mathbb{E}(A_\infty) = \mathbb{E}(\lim_{t \rightarrow \infty} A_t) < \infty.$$

Definition 1.4.4. An increasing process A is called natural if for every bounded, right-continuous martingale $(M_t)_{t \geq 0}$ and every $t > 0$,

$$\mathbb{E} \int_0^t M_s dA_s = \mathbb{E} \int_0^t M_{s-} dA_s.$$

Remark. If $s \mapsto A_s$ is a.s. continuous, then A is natural.

Definition 1.4.5. An increasing sequence $(A_n, \mathcal{F}_n)$ is called natural if for every bounded martingale $\{M_n, \mathcal{F}_n\}$,

$$\mathbb{E}(M_n A_n) = \mathbb{E} \left(\sum_{k=1}^n M_{k-1} (A_k - A_{k-1}) \right).$$

Proposition 1.4.5. In discrete time, predictability is equivalent to naturality.

Proof. “ $\Rightarrow$ ” Assume that A is predictable, i.e. $A_n \in \mathcal{F}_{n-1}$. We show that A is natural.

Indeed,

$$\begin{aligned}
 \mathbb{E}(M_n A_n) &= \mathbb{E} \left(\sum_{k=1}^n M_k A_k - \sum_{k=1}^n M_{k-1} A_{k-1} \right) \\
 &= \sum_{k=1}^n \mathbb{E}(M_k A_k) - \sum_{k=1}^n \mathbb{E}(M_{k-1} A_{k-1}) \\
 &\iff \sum_{k=2}^n \mathbb{E}(M_{k-1} A_{k-1}) + \mathbb{E}(M_n A_n) = \sum_{k=1}^n \mathbb{E}(M_{k-1} A_k) \\
 &\iff \sum_{k=1}^n \mathbb{E}(M_k A_k) = \sum_{k=1}^n \mathbb{E}(M_{k-1} A_k) \\
 &\iff \sum_{k=1}^n \mathbb{E}[(M_k - M_{k-1}) A_k] = 0.
 \end{aligned}$$

Since $A_n \in \mathcal{F}_{n-1}$,

$$\mathbb{E}(A_n(M_n - M_{n-1})) = \mathbb{E} \left[A_n \underbrace{\mathbb{E}(M_n - M_{n-1} \mid \mathcal{F}_{n-1})}_{=0} \right] = 0.$$

Therefore A is natural.

“ $\Leftarrow$ ” Now assume that A is natural. Then for every bounded martingale (M_n) ,

$$\mathbb{E}[A_n(M_n - M_{n-1})] = 0, \quad \forall n \geq 1.$$

Our goal is to prove that

$$A_n = \mathbb{E}(A_n \mid \mathcal{F}_{n-1}) \quad \text{a.s.},$$

which is equivalent to $A_n \in \mathcal{F}_{n-1}$.

$$\begin{aligned}
 \mathbb{E}[M_n(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1}))] &= \mathbb{E}(M_n A_n) - \mathbb{E}[M_n \mathbb{E}(A_n \mid \mathcal{F}_{n-1})] \\
 &= \mathbb{E}(M_n A_n) - \mathbb{E}[\mathbb{E}(A_n \mid \mathcal{F}_{n-1}) \mathbb{E}(M_n \mid \mathcal{F}_{n-1})] \\
 &= \mathbb{E}(M_n A_n) - \mathbb{E}[\mathbb{E}(A_n \mid \mathcal{F}_{n-1}) M_{n-1}] \\
 &= \mathbb{E}(M_n A_n) - \mathbb{E}(A_n M_{n-1}) \\
 &= \mathbb{E}[A_n(M_n - M_{n-1})] = 0.
 \end{aligned}$$

Fix $n \geq 1$, and let

$$X = \text{sgn}(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1}))$$

Then $X(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})) = |A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})|$.

Define

$$M_k = \begin{cases} \mathbb{E}(X \mid \mathcal{F}_k), & 0 \leq k \leq n-1 \\ X, & k = n \end{cases}$$

Then (M_k) is a bounded martingale.

$$\begin{aligned} 0 &= \mathbb{E}[M_n(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1}))] \\ &= \mathbb{E}[A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})] \end{aligned}$$

Therefore $A_n = \mathbb{E}(A_n \mid \mathcal{F}_{n-1})$ a.s.

Hence $A_n \in \mathcal{F}_{n-1}$, so A is predictable. $\square$

The previous proposition gives the discrete-time characterization of natural increasing processes. We now explain why the continuous-time definition is the correct analogue.

$$\begin{aligned} \mathbb{E}(M_n A_n) &= \sum_{k=1}^n \mathbb{E}(M_{k-1}(A_k - A_{k-1})) \\ \iff \mathbb{E}(M_t A_t) &= \mathbb{E}\left(\int_0^t M_{s-} dA_s\right) = \mathbb{E}\left(\int_0^t M_s dA_s\right). \end{aligned}$$

Indeed, the discrete sum should be viewed as a Riemann–Stieltjes approximation of the corresponding continuous-time integral.

$$\int_0^t M_s dA_s = \lim_{\|\Pi\| \rightarrow 0} \int_0^t M_s^\Pi dA_s.$$

where $\Pi = \{t_0, t_1, \dots, t_n\} \subseteq [0, t]$ with $0 = t_0 < t_1 < \dots < t_n = t$ and $\|\Pi\| = \max_{1 \leq k \leq n} (t_k - t_{k-1})$. By right-continuity,

$$M_s^\Pi = \sum_{k=1}^n M_{t_k} \mathbf{1}_{(t_{k-1}, t_k]}(s) \xrightarrow{a.s.} M_s.$$

For each Π ,

$$\begin{aligned} \mathbb{E} \int_0^t M_s^\Pi dA_s &= \mathbb{E} \sum_{k=1}^n M_{t_k} (A_{t_k} - A_{t_{k-1}}) \\ &= \sum_{k=1}^n \mathbb{E}(M_{t_k} A_{t_k}) - \sum_{k=1}^n \mathbb{E}(M_{t_k} A_{t_{k-1}}) \\ &= \mathbb{E}(M_{t_n} A_{t_n}) + \sum_{k=1}^{n-1} \mathbb{E}(M_{t_k} A_{t_k}) - \sum_{k=2}^n \mathbb{E}(M_{t_k} A_{t_{k-1}}) \\ &= \mathbb{E}(M_t A_t) + \sum_{k=1}^{n-1} \mathbb{E}(M_{t_k} A_{t_k}) - \sum_{k=1}^{n-1} \mathbb{E}(M_{t_{k+1}} A_{t_k}) \\ &= \mathbb{E}(M_t A_t) - \underbrace{\sum_{k=1}^{n-1} \mathbb{E}(A_{t_k} (M_{t_{k+1}} - M_{t_k}))}_{=0}. \end{aligned}$$

Thus the continuous-time identity is exactly the limit of the discrete-time naturality relation.

Definition 1.4.6. Let

$$S = \{\text{stopping time } T \text{ with } T < \infty \text{ a.s.}\}.$$

We say that a right-continuous process $\{X_t\}_{t \geq 0}$ is of class D if $\{X_T\}_{T \in S}$ is uniformly integrable. In particular, this implies that $\{X_t\}_{t \geq 0}$ itself is uniformly integrable.

Fix $a > 0$, and let

$$S_a = \{\text{stopping time } T \text{ with } T \leq a \text{ a.s.}\}.$$

We say that $\{X_t\}_{t \geq 0}$ is of class DL if $\{X_T\}_{T \in S_a}$ is uniformly integrable for every $a > 0$.

Theorem 1.4.6 (Doob-Meyer Decomposition). If $\{X_t\}$ is of class DL , then $X_t = M_t + A_t$, where $\{M_t\}$ is a martingale and $\{A_t\}$ is increasing.

If (A_t) is natural, then (M_t, A_t) is unique.

Proof. Assume that

$$X_t = M_t + A_t = M'_t + A'_t,$$

where (M_t) and (M'_t) are martingales, and (A_t) and (A'_t) are natural increasing processes.

Let

$$B_t = M_t - M'_t = A'_t - A_t.$$

Then (B_t) is a martingale, and also a process of bounded variation. Moreover, it is natural.

To prove uniqueness, let $(\xi_t)_{t \geq 0}$ be any bounded right-continuous martingale. Then

$$\begin{aligned} \mathbb{E}[\xi_t(A_t - A'_t)] &= \mathbb{E} \int_0^t \xi_{s-} (dA_s - dA'_s) \\ &= \mathbb{E} \left[\int_0^t \xi_{s-} dB_s \right] \\ &= \lim_{n \rightarrow \infty} \mathbb{E} \sum_{j=1}^{m_n} \xi_{t_{j-1}^{(n)}} \left[B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}} \right], \end{aligned}$$

where $\Pi_n = \{t_0^{(n)}, \dots, t_{m_n}^{(n)}\}$ is a partition of $[0, t]$.

For each j ,

$$\mathbb{E} \left[\xi_{t_{j-1}^{(n)}} (B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}}) \right] = \mathbb{E} \left[\underbrace{\xi_{t_{j-1}^{(n)}} \mathbb{E}(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}} \mid \mathcal{F}_{t_{j-1}^{(n)}})}_{=0} \right].$$

Hence $\mathbb{E}[\xi_t(A_t - A'_t)] = 0$.

Now let ξ be any bounded $\mathcal{F}_t$ -measurable random variable. Applying the preceding identity with the martingale

$$\xi_s = \mathbb{E}(\xi \mid \mathcal{F}_s), \quad 0 \leq s \leq t,$$

we get

$$\mathbb{E} [\xi(A_t - A'_t)] = 0.$$

Taking $\xi = \text{sgn}(A_t - A'_t)$, we obtain

$$\mathbb{E} |A_t - A'_t| = 0.$$

Hence $A_t = A'_t$ a.s. for every $t > 0$. In particular, this holds for all $q \in \mathbb{Q}_+$. By right-continuity, we conclude that $A_t = A'_t$ for all $t \geq 0$ a.s.

Existence: Fix $a > 0$.

It suffices to construct the decomposition on $[0, a]$.

Consider

$$Y_t \triangleq X_t - \mathbb{E}(X_a | \mathcal{F}_t), \quad 0 \leq t \leq a.$$

Then $(Y_t)_{0 \leq t \leq a}$ is a submartingale, because

$$\begin{aligned} \mathbb{E}(Y_t | \mathcal{F}_s) &= \mathbb{E}(X_t | \mathcal{F}_s) - \mathbb{E}(X_a | \mathcal{F}_s) \\ &\geq X_s - \mathbb{E}(X_a | \mathcal{F}_s) = Y_s, \quad 0 \leq s \leq t \leq a. \end{aligned}$$

Let $\Pi_n = \{t_0^{(n)}, \dots, t_{2^n}^{(n)}\}$ with $t_j^{(n)} = \frac{j}{2^n}a$, $0 \leq j \leq 2^n$. For each n , the discrete-time Doob decomposition gives

$$Y_{t_j^{(n)}} = M_{t_j^{(n)}}^{(n)} + A_{t_j^{(n)}}^{(n)}, \quad 0 \leq j \leq 2^n,$$

where

$$A_{t_j^{(n)}}^{(n)} = \sum_{k=0}^{j-1} \mathbb{E} \left(Y_{t_{k+1}^{(n)}} - Y_{t_k^{(n)}} \middle| \mathcal{F}_{t_k^{(n)}} \right).$$

Since

$$\begin{cases} Y_0 = X_0 - \mathbb{E}(X_a | \mathcal{F}_0) \leq 0, \\ Y_a = X_a - \mathbb{E}(X_a | \mathcal{F}_a) = 0, \end{cases}$$

we have

$$0 = Y_a = M_a^{(n)} + A_a^{(n)}.$$

Hence

$$M_{t_j^{(n)}}^{(n)} = \mathbb{E}(M_a^{(n)} | \mathcal{F}_{t_j^{(n)}}) = -\mathbb{E}(A_a^{(n)} | \mathcal{F}_{t_j^{(n)}}),$$

and therefore

$$Y_{t_j^{(n)}} = A_{t_j^{(n)}}^{(n)} - \mathbb{E}(A_a^{(n)} | \mathcal{F}_{t_j^{(n)}}).$$

We will show that $\{A_a^{(n)}\}_{n \geq 1}$ is uniformly integrable. Then, by the Dunford-Pettis compactness criterion, there exists an integrable random variable A_a such that

$$A_a^{(n_k)} \rightharpoonup A_a \quad \text{weakly in } L^1$$

along some subsequence. Equivalently, for every bounded random variable ξ ,

$$\lim_{k \rightarrow \infty} \mathbb{E}(\xi A_a^{(n_k)}) = \mathbb{E}(\xi A_a).$$

Define

$$A_t = Y_t + \mathbb{E}(A_a \mid \mathcal{F}_t), \quad 0 \leq t \leq a.$$

By Problem 4.11, the convergence $\mathbb{E}(\xi A_a^{(n_k)}) \rightarrow \mathbb{E}(\xi A_a)$ for every bounded random variable ξ implies

$$\mathbb{E}(\xi \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)) \rightarrow \mathbb{E}(\xi \mathbb{E}(A_a \mid \mathcal{F}_t)),$$

that is, $\mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)$ converges weakly in L^1 to $\mathbb{E}(A_a \mid \mathcal{F}_t)$.

Fix a dyadic time t . Then, for all sufficiently large k , we have

$$\begin{cases} A_t = Y_t + \mathbb{E}(A_a \mid \mathcal{F}_t), \\ A_t^{(n_k)} = Y_t + \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t) \end{cases}$$

because t belongs to the dyadic partition Π_{n_k} once k is large enough. Therefore

$$A_t^{(n_k)} - A_t = \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t) - \mathbb{E}(A_a \mid \mathcal{F}_t).$$

Since $\mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)$ converges weakly in L^1 to $\mathbb{E}(A_a \mid \mathcal{F}_t)$, it follows that

$$\mathbb{E}(\xi A_t^{(n_k)}) \rightarrow \mathbb{E}(\xi A_t)$$

for every bounded random variable ξ .

For any dyadic times $0 \leq s < t \leq a$,

$$\mathbb{E}(\xi(A_t - A_s)) = \lim_{k \rightarrow \infty} \mathbb{E}(\xi(A_t^{(n_k)} - A_s^{(n_k)})) \geq 0$$

whenever $\xi \geq 0$.

Taking $\xi = 1_{A_s > A_t}$, we get

$$\mathbb{E}[(A_t - A_s)1_{A_s > A_t}] \geq 0.$$

Therefore $A_t - A_s \geq 0$ a.s.

Since

$$A_0^{(n)} = 0 = Y_0 + \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_0),$$

passing to the limit gives

$$A_0 = Y_0 + \mathbb{E}(A_a \mid \mathcal{F}_0) = 0.$$

Moreover, for any dyadic time t ,

$$X_t - \mathbb{E}(X_a \mid \mathcal{F}_t) = Y_t = A_t^{(n_k)} - \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)$$

for all sufficiently large k .

Passing to the limit along the subsequence gives

$$Y_t = A_t - \mathbb{E}(A_a \mid \mathcal{F}_t) = X_t - \mathbb{E}(X_a \mid \mathcal{F}_t).$$

Therefore

$$X_t = A_t + \mathbb{E}(X_a - A_a \mid \mathcal{F}_t)$$

for every dyadic time $t \in [0, a]$. By right-continuity, the same identity then holds for all $t \in [0, a]$.

Define

$$M_t = \mathbb{E}(X_a - A_a \mid \mathcal{F}_t), \quad 0 \leq t \leq a.$$

Then $(M_t)_{0 \leq t \leq a}$ is a martingale and

$$X_t = A_t + M_t.$$

Finally, we prove that (A_t) is natural. It is enough to show that for every bounded right-continuous martingale (ξ_t) ,

$$\mathbb{E} \int_0^a \xi_s dA_s = \mathbb{E} \int_0^a \xi_{s-} dA_s.$$

We first note that

$$\mathbb{E} \int_0^a \xi_s dA_s = \mathbb{E}(\xi_a A_a) = \lim_{n \rightarrow \infty} \mathbb{E}(\xi_a A_a^{(n)}).$$

On the other hand,

$$\mathbb{E} \int_0^a \xi_{s-} dA_s = \lim_{n \rightarrow \infty} \mathbb{E} \sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}}).$$

Since $\{A_{t_j^{(n)}}^{(n)}\}_{0 \leq j \leq 2^n}$ is predictable, and $t \mapsto \mathbb{E}(A_a \mid \mathcal{F}_t)$ is a martingale,

$$\begin{aligned} \mathbb{E}(\xi_a A_a^{(n)}) &= \mathbb{E} \left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (A_{t_j^{(n)}}^{(n)} - A_{t_{j-1}^{(n)}}^{(n)}) \right) \\ &= \mathbb{E} \left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (Y_{t_j^{(n)}} - Y_{t_{j-1}^{(n)}}) \right) \\ &= \mathbb{E} \left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} \left[(A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}}) - (\mathbb{E}(A_a \mid \mathcal{F}_{t_j^{(n)}}) - \mathbb{E}(A_a \mid \mathcal{F}_{t_{j-1}^{(n)}})) \right] \right) \\ &= \mathbb{E} \left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}}) \right). \end{aligned}$$

Here the last equality follows because $t \mapsto \mathbb{E}(A_a \mid \mathcal{F}_t)$ is a martingale. Letting $n \rightarrow \infty$, we obtain

$$\mathbb{E} \int_0^a \xi_s dA_s = \mathbb{E} \int_0^a \xi_{s-} dA_s.$$

Therefore (A_t) is natural. □

1.5 Square Integrable

Definition 1.5.1. Let $(X_t)_{t \geq 0}$ be a right-continuous martingale. We say that X is square-integrable if

$$\mathbb{E} X_t^2 < \infty, \quad \forall t \geq 0.$$

Throughout this section, we assume $X_0 = 0$ and write $X \in \mathcal{M}_2$.

Then $Y_t = X_t^2$ is a submartingale. Recall that a process is of class DL if, for every $a > 0$, the family $\{Y_T : T \leq a \text{ is a stopping time}\}$ is uniformly integrable. By the Doob–Meyer decomposition,

$$Y_t = X_t^2 = M_t + A_t, \quad t \geq 0.$$

If $X \in \mathcal{M}_2^c$ is continuous, then we may choose M_t and A_t to be continuous as well.

Definition 1.5.2. For $X \in \mathcal{M}_2$, the quadratic variation of X is defined by $\langle X \rangle_t \triangleq A_t$, that is, $\langle X \rangle_t$ is the unique adapted natural increasing process such that $\langle X \rangle_0 = 0$ and

$$X_t^2 - \langle X \rangle_t$$

is a martingale.

Remark. The notation $\langle X \rangle_t$ is introduced here through the Doob–Meyer decomposition of the submartingale X_t^2 .

If $X, Y \in \mathcal{M}_2$, then

$$\begin{cases} (X_t + Y_t)^2 - \langle X + Y \rangle_t \text{ is a martingale,} \\ (X_t - Y_t)^2 - \langle X - Y \rangle_t \text{ is a martingale.} \end{cases}$$

Subtracting the second relation from the first, we see that

$$4X_t Y_t - (\langle X + Y \rangle_t - \langle X - Y \rangle_t)$$

is a martingale.

Definition 1.5.3. For $X, Y \in \mathcal{M}_2$, we define the cross variation by

$$\langle X, Y \rangle_t \triangleq \frac{1}{4} (\langle X + Y \rangle_t - \langle X - Y \rangle_t),$$

so that

$$X_t Y_t - \langle X, Y \rangle_t$$

is a martingale.

Remark. $\langle X, X \rangle_t = \langle X \rangle_t$.

Definition 1.5.4. Let $\Pi = \{t_0, \dots, t_m\}$ be a partition of $[0, t]$. The p -th variation of X over Π is defined by

$$V_t^{(p)}(\Pi) = \sum_{k=1}^m |X_{t_k} - X_{t_{k-1}}|^p.$$

When $p = 2$, we call $V_t^{(2)}(\Pi)$ the quadratic variation along the partition Π .

Theorem 1.5.1. Let $X \in \mathcal{M}_2^c$. For any partition Π of $[0, t]$, we have

$$V_t^{(2)}(\Pi) \xrightarrow{P} \langle X \rangle_t$$

as $\|\Pi\| \rightarrow 0$.

Proof. Let $\Pi = \{t_0, \dots, t_m\}$ and write

$$\Delta_k X = X_{t_k} - X_{t_{k-1}}, \quad \Delta_k \langle X \rangle = \langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}.$$

Then

$$V_t^{(2)}(\Pi) - \langle X \rangle_t = \sum_{k=1}^m ((\Delta_k X)^2 - \Delta_k \langle X \rangle).$$

Hence

$$\begin{aligned} & \mathbb{E} \left(V_t^{(2)}(\Pi) - \langle X \rangle_t \right)^2 \\ &= \mathbb{E} \left[\sum_{k=1}^m ((\Delta_k X)^2 - \Delta_k \langle X \rangle) \right]^2 \\ &= \sum_{k=1}^m \mathbb{E} \left[((\Delta_k X)^2 - \Delta_k \langle X \rangle)^2 \right] \\ & \quad + 2 \sum_{1 \leq j < k \leq m} \mathbb{E} \left[((\Delta_k X)^2 - \Delta_k \langle X \rangle) ((\Delta_j X)^2 - \Delta_j \langle X \rangle) \right]. \end{aligned}$$

The cross terms vanish. Indeed, for $j < k$, the random variable

$$(\Delta_j X)^2 - \Delta_j \langle X \rangle$$

is $\mathcal{F}_{t_{k-1}}$ -measurable, while

$$\mathbb{E} ((\Delta_k X)^2 - \Delta_k \langle X \rangle | \mathcal{F}_{t_{k-1}}) = 0$$

because $X_s^2 - \langle X \rangle_s$ is a martingale. Therefore

$$\mathbb{E} \left[((\Delta_k X)^2 - \Delta_k \langle X \rangle) ((\Delta_j X)^2 - \Delta_j \langle X \rangle) \right] = 0.$$

Consequently,

$$\begin{aligned} \mathbb{E} \left(V_t^{(2)}(\Pi) - \langle X \rangle_t \right)^2 &= \sum_{k=1}^m \mathbb{E} \left[((\Delta_k X)^2 - \Delta_k \langle X \rangle)^2 \right] \\ &\leq 2 \sum_{k=1}^m \mathbb{E} (\Delta_k X)^4 + 2 \sum_{k=1}^m \mathbb{E} (\Delta_k \langle X \rangle)^2 \\ &= I + II \\ &= 2\mathbb{E} V_t^{(4)}(\Pi) + II \end{aligned}$$

We first consider the bounded case. Assume that

$$|X_s| \leq K, \quad \langle X \rangle_s \leq K, \quad 0 \leq s \leq t.$$

Lemma 1.5.1. Let $X \in \mathcal{M}_2$ satisfy $|X_s| \leq K < \infty$ for all $0 \leq s \leq t$. Then

$$\mathbb{E} \left[V_t^{(2)}(\Pi)^2 \right] \leq 6K^4.$$

Assuming this lemma, we obtain

$$V_t^{(4)}(\Pi) \leq V_t^{(2)}(\Pi) m_t^2(X; \|\Pi\|),$$

where

$$m_t(X, \delta) = \sup\{|X_r - X_s| : |s - r| \leq \delta, 0 \leq r < s \leq t\}.$$

Hence, by Cauchy–Schwarz,

$$\begin{aligned} \mathbb{E} \left[V_t^{(4)}(\Pi) \right] &\leq \mathbb{E} \left[V_t^{(2)}(\Pi) m_t^2(X, \|\Pi\|) \right] \\ &\leq \left(\mathbb{E} \left[V_t^{(2)}(\Pi)^2 \right] \right)^{1/2} (\mathbb{E} m_t^4(X, \|\Pi\|))^{1/2} \\ &\leq (6K^4)^{1/2} (\mathbb{E} m_t^4(X, \|\Pi\|))^{1/2}. \end{aligned}$$

Since X is continuous and bounded on $[0, t]$, bounded convergence yields

$$\mathbb{E} m_t^4(X, \|\Pi\|) \rightarrow 0$$

as $\|\Pi\| \rightarrow 0$. Thus $I \rightarrow 0$.

For II , note that $\langle X \rangle$ is continuous and increasing, so

$$\sum_{k=1}^m (\Delta_k \langle X \rangle)^2 \leq \sup_{1 \leq k \leq m} |\Delta_k \langle X \rangle| \langle X \rangle_t.$$

Since $\langle X \rangle$ is uniformly continuous on $[0, t]$ and bounded by K , it follows that

$$\mathbb{E} \left[\sup_{1 \leq k \leq m} |\Delta_k \langle X \rangle| \langle X \rangle_t \right] \rightarrow 0$$

as $\|\Pi\| \rightarrow 0$. Hence $II \rightarrow 0$ as well.

Therefore, in the bounded case,

$$\mathbb{E} \left[\left(V_t^{(2)}(\Pi) - \langle X \rangle_t \right)^2 \right] \rightarrow 0$$

as $\|\Pi\| \rightarrow 0$, and hence

$$V_t^{(2)}(\Pi) \xrightarrow{P} \langle X \rangle_t.$$

For a general $X \in \mathcal{M}_2^c$, we use localization. For $n \geq 1$, define

$$T_n = \inf \{ t \geq 0 : |X_t| \geq n \text{ or } \langle X \rangle_t \geq n \}.$$

Then $T_n \uparrow \infty$ a.s., and

$$X_t^{(n)} = X_{t \wedge T_n}$$

is a bounded martingale. Moreover,

$$X_{t \wedge T_n}^2 - \langle X \rangle_{t \wedge T_n}$$

is a martingale, so by uniqueness of the quadratic variation,

$$\langle X^{(n)} \rangle_t = \langle X \rangle_{t \wedge T_n}.$$

Hence

$$\begin{cases} |X_s^{(n)}| \leq n, \\ \langle X^{(n)} \rangle_s \leq n, \end{cases} \quad 0 \leq s \leq t,$$

and the bounded case gives

$$\mathbb{E} \left[\left(V_{t \wedge T_n}^{(2)}(X) - \langle X \rangle_{t \wedge T_n} \right)^2 \right] \rightarrow 0$$

as $\|\Pi\| \rightarrow 0$.

Now fix $\varepsilon > 0$ and $\eta > 0$. Choose n so large that

$$\mathbb{P}(T_n < t) \leq \frac{\eta}{2}.$$

Then

$$\begin{aligned} \mathbf{P}\left(\left|V_t^{(2)}(X) - \langle X \rangle_t\right| > \varepsilon\right) &\leq \mathbf{P}(T_n < t) + \mathbf{P}\left(T_n \geq t, \left|V_{t \wedge T_n}^{(2)}(X) - \langle X \rangle_{t \wedge T_n}\right| > \varepsilon\right) \\ &\leq \frac{\eta}{2} + \frac{1}{\varepsilon^2} \mathbb{E}\left[\left(V_{t \wedge T_n}^{(2)}(X) - \langle X \rangle_{t \wedge T_n}\right)^2\right]. \end{aligned}$$

Taking $\|\Pi\|$ small enough, the second term is at most $\eta/2$. Therefore

$$\mathbf{P}\left(\left|V_t^{(2)}(X) - \langle X \rangle_t\right| > \varepsilon\right) \leq \eta.$$

This proves that

$$V_t^{(2)}(\Pi) \xrightarrow{P} \langle X \rangle_t.$$

□

Corollary 1.5.1. If $V_t^{(2)}(\Pi) \xrightarrow{P} \langle X \rangle_t$, then

$$\begin{cases} \forall q > 2, & V_t^{(q)}(\Pi) \xrightarrow{P} 0, \\ \forall q < 2, & V_t^{(q)}(\Pi) \xrightarrow{P} \infty \end{cases}$$

on $\{\langle X \rangle_t > 0\}$.

We use the following notation:

$$\mathcal{M}_2 = \{X : \mathbb{E}X_t^2 < \infty, \forall t \geq 0\},$$

$$\mathcal{M}^2 = \left\{X : \sup_{t \geq 0} \mathbb{E}X_t^2 < \infty\right\},$$

and

$$\mathcal{M}_0^2 = \{X \in \mathcal{M}^2 : X_0 = 0\}.$$

1.6 L^2 -Theory

The space $\mathcal{M}^2$ is naturally identified with $L^2(\mathcal{F}_\infty)$ through the correspondence

$$X \longmapsto X_\infty.$$

Indeed, for every $X \in \mathcal{M}^2$, we have

$$\begin{cases} X_\infty = \lim_{t \rightarrow \infty} X_t & \text{a.s.}, \\ X_t \xrightarrow{L^2} X_\infty. \end{cases}$$

Conversely, if $X_\infty \in L^2(\mathcal{F}_\infty)$, then

$$X_t \triangleq \mathbb{E}(X_\infty | \mathcal{F}_t)$$

defines a martingale in $\mathcal{M}^2$. Moreover,

$$\mathbb{E}(X_t^2) = \mathbb{E}\left[\mathbb{E}(X_\infty|\mathcal{F}_t)^2\right] \leq \mathbb{E}\left[\mathbb{E}(X_\infty^2|\mathcal{F}_t)\right] = \mathbb{E}[X_\infty^2].$$

Thus, for $X, Y \in \mathcal{M}^2$, it is natural to define

$$\langle X, Y \rangle \triangleq \mathbb{E}[X_\infty Y_\infty].$$

Remark. Here $\langle X, Y \rangle$ denotes the inner product on the Hilbert space $\mathcal{M}^2$. It is different from the time-dependent bracket process $\langle X \rangle_t$ or $\langle X, Y \rangle_t$ introduced in the previous section.

Theorem 1.6.1 (Doob's L^2 inequality).

$$\mathbb{E}\left[\sup_{0 \leq s \leq t} |X_s|^2\right] \leq 4\mathbb{E}[X_t^2].$$

Letting $t \rightarrow \infty$, we obtain

$$\mathbb{E}\left[\sup_{s \geq 0} |X_s|^2\right] \leq 4\mathbb{E}[X_\infty^2].$$

The space $\mathcal{M}^2$ is complete. Indeed, if $\{M^{(n)}\}$ is a Cauchy sequence in $\mathcal{M}^2$, then $\{M_\infty^{(n)}\}$ is Cauchy in $L^2(\mathcal{F}_\infty)$. Hence there exists $M_\infty \in L^2(\mathcal{F}_\infty)$ such that

$$M_\infty^{(n)} \xrightarrow{L^2} M_\infty.$$

Let

$$M_t = \mathbb{E}(M_\infty | \mathcal{F}_t).$$

Then

$$\sup_{t \geq 0} |M_t^{(n)} - M_t| \xrightarrow{L^2} 0.$$

Proof. By Doob's L^2 inequality,

$$\begin{aligned} \mathbb{E}\left[\sup_{t \geq 0} |M_t^{(n)} - M_t|^2\right] &= \mathbb{E}\left[\sup_{t \geq 0} \left|\mathbb{E}(M_\infty^{(n)} - M_\infty | \mathcal{F}_t)\right|^2\right] \\ &\leq 4\mathbb{E}\left[\left(M_\infty^{(n)} - M_\infty\right)^2\right] \rightarrow 0. \end{aligned}$$

□

Corollary 1.6.1. The space $c\mathcal{M}^2$ of continuous martingales in $\mathcal{M}^2$ is closed in the topology of $\mathcal{M}^2$.

Proof. Let $M^{(n)} \rightarrow M$ in $\mathcal{M}^2$, with each $M^{(n)} \in c\mathcal{M}^2$. Then

$$\sup_{t \geq 0} |M_t^{(n)} - M_t| \xrightarrow{L^2} 0.$$

Hence, after passing to a subsequence if necessary,

$$\sup_{t \geq 0} |M_t^{(n_k)} - M_t| \xrightarrow{a.s.} 0.$$

Therefore $M^{(n_k)}$ converges uniformly almost surely to M , and since each $M^{(n_k)}$ has continuous sample paths, the limit process M is also continuous. $\square$

Definition 1.6.1 (Weakly orthogonal). Two martingales $M, N \in \mathcal{M}_0^2$ are said to be weakly orthogonal if

$$\mathbb{E}[M_\infty N_\infty] = 0.$$

Definition 1.6.2 (Strongly orthogonal). Two martingales $M, N \in \mathcal{M}_0^2$ are said to be strongly orthogonal if, for every stopping time T ,

$$\mathbb{E}[M_T N_T] = 0.$$

Lemma 1.6.1. Two martingales are strongly orthogonal if and only if MN is uniformly integrable.

Lemma 1.6.2. Let M be a progressively measurable process such that for every stopping time T ,

$$\begin{cases} \mathbb{E} |M_T| < \infty, \\ \mathbb{E} M_T = 0. \end{cases}$$

Then M is uniformly integrable.

Definition 1.6.3 (Stable subspaces of $\mathcal{M}_0^2$). A subspace $\mathcal{N}_0 \subseteq \mathcal{M}_0^2$ is called stable if

- ① $\mathcal{N}_0$ is a closed subspace;
- ② for every $N \in \mathcal{N}_0$ and every stopping time T , the stopped process N^T , defined by $N_t^T = N_{t \wedge T}$, also belongs to $\mathcal{N}_0$.

Example . $c\mathcal{M}_0^2$ is a stable subspace.

Theorem 1.6.2. Let $\mathcal{N}_0$ be a stable subspace of $\mathcal{M}_0^2$. Let $\mathcal{N}_0^\perp$ be the set of $Z \in \mathcal{M}_0^2$ which are weakly orthogonal to every element of $\mathcal{N}_0$, i.e.

$$\mathcal{N}_0^\perp = \{Z \in \mathcal{M}_0^2 : \langle Z, N \rangle = 0, \forall N \in \mathcal{N}_0\}.$$

Then $\mathcal{N}_0^\perp$ is a stable subspace of $\mathcal{M}_0^2$. Moreover, if $Z \in \mathcal{N}_0^\perp$ and $N \in \mathcal{N}_0$, then Z and N are strongly orthogonal.

Finally, every $M \in \mathcal{M}_0^2$ admits a decomposition

$$M = N + Z$$

with $N \in \mathcal{N}_0$ and $Z \in \mathcal{N}_0^\perp$.

Proof. Since $\mathcal{N}_0$ is a closed subspace of the Hilbert space $\mathcal{M}_0^2$, its orthogonal complement $\mathcal{N}_0^\perp$ is also a closed subspace, and every $M \in \mathcal{M}_0^2$ admits an orthogonal decomposition

$$M = N + Z$$

with $N \in \mathcal{N}_0$ and $Z \in \mathcal{N}_0^\perp$.

It remains to prove that $\mathcal{N}_0^\perp$ is stable and that orthogonality is in fact strong. Let $Z \in \mathcal{N}_0^\perp$, $N \in \mathcal{N}_0$, and let T be a stopping time. Then

$$\begin{aligned} \mathbb{E}[Z_\infty^T N_\infty] &= \mathbb{E}[Z_T N_\infty] \\ &= \mathbb{E}[Z_T \mathbb{E}(N_\infty | \mathcal{F}_T)] \\ &= \mathbb{E}[Z_T N_T] \\ &= \mathbb{E}[\mathbb{E}(Z_\infty | \mathcal{F}_T) N_T] \\ &= \mathbb{E}[Z_\infty N_T] \\ &= \mathbb{E}[Z_\infty N_\infty^T]. \end{aligned}$$

Since $\mathcal{N}_0$ is stable, we have $N^T \in \mathcal{N}_0$, and hence

$$\mathbb{E}[Z_\infty N_\infty^T] = 0.$$

Therefore

$$\mathbb{E}[Z_\infty^T N_\infty] = 0$$

for every $N \in \mathcal{N}_0$, so $Z^T \in \mathcal{N}_0^\perp$. This proves that $\mathcal{N}_0^\perp$ is stable.

In particular, since $Z^T \in \mathcal{N}_0^\perp$, taking the same $N \in \mathcal{N}_0$ in the preceding identity yields

$$\mathbb{E}[Z_T N_T] = \mathbb{E}[Z_\infty^T N_\infty] = 0.$$

Thus Z and N are strongly orthogonal. □

Definition 1.6.4. Define

$$d\mathcal{M}_0^2 \triangleq (c\mathcal{M}_0^2)^\perp.$$

Then every $M \in \mathcal{M}_0^2$ can be written as

$$M = M^c + M^d,$$

where $M^c \in c\mathcal{M}_0^2$, $M^d \in d\mathcal{M}_0^2$, and M^c and M^d are strongly orthogonal. In particular, $M^c M^d$ is uniformly integrable.

For a càdlàg function f , define

$$\Delta f(t) = f(t) - f(t-).$$

We now extend the quadratic variation from the continuous case to a general martingale in $\mathcal{M}_0^2$. In the continuous case, the bracket $\langle M \rangle$ is characterized by the fact that $M^2 - \langle M \rangle$ is a martingale. When jumps are present, however, the increasing process must also record the jump contribution. The next theorem shows that there is a unique increasing process $[M]$ which plays this role and satisfies $\Delta[M] = (\Delta M)^2$.

Theorem 1.6.3 (Meyer). Let $M \in \mathcal{M}_0^2$. Then there exists a unique increasing process $[M]$, null at 0, such that

- ① $M^2 - [M]$ is uniformly integrable;
- ② $\Delta[M] = (\Delta M)^2$.

Proof. Write

$$M = M^c + M^d,$$

where $M^c \in c\mathcal{M}_0^2$ and $M^d \in d\mathcal{M}_0^2$. Then

$$M_t^2 = (M_t^c + M_t^d)^2 = (M_t^c)^2 + 2M_t^c M_t^d + (M_t^d)^2.$$

The theorem is obtained by combining the continuous quadratic variation of M^c , the purely discontinuous part M^d , and the strong orthogonality between M^c and M^d . $\square$

For $M, N \in \mathcal{M}_0^2$, define

$$[M, N] \triangleq \frac{1}{4} ([M + N] - [M - N]).$$

Theorem 1.6.4 (Kunita & Watanabe). The process $[M, N]$ is the unique finite-variation process, null at 0, such that

- ① $MN - [M, N]$ is uniformly integrable;
- ② $\Delta[M, N] = \Delta M \cdot \Delta N$.

1.7 Itô Integral

Let $M \in \mathcal{M}_0^2$. We begin with the simplest class of integrands.

If $H_t = Z1_{(S, T]}(t)$, where $Z \in \mathcal{F}_S$ is bounded and $S \leq T$ are stopping times, define

$$\int_0^t H_s dM_s \equiv (H \cdot M)_t \equiv Z(M_{T \wedge t} - M_{S \wedge t}).$$

We now show that $(H \cdot M)$ belongs to $\mathcal{M}_0^2$. As a first step, we prove that $(H \cdot M)$ is a martingale. By Lemma 1.6.2, it suffices to verify that for every stopping time R ,

$$\begin{cases} \mathbb{E} |(H \cdot M)_R| < \infty, \\ \mathbb{E}(H \cdot M)_R = 0. \end{cases}$$

Proof. Let R be any stopping time. Since Z is bounded, say $|Z| \leq K$, we have

$$\mathbb{E} |(H \cdot M)_R| \leq K \mathbb{E} |M_{T \wedge R}| + K \mathbb{E} |M_{S \wedge R}|.$$

Because $M \in \mathcal{M}_0^2$, the martingale M is uniformly integrable, and

$$M_{T \wedge R} = \mathbb{E}(M_\infty \mid \mathcal{F}_{T \wedge R}).$$

Hence

$$\mathbb{E} |M_{T \wedge R}| \leq \mathbb{E} [\mathbb{E}(|M_\infty| \mid \mathcal{F}_{T \wedge R})] = \mathbb{E} |M_\infty|,$$

and similarly for $M_{S \wedge R}$. Therefore

$$\mathbb{E} |(H \cdot M)_R| < \infty.$$

It remains to show that $\mathbb{E}(H \cdot M)_R = 0$. By linearity, it is enough to treat the case $Z = 1_A$ with $A \in \mathcal{F}_S$. Then

$$\mathbb{E}(H \cdot M)_R = \mathbb{E} [1_A(M_{T \wedge R} - M_{S \wedge R})].$$

Define the stopping times

$$S_A \equiv \begin{cases} S, & \text{on } A, \\ \infty, & \text{on } A^c. \end{cases} \quad T_A \equiv \begin{cases} T, & \text{on } A, \\ \infty, & \text{on } A^c. \end{cases}$$

Then

$$1_A M_{S \wedge R} = M_{S_A \wedge R}, \quad 1_A M_{T \wedge R} = M_{T_A \wedge R}.$$

Hence the expectation above is the difference of expectations of stopped values of the martingale M , and therefore equals 0 by optional sampling. Thus $(H \cdot M)$ is a martingale. $\square$

Lemma 1.7.1. If $Z \in b\mathcal{F}_S$ and M is a uniformly integrable martingale, then $H \cdot M$ is a uniformly integrable martingale.

More generally, consider a finite linear combination of such elementary processes:

$$H_t = \sum_{i=1}^n Z_i 1_{(S_i, T_i]}(t),$$

where each $Z_i \in b\mathcal{F}_{S_i}$.

We define

$$(H \cdot M)_t \triangleq \sum_{i=1}^n Z_i (M_{T_i \wedge t} - M_{S_i \wedge t}).$$

Corollary 1.7.1. If $Z_i \in b\mathcal{F}_{S_i}$ and M is U.I. martingale, then $(H \cdot M)_t$ is U.I. martingale.

We next introduce the predictable σ -algebra, which is the natural measurability class for Itô integrands.

Definition 1.7.1 (Previsible process). The previsible σ -algebra $\mathcal{P}$ on $(0, \infty) \times \Omega$ is defined to be the smallest σ -algebra for which every adapted L -process is $\mathcal{P}$ -measurable.

An L -process is a process $(X_t)_{t>0}$ such that $X_t = X_{t-}$ and X_{t+} exists.

We say that a process $(X_t)_{t>0}$ is previsible if it is $\mathcal{P}$ -measurable.

Lemma 1.7.2. Suppose S and T are stopping times with $S \leq T \leq \infty$. Let $Z \in b\mathcal{F}_S$ and $H = Z1_{(S, T]}$. Then H is previsible.

Proof. Fix $t > 0$ and let B be a Borel set. We may assume that $0 \notin B$. Then

$$\{H_t \in B\} = \{Z \in B\} \cap \{S < t\} \cap \{t \leq T\}.$$

Moreover,

$$\{S < t\} = \bigcup_{n=1}^{\infty} \left\{ S \leq t - \frac{1}{n} \right\}.$$

Since $\{Z \in B\} \in \mathcal{F}_S$, we have

$$\{Z \in B\} \cap \left\{ S \leq t - \frac{1}{n} \right\} \in \mathcal{F}_{t - \frac{1}{n}} \subseteq \mathcal{F}_t.$$

Also, $\{t \leq T\} \in \mathcal{F}_t$. Therefore $\{H_t \in B\} \in \mathcal{F}_t$, so H is adapted. Since H is clearly an L -process, it follows from the definition of $\mathcal{P}$ that H is previsible. $\square$

Definition 1.7.2. Let

$$b\varepsilon = \left\{ \sum_{i=1}^n Z_i 1_{(S_i, T_i]} : n \geq 1, Z_i \in b\mathcal{F}_{S_i}, S_1 \leq T_1 \leq S_2 \leq \dots \right\}.$$

Claim. $\sigma(b\varepsilon) = \mathcal{P}$.

Proof. “ $\subseteq$ ” The inclusion $\sigma(b\varepsilon) \subseteq \mathcal{P}$ follows from the previous lemma.

“ $\supseteq$ ” For any bounded left-continuous process,

$$H_t = \lim_{k \uparrow \infty} \lim_{n \uparrow \infty} \sum_{i=2}^{n_k} H_{\frac{i-1}{n}} 1_{(\frac{i-1}{n}, \frac{i}{n}]}(t).$$

Since H is adapted, we get $H_{\frac{i-1}{n}} \in b\mathcal{F}_{\frac{i-1}{n}}$.

Hence

$$\sum_{i=2}^{n_k} H_{\frac{i-1}{n}} 1_{(\frac{i-1}{n}, \frac{i}{n}]}(t) \in b\varepsilon.$$

$$\implies H_t \in \sigma(b\varepsilon).$$

$\square$

Proof.

$\square$

Lemma 1.7.3. Let $H \in b\mathcal{E}$ and $M \in \mathcal{M}_0^2$. Then $H \cdot M \in \mathcal{M}_0^2$, and if

$$H = \sum_{i=1}^n Z_{i-1} \mathbf{1}_{(T_{i-1}, T_i]},$$

then

$$\mathbb{E}(H \cdot M)_\infty^2 = \sum_{i=1}^n \mathbb{E} \left[Z_{i-1}^2 (M_{T_i} - M_{T_{i-1}})^2 \right].$$

Proof.

$$(H \cdot M)_t = \sum_{i=1}^n Z_{i-1} (M_{T_i \wedge t} - M_{T_{i-1} \wedge t}).$$

By the previous corollary 1.7.1, $(H \cdot M)$ is a U.I. martingale.

Hence $(H \cdot M)_t \rightarrow (H \cdot M)_\infty$ a.s. and in L^1 , where

$$(H \cdot M)_\infty = \sum_{i=1}^n Z_{i-1} \Delta_i, \quad \Delta_i = M_{T_i} - M_{T_{i-1}}.$$

Therefore

$$\mathbb{E} [(H \cdot M)_\infty^2] = \mathbb{E} \sum_{i=1}^n Z_{i-1}^2 \Delta_i^2 + 2 \mathbb{E} \left(\sum_{i < j} Z_{i-1} \Delta_i Z_{j-1} \Delta_j \right).$$

For $i < j$, the random variable $Z_{i-1} \Delta_i Z_{j-1}$ is $\mathcal{F}_{T_{j-1}}$ -measurable, so

$$\mathbb{E} [Z_{i-1} \Delta_i Z_{j-1} \Delta_j] = \mathbb{E} [Z_{i-1} \Delta_i Z_{j-1} \mathbb{E} (\Delta_j | \mathcal{F}_{T_{j-1}})] = 0.$$

Thus all cross terms vanish.

By Doob's L^2 inequality,

$$\mathbb{E} \left[\sup_{t \geq 0} (H \cdot M)_t^2 \right] \leq 4 \mathbb{E} [(H \cdot M)_\infty^2] < \infty.$$

Therefore $H \cdot M \in \mathcal{M}_0^2$. □

Recall from Meyer's theorem that for $M \in \mathcal{M}_0^2$ there exists a quadratic variation $[M]$ such that $M^2 - [M]$ is a uniformly integrable martingale. Applying this to the previous lemma 1.7.3, we get

$$\mathbb{E}(H \cdot M)_\infty^2 = \sum_{i=1}^n \mathbb{E} \left[Z_{i-1}^2 (M_{T_i} - M_{T_{i-1}})^2 \right].$$

Moreover,

$$\begin{aligned} \mathbb{E} \left[(M_{T_i} - M_{T_{i-1}})^2 \middle| \mathcal{F}_{T_{i-1}} \right] &= \mathbb{E} \left[M_{T_i}^2 - M_{T_{i-1}}^2 \middle| \mathcal{F}_{T_{i-1}} \right] \\ &= \mathbb{E} \left[[M]_{T_i} - [M]_{T_{i-1}} \middle| \mathcal{F}_{T_{i-1}} \right], \end{aligned}$$

because $M^2 - [M]$ is a martingale and $\mathbb{E}(M_{T_i} \mid \mathcal{F}_{T_{i-1}}) = M_{T_{i-1}}$. Therefore

$$\mathbb{E}(H \cdot M)_\infty^2 = \sum_{i=1}^n \mathbb{E} [Z_{i-1}^2 ([M]_{T_i} - [M]_{T_{i-1}})] = \mathbb{E} \left[\int_0^\infty H_s^2 d[M]_s \right].$$

This shows that, for simple predictable integrands, the map

$$H \mapsto (H \cdot M)_\infty$$

is an isometry from $(L^2(M) \cap b\varepsilon, \|\cdot\|_M)$ into $L^2(\mathcal{F}_\infty)$ equipped with the usual L^2 norm.

For fixed $M \in \mathcal{M}_0^2$, define

$$\|H\|_M = \left(\mathbb{E} \int_0^\infty H_s^2 d[M]_s \right)^{1/2}.$$

Let

$$L^2(M) = \{\text{previsible } H : \|H\|_M < \infty\}.$$

Then the map

$$\begin{aligned} I : L^2(M) \cap b\varepsilon &\rightarrow L^2(\mathcal{F}_\infty), \\ H &\mapsto (H \cdot M)_\infty \end{aligned}$$

is an isometry, since

$$\mathbb{E} \left[\int_0^\infty H_s^2 d[M]_s \right] = \mathbb{E}(H \cdot M)_\infty^2.$$

Hence I extends uniquely from simple predictable processes to all of $L^2(M)$.

Definition 1.7.3 (Itô Integral). Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$. Then Itô Integral $(H \cdot M)_t$ is the image of H under extension of $I : L^2(M) \rightarrow L^2(\mathcal{F}_\infty)$.

Localization.

Definition 1.7.4. Let T be a stopping time and let H be any process. Define

$$H(0, T](t) = \begin{cases} H_t, & 0 < t \leq T, \\ 0, & t > T. \end{cases}$$

Remark. If H is previsible, then so is $H(0, T]$.

Definition 1.7.5. Let T be a stopping time and let X be any process. Define

$$X^T(t) = \begin{cases} X(t), & 0 \leq t \leq T, \\ X(T), & t > T. \end{cases}$$

For an elementary integrand, the localization identity can be checked directly. Indeed,

$$(H \cdot M)_t^T = (H(0, T] \cdot M^T)_t,$$

and

$$\left[Z_i (M_{T_i \wedge t} - M_{T_{i-1} \wedge t}) \right]^T = \left(Z_i 1_{(T_{i-1} \wedge T, T_i \wedge T]} \cdot M^T \right)_t = Z_i (M_{T_i \wedge T \wedge t} - M_{T_{i-1} \wedge T \wedge t}).$$

Theorem 1.7.1. Fix $M \in \mathcal{M}_0^2$. For any $H \in L^2(M)$ and any stopping time T :

- (i) $(H \cdot M)^T = H(0, T] \cdot M = H \cdot M^T$.
- (ii) $(H \cdot M)_t^2 - \int_0^t H_s^2 d[M]_s$ is a uniformly integrable martingale. In particular,

$$[H \cdot M]_t = \int_0^t H_s^2 d[M]_s.$$

- (iii) If $H, K \in b\mathcal{P}$, then

$$(HK) \cdot M = H \cdot (K \cdot M) = K \cdot (H \cdot M).$$

- (iv) $\Delta(H \cdot M) = H \Delta M$.

- (v) If M is continuous, then $H \cdot M$ is continuous.

Proof. We first prove the identities for elementary processes and then extend them by density.

For (i), let $H = Z1_{(U, V]}$. Then

$$(H \cdot M)_t = Z(M_{V \wedge t} - M_{U \wedge t}),$$

so

$$(H \cdot M)_t^T = (H \cdot M)_{t \wedge T} = Z(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t}).$$

On the other hand,

$$H(0, T] = Z1_{(U \wedge T, V \wedge T]},$$

and therefore

$$(H(0, T] \cdot M)_t = Z(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t}).$$

Similarly,

$$(H \cdot M^T)_t = Z(M_{V \wedge t}^T - M_{U \wedge t}^T) = Z(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t}).$$

Hence (i) holds for elementary processes, and then for all $H \in L^2(M)$ by continuity of the integral map.

For (ii), set

$$N_t = (H \cdot M)_t^2 - \int_0^t H_s^2 d[M]_s.$$

Since

$$\mathbb{E} \left[\sup_{t \geq 0} (H \cdot M)_t^2 \right] \leq 4\mathbb{E} \left[\int_0^\infty H_s^2 d[M]_s \right] < \infty,$$

the family $\{N_t : t \geq 0\}$ is uniformly integrable. It therefore suffices to show that $\mathbb{E}N_T = 0$ for every stopping time T .

By part (i),

$$(H \cdot M)_T = (H(0, T] \cdot M)_\infty.$$

Applying the isometry to the integrand $H(0, T]$, we obtain

$$\begin{aligned} \mathbb{E}(H \cdot M)_T^2 &= \mathbb{E}(H(0, T] \cdot M)_\infty^2 \\ &= \mathbb{E} \int_0^\infty H(0, T]_s^2 d[M]_s \\ &= \mathbb{E} \int_0^T H_s^2 d[M]_s. \end{aligned}$$

Hence $\mathbb{E}N_T = 0$, and (ii) follows.

For (iii) First take $H = Z_1 \mathbf{1}_{(S_1, T_1]}$ and $K = Z_2 \mathbf{1}_{(S_2, T_2]}$. Then

$$HK = Z_1 Z_2 \mathbf{1}_{(S, T]},$$

where

$$S = S_1 \vee S_2, \quad T = (T_1 \wedge T_2) \vee (S_1 \vee S_2).$$

Hence

$$((HK) \cdot M)_\infty = Z_1 Z_2 (M_T - M_S).$$

On the other hand,

$$\begin{aligned} (H \cdot (K \cdot M))_\infty &= Z_1 \left((K \cdot M)_{T_1} - (K \cdot M)_{S_1} \right) \\ &= Z_1 \left((K \cdot M^{T_1})_\infty - (K \cdot M^{S_1})_\infty \right) \\ &= Z_1 Z_2 \left[(M_{T_2 \wedge T_1} - M_{S_2 \wedge T_1}) - (M_{T_2 \wedge S_1} - M_{S_2 \wedge S_1}) \right] \\ &= Z_1 Z_2 (M_T - M_S). \end{aligned}$$

Therefore, for every $H, K \in b\varepsilon$,

$$H \cdot (K \cdot M) = (HK) \cdot M.$$

The identity

$$K \cdot (H \cdot M) = (HK) \cdot M$$

is proved in the same way.

Now let $K \in b\mathcal{P}$. Choose $(K^n)_{n \geq 1} \subset b\varepsilon$ such that

$$K^n \xrightarrow{L^2(M)} K.$$

Fix $H \in b\varepsilon$. By the isometry,

$$\begin{aligned} \mathbb{E} \left[(H \cdot (K \cdot M) - H \cdot (K^n \cdot M))_\infty^2 \right] &= \mathbb{E} \left[(H \cdot ((K - K^n) \cdot M))_\infty^2 \right] \\ &= \mathbb{E} \left[\int_0^\infty H_s^2 d[(K - K^n) \cdot M]_s \right] \\ &= \mathbb{E} \left[\int_0^\infty H_s^2 (K_s - K_s^n)^2 d[M]_s \right] \\ &\leq \|H\|_\infty^2 \mathbb{E} \left[\int_0^\infty (K_s - K_s^n)^2 d[M]_s \right] \rightarrow 0. \end{aligned}$$

Hence

$$H \cdot (K^n \cdot M) \xrightarrow{L^2(\mathcal{F}_\infty)} H \cdot (K \cdot M).$$

On the other hand, for each n we already know that

$$H \cdot (K^n \cdot M) = (HK^n) \cdot M.$$

Since $H \in b\varepsilon$, we also have

$$\mathbb{E} \left[\int_0^\infty (H_s K_s^n - H_s K_s)^2 d[M]_s \right] \leq \|H\|_\infty^2 \mathbb{E} \left[\int_0^\infty (K_s^n - K_s)^2 d[M]_s \right] \rightarrow 0,$$

so $HK^n \xrightarrow{L^2(M)} HK$. Applying the isometry once again,

$$(HK^n) \cdot M \xrightarrow{L^2(\mathcal{F}_\infty)} (HK) \cdot M.$$

By uniqueness of the $L^2(\mathcal{F}_\infty)$ limit,

$$H \cdot (K \cdot M) = (HK) \cdot M.$$

(iv) First let $H = Z1_{(S,T]}$. Then

$$(H \cdot M)_t = Z(M_{t \wedge T} - M_{t \wedge S}),$$

so

$$\Delta(H \cdot M)_t = Z1_{(S,T]}(t) \Delta M_t = H_t \Delta M_t.$$

Hence the identity holds for all $H \in b\varepsilon$.

Now let $H \in L^2(M)$, and choose $H^n \in b\epsilon$ such that

$$H^n \xrightarrow{L^2(M)} H.$$

By the isometry,

$$H^n \cdot M \xrightarrow{L^2(\mathcal{F}_\infty)} H \cdot M,$$

and hence, after passing to a subsequence,

$$\sup_{t \geq 0} |(H^{n_k} \cdot M)_t - (H \cdot M)_t| \xrightarrow{a.s.} 0.$$

Also,

$$\int_0^\infty (H_s^{n_k} - H_s)^2 d[M]_s \xrightarrow{a.s.} 0$$

along a further subsequence.

Fix $t \geq 0$. Since

$$\Delta(H^{n_k} \cdot M)_t = H_t^{n_k} \Delta M_t,$$

we distinguish two cases.

Case 1. If $\Delta M_t = 0$, then $\Delta(H^{n_k} \cdot M)_t = 0$ for every k , and therefore

$$\Delta(H \cdot M)_t = 0 = H_t \Delta M_t.$$

Case 2. If $\Delta M_t \neq 0$, then $\Delta[M]_t = (\Delta M_t)^2 > 0$. Hence

$$\int_0^\infty (H_s^{n_k} - H_s)^2 d[M]_s \geq (H_t^{n_k} - H_t)^2 \Delta[M]_t,$$

and so $H_t^{n_k} \rightarrow H_t$. Letting $k \rightarrow \infty$ in

$$\Delta(H^{n_k} \cdot M)_t = H_t^{n_k} \Delta M_t,$$

we obtain

$$\Delta(H \cdot M)_t = H_t \Delta M_t.$$

□

Theorem 1.7.2 (Kunita-Watanabe). Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$. Then $\forall N \in \mathcal{M}_0^2$,

$$\mathbb{E}[(H \cdot M)_\infty \cdot N_\infty] = \mathbb{E}[(H \cdot [M, N])_\infty].$$

Moreover,

$$[H \cdot M, N] = H \cdot [M, N] \stackrel{def}{=} \int H_s d[M, N]_s.$$

Theorem 1.7.3. Let $M, N \in \mathcal{M}_0^2$. Then

$$\left(\int_0^\infty |H_s K_s| d[M, N]_s \right)^2 \leq \int_0^\infty H_s^2 d[M]_s \cdot \int_0^\infty K_s^2 d[N]_s. \quad \text{a.s.}$$

Proof. $|H_s K_s| = H_s K_s \cdot \text{sgn}(H_s K_s)$.

$$|d[M, N]_s| = d[M, N]_s \cdot \text{sgn}([M, N]_s).$$

$$\begin{aligned} LHS &= \int_0^\infty H_s K_s \cdot \text{sgn}(H_s K_s) \cdot \text{sgn}([M, N]_s) d[M, N]_s \\ &= \int_0^\infty \tilde{H}_s K_s d[M, N]_s \end{aligned}$$

$$RHS = \int_0^\infty \tilde{H}_s d[M]_s \cdot \int_0^\infty K_s^2 d[N]_s$$

$$\Leftrightarrow \left(\int_0^\infty \tilde{H}_s K_s d[M, N]_s \right)^2 \leq \int_0^\infty \tilde{H}_s d[M]_s \int_0^\infty K_s^2 d[N]_s.$$

$$\begin{aligned} H_n &= \sum_{i=1}^n U_i(t_i, t_{i+1}] \\ \Leftrightarrow \\ K_n &\equiv \sum_{i=1}^n V_i(t_i, t_{i+1}]. \end{aligned}$$

$$\Leftrightarrow \left(\sum_{i=1}^n U_i V_i ([M, N]_{t_{i+1}} - [M, N]_{t_i}) \right)^2 \leq \sum_{i=1}^2 U_i^2 ([M]_{t_{i+1}} - [M]_{t_i}) \cdot \sum_{i=1}^n V_i^2 ([N]_{t_{i+1}} - [N]_{t_i}).$$

It suffices to prove

$$|[M, N]_{t_{i+1}} - [M, N]_{t_i}| \leq ([M]_{t_{i+1}} - [M]_{t_i})^{\frac{1}{2}} ([N]_{t_{i+1}} - [N]_{t_i})^{\frac{1}{2}}.$$

like $(\sum_i a_i b_i)^2 \leq \sum_i a_i^2 \sum_i b_i^2$.

To prove this, note that $\forall \lambda \in \mathbb{Q}$, $[M + \lambda N]$ is increasing and so

$$[M + \lambda N]_{t_{i+1}} - [M + \lambda N]_{t_i} \geq 0 \quad \text{a.s.}$$

Also

$$[M + \lambda N]_t = [M]_t + 2\lambda[M, N]_t + \lambda^2[N]_t.$$

because $(M + \lambda N)_t^2 - [M + \lambda N]_t$ is U.I. and by the uniqueness (?)

$$M_t^2 + 2\lambda M_t N_t + \lambda^2 N_t^2 - ([M]_t + 2\lambda[M, N]_t + \lambda^2[N]_t).$$

Hence

$$([M]_{t_{i+1}} - [M]_{t_i}) + 2\lambda([M, N]_{t_{i+1}} - [M, N]_{t_i}) + \lambda^2([N]_{t_{i+1}} - [N]_{t_i}) \geq 0$$

$$\Leftrightarrow b^2 - 4ac \leq 0.$$

So

$$([M, N]_{t_{i+1}} - [M, N]_{t_i})^2 \leq ([M]_{t_{i+1}} - [M]_{t_i}) ([N]_{t_{i+1}} - [N]_{t_i}).$$

$$\mathbb{E}[(H \cdot M)_\infty N_\infty] = \mathbb{E}[(H \cdot [M, N])_\infty].$$

If $H = Z(S, T)$,

$$\begin{cases} LHS = \mathbb{E}[Z(M_T - M_S) \cdot N_\infty] \\ RHS = \mathbb{E}[Z([M, N]_T - [M, N]_S)] \end{cases}$$

$$\begin{aligned} LHS &= \mathbb{E}[\mathbb{E}(N_\infty \mid \mathcal{F}_T) \cdot Z(M_T - M_S)] \\ &= \mathbb{E}[Z(M_T N_T - M_S N_T)] \\ &= \mathbb{E}[Z M_T N_T] - \mathbb{E}[\mathbb{E}(N_T \mid \mathcal{F}_S) \cdot Z M_S] \\ &= \mathbb{E}[Z M_T N_T] - \mathbb{E}[Z M_S N_S]. \end{aligned}$$

Because $M_t N_t - [M, N]_t$ is U.I.

$$\mathbb{E}[M_T N_T] = \mathbb{E}[[M, N]_T] \implies \mathbb{E}[Z(M_T N_T - [M, N]_T)] = 0.$$

And

$$\begin{aligned} LHS &= \mathbb{E}[Z(M_T N_T - M_S N_S)] \\ &= \mathbb{E}[Z \mathbb{E}[M_T N_T - M_S N_S \mid \mathcal{F}_S]] \\ &= \mathbb{E}[Z \mathbb{E}([M, N]_T - [M, N]_S \mid \mathcal{F}_S)] \\ &= \mathbb{E}[Z([M, N]_T - [M, N]_S)]. \end{aligned}$$

Hence $\forall H \in b\mathcal{E}$

$$\mathbb{E}[(H \cdot M)_\infty \cdot N_\infty] = \mathbb{E}[(H \cdot [M, N])_\infty].$$

If $H_n \xrightarrow{L^2(M)} H$, i.e.

$$\mathbb{E} \left(\int_0^\infty (H_n(s) - H_s)^2 d[M]_s \right) \rightarrow 0$$

$\implies (H_n \cdot M)_\infty \xrightarrow{L^2} (H \cdot M)_\infty$ by

$$\mathbb{E}[(H_n \cdot M - H \cdot M)_\infty^2] = \mathbb{E} \left[\int_0^\infty (H_n - H) d[M]_s \right] \rightarrow 0.$$

Hence $\mathbb{E}[(H_n \cdot M)_\infty \cdot N_\infty] \rightarrow \mathbb{E}[(H \cdot M)_\infty N_\infty]$.

Next,

$$\begin{aligned} \mathbb{E}[|(H_n \cdot [M, N])_\infty - (H \cdot [M, N])_\infty|] &= \mathbb{E}[|(H_n - H) \cdot [M, N])_\infty|] \\ &\leq \left(\mathbb{E} \left(\int_0^\infty (H_n - H) d[M]_s \right) \right)^{\frac{1}{2}} (\mathbb{E}[N]_\infty)^{\frac{1}{2}} \rightarrow 0 \end{aligned}$$

by using

$$\left(\int_0^\infty |H_s| d[M, N]_s \right) \leq \left(\int_0^\infty H_s^2 d[M]_s \right)^{\frac{1}{2}} \left(\int_0^\infty 1 d[N]_s \right)^{\frac{1}{2}}.$$

Hence $\mathbb{E}[(H_n \cdot [M, N])_\infty] \rightarrow \mathbb{E}[(H \cdot [M, N])_\infty]$. $\square$

Remark. proof of $[H \cdot M, N] = H \cdot [M, N] \Leftrightarrow$ It suffices to. prove $(H \cdot M)_t N_t - \int_0^\infty H_s d[M, N]_s$ is a U.I. martingale.

$\Leftrightarrow \forall$ stopping time R

$$\mathbb{E}[(H \cdot M)_R \cdot N_R] = \mathbb{E} \left[\int_0^R H_s d[M, N]_s \right].$$

$$\begin{aligned} LHS &= \mathbb{E} [(H \cdot M)_\infty^R \cdot N_\infty^R] \\ &= \mathbb{E} [(H^R \cdot M)_\infty \cdot N_\infty^R] \\ &= \mathbb{E} [(H^R \cdot [M, N^R])_\infty] \\ &= \mathbb{E} \left[\int_0^\infty H^R(s) d[M, N^R]_s \right] \\ &= \mathbb{E} \left[\int_0^R H(s) d[M, N]_s \right] = RHS. \end{aligned}$$

Definition 1.7.6 (local martingale). $\mathcal{M}_{loc} = \{\text{local martingale}\}$.

$\exists$ stopping times $T_n \uparrow \infty$ s.t. M^{T_n} is a martingale.

Definition 1.7.7 (continuous local martingale). $c\mathcal{M}_{loc} = \{\text{continuous local martingale}\}$.

Quadratic variation of $c\mathcal{M}_{loc}$.

If $M \in c\mathcal{M}_{loc}$, then $\exists [M]$ increasing process such that $M^2 - [M] \in c\mathcal{M}_{loc}$.

Lemma 1.7.4. If $M \in c\mathcal{M}_{loc}$. Fix $n \geq 1$, define

$$T_n = \inf\{t \geq 0: |M_t| \geq n\}.$$

Then M^{T_n} is a bounded martingale.

Proof. By definition, $\exists$ stopping $(S_k) \uparrow \infty$ s.t. M^{S_k} is a martingale.

$\implies M^{S_k \wedge T_n}$ is also a martingale.

So $\forall s < t$,

$$\mathbb{E}(M_t^{S_k \wedge T_n} | \mathcal{F}_s) = M_s^{S_k \wedge T_n}.$$

Let $k \rightarrow \infty$ to get $\mathbb{E}(M_t^{T_n} | \mathcal{F}_s) = M_s^{T_n}$.

pf of Quadratic variation:

(Existence.) $\forall n \geq 1$, we let

$$T_n = \inf\{t \geq 0: |M_t| \geq n\}.$$

Then define $\forall t < T_n$,

$$[M]_t = [M^{T_n}]_t.$$

We need to check that $[M^{T_n}]_t$ is well-defined.

$$[M^{T_2}]^{T_1} = [M^{T_2}]_{t \wedge T_1}$$

Because $[M]^T = [M^T]$,

$$[M^{T_2}]^{T_1} = [M^{T_2 \wedge T_1}] = [M^{T_1}].$$

By $T_n \uparrow \infty$, we have $[M]_t$ is defined for all $t \geq 0$.

(uniqueness.) If $\exists A, A'$ increasing such that $N = M^2 - A$ and $N' = M^2 - A' \in c\mathcal{M}_{loc}$. Then $N - N' = A' - A \in c\mathcal{M}_{loc}$ is of finite variation.

Lemma 1.7.5. If $M \in c\mathcal{M}_{loc}$ and M is of finite variation, then $M_t \equiv 0$.

Proof. Let $V_M(t)$ = total variation of M on $[0, t]$. Define $T_n = \inf\{t \geq 0: V_M(t) > n\}$.

Then M^{T_n} is a bounded martingale and is of finite variation. Next, define

$$A_t^n \equiv \sum_{k=1}^{2^n} (M(t_k^n) - M(t_{k-1}^n))^2$$

with $t_k^n = \frac{k}{2^n}t$.

So

$$A_t^n \leq \sup_{1 \leq k \leq 2^n} |M(t_k^n) - M(t_{k-1}^n)| \cdot V_M(t) \xrightarrow{a.s.} 0$$

by continuity of M .

Thus $A_t^n \rightarrow 0, \forall t \geq 0$.

We will prove $A_t^n \rightarrow A_t$ such that $M_t^2 - A_t$ is martingale.

$$\implies A_t = 0 \text{ and } \mathbb{E}M_t^2 = \mathbb{E}M_0^2 = 0.$$

$$\implies M_t = 0 \text{ a.s. } \forall t \geq 0.$$

By a.s. continuity, $M_t \equiv 0$, a.s. □

□

1.7.1 Itô Integral of local martingale

Theorem 1.7.4. Let $M \in c\mathcal{M}_{loc}$ with $M_0 = 0$. Let $H \in \mathcal{P}$ such that W.P.1

$$\int_0^t H_s^2 d[M]_s < \infty$$

for all $t > 0$. Then $H \cdot M$ is well-defined such that $H \cdot M \in c\mathcal{M}_{0,loc}$.

Proof. Let $U_n = \inf\{t: |M_t| > n\}$, $V_n = \inf\{t: \int_0^t H_s^2 d[M]_s > n\}$.

(Recall $L^2(M) \implies \mathbb{E} \int_0^\infty H_s^2 d[M]_s < \infty$).

$\implies T_n = U_n \wedge V_n$. In this way, $M^{T_n} \in c\mathcal{M}_0^2$ and $H^{T_n} \in L^2(M^{T_n})$.

So $H^{T_n} \cdot M^{T_n}$ is defined by previous L^2 theory.

$\implies (H \cdot M)^{T_n} = H^{T_n} \cdot M^{T_n}$ ($t < T_n$) is a L^2 -bounded martingale.

$\implies H \cdot M \in c\mathcal{M}_{0,loc}$. □

Definition 1.7.8 (semimartingale). We say X is a semimartingale if

$$X_t = X_0 + M_t + A_t.$$

M_t is local mart, A_t is finite variation.

Definition 1.7.9. Let X be a continuous semimartingale. Then $\exists$ unique

$$\begin{cases} M \in c\mathcal{M}_{0,loc} \\ A \in cFV_0 \end{cases}$$

such that $X_t = X_0 + M_t + A_t, \forall t \geq 0$.

Proof. If $\exists M'$ and A' such that

$$X = X_0 + M' + A' = X_0 + M + A.$$

Then $M - M' = A' - A \in c\mathcal{M}_{0,loc} \cap cFV = \{0\}$. □

We write $M = X^{cm}$ and

$$[X, X] = [M, M] = [X^{cm}, X^{cm}].$$

Similarly, $[X, Y] = [X^{cm}, Y^{cm}]$.

For any $H \in lb\mathcal{P}$ define

$$H \cdot X = H \cdot M + H \cdot A.$$

Obviously $H \cdot M \in c\mathcal{M}_{0,loc}$ and $H \cdot A \in FV$.

And

$$\begin{aligned} [H \cdot X, K \cdot Y] &= [H \cdot X^{cm}, K \cdot Y^{cm}] \\ &= H \cdot [X^{cm}, K \cdot Y^{cm}] \\ &= H \cdot K[X^{cm}, Y^{cm}] \\ &= (HK) \cdot [X^{cm}, Y^{cm}] = (HK) \cdot [X, Y]. \end{aligned}$$

1.7.2 Itô's Formula

Theorem 1.7.5 (Itô's Formula). Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 . Let

$$X_t \equiv (X_t^1, \dots, X_t^n)$$

be a continuous semimartingale.

Then

$$f(x_t) - f(x_0) = \int_0^t \nabla f(X_s) dX_s + \frac{1}{2} \int_0^t \sum_{i,j} D_{ij} f(X_s) d[X^i, X^j]_s$$

where $\nabla f(X_s) = (D_1 f(X_s), \dots, D_n f(X_s))$. $D_i f(x) = \frac{\partial}{\partial x_i} f(x)$, $D_{ij} = D_i D_j = D_j D_i$.

Theorem 1.7.6 (Integration by Parts). Let X, Y be continuous semimartingales. Then

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X, Y]_t.$$

Proof. Assume IBP holds.

Let

$$\mathcal{A} = \left\{ f \in C^2(\mathbb{R}^n) : df(X_t) = \sum_i D_i f(X_t) dX_t + \frac{1}{2} \sum_{i,j} D_{ij} f(X_t) d[X^i, X^j]_t \right\}.$$

(i). $f, g \in \mathcal{A} \implies f + g \in \mathcal{A}$. (ii). $f, g \in \mathcal{A} \implies fg \in \mathcal{A}$.

$$\begin{aligned} d(fg(X_t)) &= d(f(X_t) \cdot g(X_t)) \\ &= f(X_t) \cdot dg(X_t) + g(X_t) df(X_t) + d[f(X_t), g(X_t)]_t \\ &= f(X_t) \left(\sum_i D_i g(X_t) dX_t + \frac{1}{2} \sum_{i,j} D_{ij} g(X_t) d[X^i, X^j]_t \right) \\ &\quad + g(X_t) \left(\sum_i D_i f(X_t) dX_t + \frac{1}{2} \sum_{i,j} D_{ij} f(X_t) d[X^i, X^j]_t \right) \\ &\quad + d \left[\int_0^t \sum_i D_i f(X_t) dX_t, \int_0^t \sum_j D_j g(X_t) dX_t^j \right] \\ &= dX_t \left(\sum_i f D_i g(X_t) + \sum_i g D_i f(X_t) \right) \\ &\quad + \frac{1}{2} \sum_{i,j} (g D_{ij} f(X_t) + f D_{ij} g(X_t) + 2 D_i f D_j g(X_t)) d[X^i, X^j]_t \\ &= \sum_i D_i (fg)(X_t) dX_t + \frac{1}{2} \sum_{i,j} D_{ij} (fg)(X_t) d[X^i, X^j]_t \\ &\implies fg \in \mathcal{A}. \end{aligned}$$

(iii). $f(x) = x \in \mathcal{A} \implies \forall \text{ polynomial } \in \mathcal{A}$. □

Lemma 1.7.6. $\forall f \in C^2(\mathbb{R}^d)$, $\exists$ polynomials $\{P_n(x)\}_{n \geq 1}$ such that

$$\begin{cases} P_n(x) \rightarrow f(x) \\ D_i P_n(x) \rightarrow D_i f(x) \\ D_{ij} P_n(x) \rightarrow D_{ij} f(x) \end{cases}$$

uniformly on compacts.

Define $U_N = \inf\{t \geq 0: |X_t| + [M]_t > N\}$ for each $N > 0$.

Then $U_N \uparrow \infty$. So it suffices to prove Itô formula holds for $\forall t \leq U_N$.

Note that

$$P_n(X_t) = P_n(X_0) + \int_0^t \sum_i D_i P_n(X_s) dX_s + \frac{1}{2} \int_0^t \sum_{i,j} D_{ij} P_n(X_s) d[X^i, X^j]_s$$

for $0 \leq t \leq U_N$.

By letting $n \rightarrow \infty$, we will prove the limits are

$$f(X_t) = f(X_0) + \int_0^t \sum_i D_i f(X_s) dX_s + \frac{1}{2} \int_0^t \sum_{i,j} D_{ij} f(X_s) d[X^i, X^j]_s.$$

$$\begin{aligned} I_{1,n} &= \int_0^t \sum_i D_i P_n(X_s) dX_s \\ &= \int_0^t \sum_i D_i P_n(X_s) dM_s + \int_0^t \sum_i D_i P_n(X_s) dA_s. \end{aligned}$$

Notice that

$$\left| \int_0^t \sum_i D_i P_n(X_s) dA_s - \int_0^t \sum_i D_i f(X_s) dA_s \right| \leq d \cdot \sup_i \underbrace{\|D_i P_n - D_i f\|}_{\text{max on compacts}} \cdot A_t \xrightarrow{n \rightarrow \infty} 0 \text{ a.s.}$$

Next,

$$\begin{aligned} &\mathbb{E} \sup_{s \leq t} \left| \int_0^t \sum_i (D_i P_n(X_s) - D_i f(X_s)) dM_s \right|^2 \\ &\leq \mathbb{E} \int_0^t \left(\sum_i D_i P_n(X_s) - D_i f(X_s) \right)^2 d[M]_s \\ &\leq d \cdot \sup_{1 \leq i \leq d} \|D_i P_n - D_i f\|^2 \cdot \underbrace{\mathbb{E}[M]_{t \wedge U_N}}_N \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

By taking subsequence, $\int_0^t \sum_i D_i P_{n_k}(X_s) dM_s \xrightarrow{\text{a.s.}} \int_0^t \sum_i D_i f(X_s) dM_s$.

$$I_{2,n} = \frac{1}{2} \int_0^t \sum_{i,j} D_{ij} P_n(X_s) d[X^i, X^j]_s.$$

Notice that

$$\left| I_{2,n} - \frac{1}{2} \int_0^t \sum_{i,j} D_{ij} f(X_s) d[X^i, X^j]_s \right| \leq C \sum_{i,j} \|D_{ij} P_n - D_{ij} f\| \cdot \underbrace{\sum_{i,j} [X^i, X^j]_t}_{\leq C_N} \xrightarrow{n \rightarrow \infty} 0.$$

Lemma 1.7.7. Let M be a bounded continuous martingale with $M_0 = 0$. Let V be continuous FV_0

$$\begin{aligned} M_t^2 &= \int_0^t 2M_s dM_s + [M]_t, \\ M_t V_t &= \int_0^t M_s dV_s + \int_0^t V_s dM_s. \end{aligned}$$

Proof.

$$\begin{aligned} M_t^2 &= \left[\sum_{k=1}^{2^n} M(t_k^n) - M(t_{k-1}^n) \right]^2 \\ &= \sum_{k=1}^{2^n} (M(t_k^n) - M(t_{k-1}^n))^2 + 2 \sum_{k=1}^{2^n} [M(t_k^n)M(t_{k-1}^n) - M(t_{k-1}^n)^2] \\ &= M(t_{2^n}^n)^2 - M(t_0^n)^2 = M(t)^2 \end{aligned}$$

where $t_k^n = \frac{k}{2^n} t$.

Note that $\sum_{k=1}^{2^n} (M(t_k^n) - M(t_{k-1}^n))^2 \rightarrow [M]_t$ and

$$2 \sum_{k=1}^{2^n} [M(t_k^n)M(t_{k-1}^n) - M(t_{k-1}^n)^2] = 2 \sum_{k=1}^{2^n} M(t_{k-1}^n)(M(t_k^n) - M(t_{k-1}^n)) \rightarrow 2 \int_0^t M_s dM_s.$$

$$\begin{aligned} M_t V_t &= M_{t_{2^n}^n} V_{t_{2^n}^n} - M_{t_0^n} V_{t_0^n} \\ &= \sum_{k=1}^{2^n} [M(t_k^n) V(t_k^n) - M(t_{k-1}^n) V(t_{k-1}^n)] \\ &= \sum_{k=1}^{2^n} M(t_k^n) (V(t_k^n) - V(t_{k-1}^n)) + \sum_{k=1}^{2^n} V(t_{k-1}^n) (M(t_k^n) - M(t_{k-1}^n)) \\ &\rightarrow \int_0^t M_s dV_s + \int_0^t V_s dM_s. \end{aligned}$$

□

Theorem 1.7.7 (Lévy Thm). Let $\{B_t^i\}_{t \geq 0} \in c\mathcal{M}_{0,loc}$ for $i = 1, 2, \dots, n$. If $\forall i, j$, we have

$$[B^i, B^j]_t = \begin{cases} t, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then $B_t = (B_t^1, \dots, B_t^n)$ is a Brownian motion on $\mathbb{R}^n$ with $\mathcal{N}(0, I_n)$.

Proof. $\forall s < t$, we will prove

$$\mathbb{E} \left[e^{i\theta(B_t - B_s)} \middle| \mathcal{F}_s \right] = e^{-\frac{1}{2}(t-s)|\theta|^2}$$

for all $\theta \in \mathbb{R}^n$.

Define $f(x, t) = e^{i\theta x + \frac{1}{2}|\theta|^2 t}$, $\mathbb{R}^{n+1} \rightarrow \mathbb{C}$.

Apply Itô's formula to $f(B_t, t) \equiv f(X_t)$.

$X_t = (B_t^1, B_t^2, \dots, B_t^n, t)$.

$$\begin{aligned} f(X_t) &= f(x_0) + \int_0^t \sum_{i=1}^n (D_i f(X_s) dB_s^i) + \int_0^t D_{n+1} f(X_s) ds \\ &\quad + \frac{1}{2} \int_0^t \sum_{1 \leq i, j \leq n} D_{ij} f(X_s) d[B^i, B^j]_s \\ &\quad + \frac{1}{2} \int_0^t \sum_{1 \leq k \leq n} D_{k, n+1} f(X_s) d[B^k, s]_s \\ \implies f(B_t, t) &= f(B_0, 0) + \int_0^t \sum_i D_i f(B_s, s) dB_s^i \\ &\quad + \int_0^t \left(e^{i\theta B_s + \frac{1}{2}|\theta|^2 s} \cdot \frac{1}{2} |\theta|^2 \right) ds \\ &\quad + \int_0^t \frac{1}{2} \left(\sum_{j=1}^n e^{i\theta B_s + \frac{1}{2}|\theta|^2 s} (i\theta_j)^2 \right) ds. \end{aligned}$$

We conclude that $f(B_t, t) = f(B_0, 0) + c\mathcal{M}_{0,loc}$

$$\implies \underbrace{e^{i\theta B_t + \frac{1}{2}|\theta|^2 t}}_{M_t^\theta \in c\mathcal{M}_{0,loc}} = 1 + c\mathcal{M}_{0,loc}$$

Since $|M_t^\theta| = e^{\frac{1}{2}|\theta|^2 t} < \infty$ and

$$\mathbb{E}(M_t^{T_n} \mid \mathcal{F}_s) = M_s^{T_n}$$

where T_n reduces M .

We may take $n \rightarrow \infty$ to get $\mathbb{E}(M_t \mid \mathcal{F}_s) = M_s$.

$$\implies \mathbb{E}(e^{i\theta B_t + \frac{1}{2}|\theta|^2 t} \mid \mathcal{F}_s) = e^{i\theta B_s + \frac{1}{2}|\theta|^2 s}.$$

$$\implies \mathbb{E}(e^{i\theta(B_t - B_s)} \mid \mathcal{F}_s) = e^{\frac{1}{2}|\theta|^2(t-s)}.$$

$$\implies B_t - B_s \sim \mathcal{N}(0, t - s)$$

Q independent of $\mathcal{F}_s$. □

Theorem 1.7.8 (Dubins and Schwarz). let $M \in c\mathcal{M}_{0,loc}$ such that $[M]_t \uparrow \infty$ as $t \rightarrow \infty$. Then $M_t = B([M]_t)$ where B is some Brownian motion.

Let $\tau_t = \inf\{u \geq 0: [M]_u > t\}$. Set $\mathcal{G}_t \equiv \mathcal{F}(\tau_t)$. Then $B_t \equiv M(\tau_t)$ is a B.M. with respect to $\mathcal{G}_t$.

Proof. (i) $M(\tau_t) \in \mathcal{F}_{\tau_t} \implies B_t \in \mathcal{G}_t$.

(ii) $t \mapsto B_t$ is continuous.

$$\begin{cases} \tau_{a-} = b, \\ \tau_{a+} = c. \end{cases}$$

We only need to prove $M(b) = M(c)$.

Assume M is bounded $\forall q \in \mathbb{Q}_+$, define

$$S_q \equiv \inf\{t > q: [M]_t > [M]_q\}.$$

We will show $M_t \equiv M_q$ for all $t \in [q, S_q)$

Note $S_q \geq q$, by OST,

$$\mathbb{E}(M_{S_q}^2 - [M]_{S_q} \mid \mathcal{F}_q) = M_q^2 - [M]_q.$$

Because $[M]_{S_q} = [M]_q$,

$$\implies \mathbb{E}[M_{S_q}^2 - M_q^2 \mid \mathcal{F}_q] = 0.$$

$$\implies \mathbb{E}((M_{S_q} - M_q)^2 \mid \mathcal{F}_q) = 0.$$

$$\implies \mathbb{E}(M_{S_q} - M_q)^2 = 0.$$

$$\implies M_{S_q} = M_q.$$

(iii) B_t and $B_t^2 - t \in c\mathcal{M}_{0,loc} \implies [B]_t = t \implies$ B.M.

Recall $B_t = M(\tau_t)$. Define

$$T_n = \inf\{t \geq 0: |M_t| > n\},$$

$$U_n = [M]_{T_n}.$$

We will check $\tau_{t \wedge U_n} = T_n \wedge \tau_t$.

$$t \wedge U_n \leq t \implies \tau_{t \wedge U_n} \leq \tau_t.$$

$$\tau_{t \wedge U_n} = \inf\{u: [M]_u > t \wedge U_n = t \wedge [M]_{T_n}\}.$$

Claim: $\tau_t \wedge T_n = \tau_{t \wedge U_n}$

case 1. $t \leq U_n$ we have

$$\tau_{t \wedge U_n} = \inf\{u: [M]_u > t\} = \tau_t,$$

and

$$t \leq U_n = [M]_{T_n} \implies T_n \geq \tau_t \implies \tau_t \wedge T_n = \tau_t = \tau_{t \wedge U_n}.$$

case 2. $t > U_n$ we have

$$\tau_{t \wedge U_n} = \tau_{U_n} = \inf\{u: [M]_u > U_n\}$$

and

$$t > U_n > [M]_{T_n} \implies T_n < \tau_t \implies \tau_t \wedge T_n = T_n.$$

It suffices to prove $T_n = \tau_{[M]_{T_n}} = \tau_{U_n}$.

One hand, $\tau_{U_n} \geq T_n$ can be proved by def of τ_{U_n} .

On the other hand, $T_n = \inf\{t: |M_t| > n\}$. Take $\forall \varepsilon > 0$, $u_\varepsilon \in (T_n, T_n + \varepsilon)$ s.t. $|M_{u_\varepsilon}| > n$.

Hence $M_{u_\varepsilon} \neq M_{T_n}$. Suppose $[M]_{u_\varepsilon} = [M]_{T_n}$, then M is constant by the property of continuous local martingale. Contradiction. So $[M]_{u_\varepsilon} > [M]_{T_n} = U_n \implies \tau_{U_n} \leq u_\varepsilon$. Let $\varepsilon \downarrow 0$ to get $\tau_{U_n} \leq T_n$.

Then $B_{t \wedge U_n} = M(\tau_{t \wedge U_n}) = M(T_n \wedge \tau_t)$, $B_t^{U_n} = M_{\tau_t}^{T_n}$.

By OST

$$\mathbb{E}(B_t^{U_n} | \mathcal{F}_{\tau_s}) = \mathbb{E}(M_{\tau_t}^{T_n} | \mathcal{F}_{\tau_s}) = M_{\tau_s}^{T_n} = B_s^{U_n}.$$

Then B^{U_n} is a martingale w.r.t. $\mathcal{G}_t = \mathcal{F}_{\tau_t} + (2)$ U_n is $\mathcal{G}_t$ -stopping time $\implies B \in c\mathcal{M}_{0,loc}$

pf of (2)

$$\{[M]_{T_n} \leq t\} = \{U_n \leq t\} \in \mathcal{F}_{\tau_t}.$$

$$\{T_n \leq \tau_t\} \in \mathcal{F}_{\tau_t}.$$

$(M^2 - [M])^{T_n}$ is martingale. Then

$$\begin{aligned} \mathbb{E}[(M_{\tau_t}^{T_n})^2 - [M]_{\tau_t}^{T_n} | \mathcal{F}_{\tau_s}] &= (M_{\tau_s}^{T_n})^2 - [M]_{\tau_s}^{T_n} \\ \implies \mathbb{E}[(B_t^{U_n})^2 - t \wedge U_n | \mathcal{F}_s] &= (B_s^{U_n})^2 - s \wedge U_n. \end{aligned}$$

$$\implies (B_t^2 - t)^{U_n} \text{ is a martingale } \implies B_t^2 - t \in \mathcal{M}_{loc} \implies [B]_t = t. \quad \square$$

If $[M]_\infty < \infty$, we get $\tau_t = \infty$, $\forall t \geq [M]_\infty$.

$$B_t = M(\infty), \quad \forall t \geq [M]_\infty.$$

So this def does not give a B.M.

Define

$$B_t = \begin{cases} M(\tau_t), & t < [M]_\infty, \\ M(\infty) + W(t) - W([M]_\infty), & t \geq [M]_\infty, \end{cases}$$

where W is a standard B.M. independent of M .

Or

$$B_t \begin{cases} M(\tau_t), & t < [M]_\infty \\ M(\infty) + W(t - [M]_\infty), & t \geq [M]_\infty. \end{cases}$$

Then $B([M]_t) = M_t$.

Lemma 1.7.8. On $\{[M]_\infty < \infty\}$, the limit $M_\infty \equiv \lim_{t \uparrow \infty} M_t$ exists a.s.

Proof. Define $S_K = \inf\{t \geq 0 : [M]_t > K\}$. Then on $\{[M]_\infty < \infty\}$, $S_K = \infty$, $\forall K > [M]_\infty$
 M^{S_K} is a L^2 -bounded martingale.

$\implies \lim_{t \uparrow \infty} M_t^{S_K}$ exists a.s.

Let $K \uparrow \infty$ to get $\lim_{t \uparrow \infty} M_t$ exists a.s.

By an enlargement of the prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

$$\begin{aligned} \implies \tilde{\Omega} &= \Omega \times \Omega^* \\ \tilde{\mathcal{F}}_t &= \mathcal{F}_t \times \mathcal{F}_t^* \\ \tilde{\mathbb{P}} &= \mathbb{P} \times \mathbb{P}^* \end{aligned}$$

where on $(\Omega^*, \mathcal{F}^*, \mathcal{F}_t^*, \mathbb{P}^*)$ we have W^* is B.M. independent of M . □

Theorem 1.7.9 (Dubins-Schwarz (general version)). Let $M \in c\mathcal{M}_{0,loc}$. Then $\exists$ B.M. B s.t. $M_t = B([M]_t)$ for all $t \geq 0$.

Theorem 1.7.10 (Doléans' Thm). Let X be a semimartingale with $X_0 = 0$. Suppose $Z_0 \in \mathcal{F}_0$. Then $\exists$ a unique continuous semimartingale Z s.t.

$$Z_t = Z_0 + \int_0^t Z_s dX_s.$$

The solution $Z_t = Z_0 e^{X_t - \frac{1}{2}[X]_t}$.

Notation: $\mathcal{E}(X_t) = e^{X_t - \frac{1}{2}[X]_t}$ is called the exponential semimartingale of X .

Proof. $\mathcal{E}(X_t) = e^{X_t - \frac{1}{2}[X]_t} = f(X_t, [X]_t)$ with $f(x, y) = e^{x - \frac{1}{2}y}$.

By apply Itô's formula,

$$df(X_t, [X]_t) = \partial_x f(X_t, [X]_t) dX_t + \partial_y f(X_t, [X]_t) d[X]_t + \frac{1}{2} (\partial_{xx} f(X_t, [X]_t) d[X]_t).$$

Because

$$\begin{cases} dX_t dX_t = d[X]_t \\ dX_t d[X]_t = 0 \\ d[X]_t d[X]_t = 0. \end{cases}$$

Hence

$$d\mathcal{E}(X_t) = \mathcal{E}(X_t) dX_t.$$

So $\mathcal{E}(X)$ is a solution to

$$dZ_t = Z_t dX_t.$$

(Uniqueness) Assume Z is another solution. We will prove

$$Z_t = Z_0 \mathcal{E}(X_t).$$

We now have $dZ_t = Z_t dX_t$.

Define

$$Y_t = e^{-X_t + \frac{1}{2}[X]_t} = (\mathcal{E}(X)_t)^{-1}.$$

It suffices to prove $d(Z_t Y_t) = 0$ which implies

$$Z_t Y_t = Z_0 Y_0 = Z_0, \quad \forall t.$$

We have

$$d(Z_t Y_t) = Z_t dY_t + Y_t dZ_t + d[Y, Z]_t.$$

Notice that by Itô's formula,

$$dY_t = (-1)Y_t dX_t + \frac{1}{2}Y_t d[X]_t + \frac{1}{2}(-1)^2 Y_t d[X]_t = -Y_t dX_t + Y_t d[X]_t.$$

$$\implies d(Z_t Y_t) = Z_t(-Y_t dX_t + Y_t d[X]_t) + Y_t(Z_t dX_t) + Y_t(Z_t dX_t) + (-Y_t Z_t) d[X]_t = 0.$$

□

Let $M \in c\mathcal{M}_{0,loc}$. Then $\mathcal{E}(\theta M)_t = e^{\theta M_t - \frac{1}{2}\theta^2[M]_t}$ is in $c\mathcal{M}_{loc}$.

By Fatou's lemma, we get

$$\begin{aligned} \mathbb{E}(\mathcal{E}(\theta M)_t \mid \mathcal{F}_s) &= \mathbb{E}(\lim_{n \rightarrow \infty} \mathcal{E}(\theta M)_{t \wedge T_n} \mid \mathcal{F}_s) \\ &\leq \liminf_{n \rightarrow \infty} \mathbb{E}(\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s) \\ &= \liminf_{n \rightarrow \infty} \mathcal{E}(\theta M)_s^{T_n} \\ &= \mathcal{E}(\theta M)_s. \end{aligned}$$

Hence $\mathcal{E}(X_t)$ is a supermartingale and $\mathbb{E}(\mathcal{E}(M)_t) \leq 1$.

Theorem 1.7.11. Suppose $\forall t > 0, \exists K_t > 0$ s.t.

$$[M]_t < K_t \quad \text{a.s.}$$

If for some $t > 0, \exists K_t > 0$ s.t.

$$[M]_t < K_t \quad \text{a.s.}$$

Then for this $t > 0$,

$$\mathbb{P} \left(\max_{s \leq t} M_s > y \right) \leq e^{-\frac{y^2}{2Kt}}, \quad \forall y > 0.$$

and $\mathcal{E}(\theta M)$ is a true martingale.

Proof. By using $\mathcal{E}(\theta M)$ is a supermartingale, by Doob's inequality,

$$\begin{aligned} \mathbb{P} \left(\max_{s \leq t} M_s > y \right) &= \mathbb{P} \left(\max_{s \leq t} \mathcal{E}(\theta M)_s > e^{\theta y - \frac{1}{2}\theta^2 Kt} \right) \\ &\leq \frac{\mathbb{E}(\mathcal{E}(\theta M)_t)}{e^{\theta y - \frac{1}{2}\theta^2 Kt}} \\ &\leq e^{-\theta y + \frac{1}{2}\theta^2 Kt}. \end{aligned}$$

Pick $\theta = \frac{y}{Kt}$ to get $e^{-\frac{y^2}{2Kt}}$.

Next, define $Z_t = \max_{s \leq t} M_s$. Then $\forall \theta > 0$,

$$\mathbb{E}e^{\theta Z_t} = \int_{\mathbb{R}} e^{\theta y} \mathbb{P}(Z_t \in dy).$$

Notice that $\forall z \in \mathbb{R}$,

$$\begin{aligned} e^{\theta Z} &= 1 + \int_0^Z \theta e^{\theta y} dy. \\ \implies \mathbb{E}(e^{\theta Z_t}) &= 1 + \mathbb{E} \left(\int_0^{Z_t} \theta e^{\theta y} dy \right) \\ &= 1 + \mathbb{E} \left(\int_0^\infty 1_{y < Z_t} \theta e^{\theta y} dy \right) \\ &= 1 + \int_0^\infty \theta e^{\theta y} dy \mathbb{P}(Z_t > y) \\ &\leq 1 + \int_0^\infty \theta e^{\theta y} e^{-\frac{y^2}{2Kt}} dy \\ &< \infty. \end{aligned}$$

By $\mathcal{E}(\theta M)_t$ is a $\mathcal{M}_{loc}$, we get $\exists T_n \uparrow \infty$ s.t. $\mathcal{E}(\theta M)_t^{T_n}$ is a martingale.

$$\implies \forall s < t, \quad \mathbb{E}[\mathcal{E}(\theta M)_t^{T_n} | \mathcal{F}_s] = \mathcal{E}(\theta M)_s^{T_n}.$$

By DCT for $\mathcal{E}(\theta M)_t^{T_n} \leq Z_t$ and $\mathbb{E}Z_t < \infty$, we get

$$\lim_{n \rightarrow \infty} \mathbb{E}(\mathcal{E}(\theta M)_t^{T_n} | \mathcal{F}_s) = \mathbb{E}(\mathcal{E}(\theta M)_t | \mathcal{F}_s).$$

□

Corollary 1.7.2. Let B be a B.M. in $\mathbb{R}^n$ and let H be a bounded previsible process. Then

$$e^{\int H dB - \frac{1}{2} \int |H_s|^2 ds}$$

is a true martingale.

Remark. $[M]_t = \int_0^t |H_s|^2 ds < K_t$.

Corollary 1.7.3. Let $M \in c\mathcal{M}_{0,loc}$. Then $\forall t > 0, y > 0, K > 0$,

$$\begin{aligned} \mathbb{P} \left(\max_{s \leq t} M_s \geq y, [M]_t \leq K \right) &\leq e^{-\frac{y^2}{2K}}. \\ \implies \mathbb{P} \left(\max_{s \leq t} |M_s| \geq y, [M]_t \leq K \right) &\leq 2e^{-\frac{y^2}{2K}}. \end{aligned}$$

Proof. Let $T = \inf\{u \geq 0: [M]_u \geq K\}$. Then

$$\mathbb{P} \left(\max_{s \leq t} M_s \geq y, [M]_t \leq K \right) \leq \mathbb{P} \left(\max_{s \leq t} M_{s \wedge T} \geq y \right) \leq e^{-\frac{y^2}{2K}}$$

since $[M^T]_t \leq K$. □

Theorem 1.7.12 (Novikov's Thm). If $\forall t > 0, \mathbb{E}(e^{\frac{1}{2}[M]_t}) < \infty$ then $\mathcal{E}(X)_t$ is a true martingale.

Proof. See P198 of [KS]. □

1.7.3 Burkholder-Davis-Gundy inequalities

Definition 1.7.10. We say a function $F : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is moderate if continuous and increasing, $F(x) = 0 \Leftrightarrow x = 0$, and $\forall \alpha > 1$,

$$\sup_{x>0} \frac{F(\alpha x)}{F(x)} < \infty.$$

Example . $F(x) = x^p, \forall p > 0$. $F(x) = x^p (\log x \vee 1), \forall p \geq 0$.

Theorem 1.7.13 (BDG inequality). Let F be moderate. Then $\exists$ universal constants $c_F \leq C_F < \infty$ s.t. $\forall M \in c\mathcal{M}_{0,loc}$,

$$c_F \mathbb{E} \left(F([M]_\infty^{\frac{1}{2}}) \right) \leq \mathbb{E} F(M_\infty^*) \leq C_F \mathbb{E} \left(F([M]_\infty^{\frac{1}{2}}) \right).$$

$M_t^* = \sup_{s \leq t} |M_s|$. Fix $t > 0$. Let $\tilde{M}_s^t = M_{s \wedge t}$. Then

$$c_F \mathbb{E} \left(F([M]_t^{\frac{1}{2}}) \right) \leq \mathbb{E} (F(M_t^*)) \leq C_F \mathbb{E} F([M]_t^{\frac{1}{2}}).$$

Example . If $F(x) = x^p$, then

$$c_p \mathbb{E}[M]_t^{\frac{p}{2}} \leq \mathbb{E} |M_t^*|^p \leq C_p \mathbb{E}[M]_t^{\frac{p}{2}}.$$

Lemma 1.7.9 (“good” λ inequality). Let X and Y be nonnegative R.V.s. Suppose $\exists \beta > 1$ s.t. $\forall \lambda > 0, \delta > 0$,

$$\mathbf{P}(X > \beta\lambda, Y \leq \delta\lambda) \leq \psi(\delta)\mathbf{P}(X > \lambda)$$

where $\psi(\delta) \downarrow 0$ as $\delta \downarrow 0$.

Then for each moderate F , $\exists C = C(\beta, \psi, F) > 0$ s.t.

$$\mathbb{E}[F(X)] \leq c\mathbb{E}[F(Y)].$$

Proof for Lemma.

- Assume good λ inequality. Then it holds by replacing X by $X \wedge a$ for any $a > 0$.

If $a > \beta\lambda > \lambda$, then

$$X \wedge a > \beta\lambda \Leftrightarrow X > \beta\lambda$$

$$X \wedge a > \lambda \Leftrightarrow X > \lambda.$$

Hence we may assume $\mathbb{E}F(X) < \infty$.

[$\mathbb{E}F(X \wedge a) \leq \mathbb{E}F(Y)$ for all $a > 0$. Then let $a \uparrow \infty$.]

Notice that $\sup_{x>0} \frac{F(x)}{F(\frac{x}{\beta})}$.

Let $\gamma > 0$ s.t. $F(x) \leq \frac{1}{\gamma}F(\frac{x}{\beta})$ for all $x > 0$.

Now

$$\int F(d\lambda)\psi(\delta)\mathbf{P}(X > \lambda) \geq \int F(d\lambda)\mathbf{P}(X > \beta\lambda, Y \leq \delta\lambda).$$

$$LHS = \psi(\delta)\mathbb{E}\left(\int_0^X F(d\lambda)\right) = \psi(\delta)\mathbb{E}(F(X)).$$

$$\begin{aligned} RHS &= \int F(d\lambda)\mathbb{E}(1_{\frac{Y}{\delta} \leq \lambda < \frac{X}{\beta}}) \\ &= \mathbb{E}\left(\int_{\frac{Y}{\delta}}^{\frac{X}{\beta}} F(d\lambda)\right) \\ &= \mathbb{E}\left(F\left(\frac{X}{\beta}\right) - F\left(\frac{Y}{\delta}\right)\right) \\ &\geq \mathbb{E}(\gamma F(X)) - \mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right). \end{aligned}$$

Let $\delta > 0$ be small such that

$$\psi(\delta) < \frac{\gamma}{2} \implies \gamma - \psi(\delta) \geq \frac{\gamma}{2}.$$

Hence

$$\mathbb{E}(F(X))(\psi(\delta) - \gamma) \geq -\mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right).$$

$$\implies \mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right) \geq (\gamma - \psi(\delta))\mathbb{E}F(X) \geq \frac{\lambda}{2}\mathbb{E}F(X).$$

Using $\sup_{y>0} \frac{F(y/\delta)}{F(y)} < \infty$, $\implies \exists \mu > 0$ s.t. $F(y/\delta) \leq \mu F(y)$ for all $y > 0$

$$\implies \mathbb{E}(F(Y/\delta)) \leq \mu \mathbb{E}F(Y)$$

$$\implies \frac{\gamma}{2} \mathbb{E}F(X) \leq \mu \mathbb{E}F(Y).$$

□

Proof. of BDF inequality.

$$\mathbb{P} \left(M_{\infty}^* > \beta\lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta\lambda \right) \leq e^{-\frac{(\beta\lambda)^2}{2(\delta\lambda)^2}} = e^{-\frac{\beta^2}{2\delta^2}}.$$

Define $\tau = \inf\{u \geq 0: |M_u| > \lambda\}$.

Then

$$\begin{aligned} \mathbb{P} \left(M_{\infty}^* > \beta\lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta\lambda \right) &= \mathbb{P} \left(M_{\infty}^* > \beta\lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta\lambda, \tau < \infty \right) \\ &= \mathbb{E} \left[1_{\tau < \infty} \mathbb{P} \left(M_{\infty}^* > \beta\lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta\lambda \mid \mathcal{F}_{\tau} \right) \right] \end{aligned}$$

Define $N_t \equiv (M_{t+\tau} - M_{\tau})^2 - ([M]_{t+\tau} - [M]_{\tau})$.

Then $N_t \in \mathcal{F}_{t+\tau}$ is a continuous local martingale

$$\mathbb{E} [M_{t+\tau}^2 - [M]_{t+\tau} \mid \mathcal{F}_{s+\tau}] = M_{s+\tau}^2 - [M]_{s+\tau}.$$

$\implies N_0 = 0$ and on $\{M_{\infty}^* > \beta\lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta\lambda\}$ we have $\exists \tilde{t} > 0$ s.t.

$$M_{\tilde{t}+\tau} \geq \beta\lambda \implies N_{\tilde{t}} = (M_{\tilde{t}+\tau} - M_{\tau})^2 - ([M]_{\tilde{t}+\tau} - [M]_{\tau}) \geq [(\beta - 1)\lambda]^2 - (\delta\lambda)^2.$$

Moreover $N_t \geq -\delta^2\lambda^2$, $\forall t \geq 0$.

Hence N_t reaches $[(\beta - 1)\lambda]^2 - (\delta\lambda)^2$ before $(-\delta^2\lambda^2)$.

By (Gambler's Ruin $\mathbb{P}_k(\tau_0 < \tau_N) = \frac{N-k}{N}$)

$$\mathbb{P}(\tau_b < \tau_a) \leq \frac{-a}{b-a}$$

$$\implies \mathbb{P} \left(M_{\infty}^* > \beta\lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta\lambda \mid \mathcal{F}_{\tau} \right) \leq \mathbb{P}(\tau_{[(\beta-1)\lambda]^2 - (\delta\lambda)^2} < \tau_{-(\delta\lambda)^2}) \leq \frac{(\delta\lambda)^2}{[(\beta-1)\lambda]^2} = \frac{\delta^2}{(\beta-1)^2}.$$

Here we take $0 < \delta < \beta - 1$. Therefore

$$\mathbb{P} \left(M_{\infty}^* > \beta\lambda, [M]_{\infty}^{\frac{1}{2}} \leq \delta\lambda \right) \leq \delta^2 \frac{1}{(\beta-1)^2} \mathbb{P}(\tau < \infty) = \delta^2 \frac{1}{(\beta-1)^2} \mathbb{P}(M_{\infty}^* > \lambda).$$

Similarly,

$$\mathbb{P} \left([M]_{\infty}^{\frac{1}{2}} > \beta\lambda, M_{\infty}^* \leq \delta\lambda \right).$$

$$\tilde{N}_t = [M]_{t+\tilde{\tau}} - [M]_{\tilde{\tau}} - (M_{\tilde{\tau}+t} - M_{\tilde{\tau}})^2$$

with $\tilde{\tau} = \inf\{u \geq 0: [M]_u^{\frac{1}{2}} > \lambda\}$.

$$\implies \mathbb{P}\left([M]_{\infty}^{\frac{1}{2}} > \beta\lambda, M_{\infty}^* \leq \delta\lambda, \tilde{\tau} < \infty\right) = \mathbb{E}[\mathbb{P}(\cdots | \mathcal{F}_{\tilde{\tau}}) \mathbf{1}_{\tilde{\tau} < \infty}].$$

Then $\tilde{N}_t$ reaches $(\beta\lambda)^2 - \lambda^2 - (\delta\lambda)^2$ before $-(\delta\lambda)^2$.

□

1.8 Semimartingale local time

For a Brownian motion $(B_t)_{t \geq 0}$. We consider

$$l_t^0 = \int_0^t \delta_0(B_s) ds,$$

where δ_0 is the Dirac delta function such that $\forall$ function f ,

$$\int f(x) \delta_0(x) dx = f(0).$$

$$\mu([a, b]) = l_b^0 - l_a^0,$$

$\text{supp}(\mu) = \{t \geq 0: B_t = 0\}$, level set $= \{t \geq 0: B_t = x\}$, using local time

$$l_t^x = \int_0^t \delta_x(B_s) ds.$$

Theorem 1.8.1 (Tanaka's formula). Let X be a continuous semimartingale. Then $\exists$ continuous increasing adapted process $\{l_t, t \geq 0\}$

$$|X_t| - |X_0| = \int_0^t \text{sgn}(X_s) dX_s + l_t,$$

where

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0. \end{cases}$$

Then $\{l_t, t \geq 0\}$ is called the local time of X at 0, and

$$\int_0^t \mathbf{1}_{\{X_s \neq 0\}} dl_s = 0.$$

Remark.

(1) $f(x) = |x| \implies f'(x) = \text{sgn}(x), \forall x \neq 0 \implies f''(x) = 2\delta(x)$. (distributional derivative)

(2) $|X_t - x| - |X_0 - x| = \int_0^t \text{sgn}(X_0 - x) dX_s + l_t^x$.

(3) Note that $|X_t| = X_t^+ + X_t^-$, $X_t = X_t^+ - X_t^-$

$$\implies 2X_t^+ = |X_t| + X_t = \int_0^t 1_{\{X_s > 0\}} dX_s - \int_0^t 1_{\{X_s \leq 0\}} dX_s + l_t + \int_0^t 1 dX_s + X_0 + |X_0|$$

$$\implies 2X_t^+ = 2 \int_0^t 1_{\{X_s > 0\}} dX_s + l_t + 2X_0^+.$$

$$\implies X_t^+ - X_0^+ = \int_0^t 1_{\{X_s > 0\}} dX_s + \frac{1}{2}l_t.$$

Similarly, $2X_t^- = |X_t| - X_t$

$$\implies X_t^- - X_0^- = - \int_0^t 1_{\{X_s \leq 0\}} dX_s + \frac{1}{2}l_t.$$

Proof. Apply Itô's formula to the function $f(x) = |x|$.

$$\begin{aligned} \implies |X_t| &= |X_0| + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X]_s \\ &= |X_0| + \int_0^t \text{sgn}(X_s) dX_s + \underbrace{\frac{1}{2} \int_0^t 2\delta(X_s) d[X]_s}_{=l_t}. \end{aligned}$$

$$\exists \varphi(x) \in C^\infty(\mathbb{R}), \text{ define } \varphi_n(x) = \begin{cases} -1, & x \leq 0 \\ \varphi(nx), & 0 < x < \frac{1}{n} \\ 1, & x \geq \frac{1}{n}, \end{cases} \text{ then } \varphi_n(x) \rightarrow \text{sgn}(x), \forall x \neq 0.$$

Let

$$f_n(x) = \begin{cases} -x, & x \leq 0 \\ \int_0^x \varphi_n(t) dt, & x > 0. \end{cases}$$

$$\implies f'_n(x) = \varphi_n(x).$$

Because $f(x) = |x| = \int_0^{|x|} 1 dt$.

$$\implies |f(x) - f_n(x)| \leq \int_0^{|x|} |\varphi_n(t) - 1| dt \leq \int_0^{\frac{1}{n}} |\varphi_n(t) - 1| dt \leq 2\frac{1}{n}.$$

Apply Itô formula to f_n to get that

$$f_n(X_t) = f_n(X_0) + \int_0^t f'_n(X_s) dX_s + \frac{1}{2} \int_0^t f''_n(X_s) d[X]_s.$$

Note

$$\int_0^t f'_n(X_s) dX_s = \int_0^t f'_n(X_s) dM_s + \int_0^t f'_n(X_s) dA_s,$$

and

$$\int_0^t f'_n(X_s) dA_s = \int_0^t \varphi_n(X_s) dA_s.$$

We have $\varphi_n(x) \uparrow \text{sgn}(x)$, $\forall x$. By Monotone convergence,

$$\left| \int_0^t \varphi_n(X_s) - \text{sgn}(X_s) dA_s \right| \leq \int_0^t |\varphi_n(X_s) - \text{sgn}(X_s)| dA_s \rightarrow 0,$$

which implies $\int_0^t \varphi_n(X_s) dA_s \rightarrow \int_0^t \text{sgn}(X_s) dA_s$ uniform in t .

Next, $T_k = \inf\{t \geq 0: |M_t| \geq k\}$. Then M^{T_k} is bounded.

We may assume M is bounded. Then

$$\left\| \int_0^\infty (\text{sgn}(X_s) - f'_n(X_s)) dM_s \right\|_2^2 = \mathbb{E} \int_0^\infty (\text{sgn}(X_s) - f'_n(X_s))^2 d[M]_s \rightarrow 0.$$

So

$$\mathbb{E} \left[\sup_{t \leq T} \left| \int_0^t (\text{sgn}(X_s) - f'_n(X_s)) dM_s \right|^2 \right] \xrightarrow{n \rightarrow \infty} 0.$$

Hence

$$\sup_{t \geq 0} \int_0^t (\text{sgn}(X_s) - f'_n(X_s)) dM_s \xrightarrow{a.s.} 0.$$

$$\implies \int_0^t f'_n(X_s) dX_s \xrightarrow{a.s.} \int_0^t \text{sgn}(X_s) dX_s, f_n(X_t) \xrightarrow{a.s.} |X_t|, f_n(X_0) \xrightarrow{a.s.} |X_0|.$$

$$\implies C_t^n \triangleq \int_0^t \frac{1}{2} f''_n(X_s) d[X]_s, C_t^n \xrightarrow{a.s.} l_t \triangleq |X_t| - |X_0| - \int_0^t \text{sgn}(X_s) dX_s.$$

$$dC_s^n = \frac{1}{2} f''_n(X_s) d[X]_s \implies \forall \delta > 0, \text{ if } \frac{1}{n} < \delta, \text{ then}$$

$$|X_s| > \delta > \frac{1}{n} \implies f''_n(X_s) = 0.$$

$$\implies \int_0^t 1_{\{|X_s| > \delta\}} dC_s^n = 0.$$

Hence

$$\int_0^t 1_{\{|X_s| > \delta\}} dl_s \leq \lim_{n \rightarrow \infty} \int_0^t 1_{\{|X_s| > \frac{\delta}{2}\}} dC_s^n = 0.$$

$$\int_0^t \varphi_\delta(X_s) dl_s = \lim_{n \rightarrow \infty} \int_0^t \varphi_\delta(X_s) dC_s^n \leq \liminf_{n \rightarrow \infty} \int_0^t 1_{\{|X_s| > \frac{\delta}{2}\}} dC_s^n.$$

$$\text{We prove } \int_0^t 1_{\{|X_s| > \delta\}} dl_s = 0, \forall \delta > 0 \implies \int_0^t 1_{\{|X_s| \neq 0\}} dl_s = 0.$$

□

Let X, Y be continuous semimartingale. Then

$$\int_0^t Y_t dX_t = \text{It\^o Integral.}$$

Stratonovich Integral

$$\int_0^t Y_t \partial X_t \triangleq \int_0^t Y_t dX_t + \frac{1}{2}[X, Y]_t.$$

Recall that

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X, Y]_t \\ &= \int_0^t X_s dY_s + \frac{1}{2}[X, Y]_t + \int_0^t Y_s dX_s + \frac{1}{2}[X, Y]_t \\ &= \int_0^t X_s \partial Y_s + \int_0^t Y_s \partial X_s. \end{aligned}$$

Theorem 1.8.2. $f(X_t) = f(X_0) + \int_0^t f'(X_s) \partial X_s.$

Proof. By It\^o formula,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X]_s.$$

By definition,

$$\int_0^t f'(X_s) \partial X_s = \int_0^t f'(X_s) dX_s + \frac{1}{2} [f'(X), X]_t.$$

$$df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) d[X]_t$$

$$\implies df'(X_t) = f''(X_t) dX_t + \frac{1}{2} f'''(X_t) d[X]_t.$$

So

$$f'(X_t) = f'(X_0) + \int_0^t f''(X_s) dX_s + \frac{1}{2} \int_0^t f'''(X_s) d[X]_s.$$

$$\implies [f'(X), X]_t = \left[\int_0^t f''(X_s) dX_s, X_t \right] = \int_0^t f''(X_s) d[X]_s.$$

$$([H \cdot M, K \cdot N]_t = \int_0^t H K d[M, N]_s).$$

So

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} [f'(X), X]_t = f(x_0) + \int_0^t f'(X_s) \partial X_s.$$

□

Theorem 1.8.3 (Riemann-Sum Approximation). Let C be a locally bounded adapted L -process.

Let X be a continuous semimartingale. Then

$$\sum_{i=1}^{\infty} c(t_{i-1}^n)(X_{t_i^n} - X_{t_{i-1}^n}) \rightarrow \int_0^t c(s) dX_s.$$

with $t_i^n = t \wedge \frac{i}{2^n}$. The convergence is uniform in probability on compacts, i.e. $\forall K > 0$,

$$\sup_{t \leq K} |I^n(t) - I(t)| \xrightarrow{P} 0.$$

Proof. Let $c^n(t) = c(t_{i-1}^n)$ for $t \in (t_{i-1}^n, t_i^n]$. Then $c^n(t) \rightarrow c(t)$ for all $t \geq 0$.

Also,

$$\begin{aligned} I^n(t) &= \sum_{i=1}^{\infty} c(t_{i-1}^n)(X(t_i^n) - X(t_{i-1}^n)) \\ &= \int_0^t c^n(s) dX_s \\ &= (c^n, X)_t = \int_0^t c^n(s) dM_s + \int_0^t c^n(s) dA_s. \end{aligned}$$

$$\forall t \leq k, |c^n(t)| \leq c_k$$

$$\implies \int_0^t c^n(s) dA_s \rightarrow \int_0^t c(s) dA_s.$$

By localization, we may assume M is bounded. If c is bounded and $M \in \mathcal{M}_0^2$, then

$$\int_0^t c^n(s) dM_s \rightarrow \int_0^t c(s) dM_s.$$

$$\mathbb{E} \left[\sup_{t \leq k} \left| \int_0^t (c^n(s) - c(s)) dM_s \right|^2 \right] \leq c_k \mathbb{E} \int_0^k (c^n(s) - c(s))^2 d[M]_s \rightarrow 0.$$

□

1.8.1 Stratonovich Calculus

Let X, Y be continuous semimartingale. Define

$$\begin{aligned} S^n(t) &\equiv \sum_{i=1}^n \frac{1}{2} (Y(t_i^n) + Y(t_{i-1}^n)) (X(t_i^n) - X(t_{i-1}^n)) \\ S(t) &= \int_0^t Y_s \partial X_s. \end{aligned}$$

Then

$$\sup_{t \leq k} |S^n(t) - S(t)| \xrightarrow{P} 0.$$

Proof.

$$\begin{aligned} S^n(t) &= \sum_{i=1}^{\infty} \left[Y(t_{i-1}^n) + \frac{1}{2}(Y(t_i^n) - Y(t_{i-1}^n)) \right] (X(t_i^n) - X(t_{i-1}^n)) \\ &= \sum_{i=1}^{\infty} Y(t_{i-1}^n)(X(t_i^n) - X(t_{i-1}^n)) + \frac{1}{2} \sum_{i=1}^{\infty} \Delta Y(t_i^n) \Delta X(t_i^n). \end{aligned}$$

It remains to prove

$$\begin{aligned} \sum_{i=1}^{\infty} \Delta Y(t_i^n) \Delta X(t_i^n) &\rightarrow [X, Y]_t = \frac{1}{4}([X + Y]_t - [X - Y]_t) \\ \Leftrightarrow \sum_{i=1}^{\infty} \Delta X(t_i^n)^2 &\rightarrow [X]_t \\ \Leftrightarrow \sum_{i=1}^{\infty} (\Delta M(t_i^n) + \Delta A(t_i^n))^2 &\rightarrow [X]_t = [M]_t. \end{aligned}$$

We have $\sum_{i=1}^{\infty} (\Delta M(t_i^n))^2 \rightarrow [M]_t$,

$$\left| \sum_{i=1}^{\infty} \Delta M(t_i^n) \Delta A(t_i^n) \right| \leq \sup_{i \geq 1} |\Delta M(t_i^n)| \sum_{i=1}^{\infty} |\Delta A(t_i^n)| \leq \sup_{\substack{t \leq k \\ i \geq 1}} |\Delta M(t_i^n)| \text{Var}_A(t) \xrightarrow{a.s.} 0.$$

□

1.9 Stochastic Differential Equation

$$\begin{aligned} dX_t &= b(t, X_t) dt + \sigma(t, X_t) dW_t, \\ X_t &= X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s. \end{aligned}$$

d-dimension SDE ...

Definition 1.9.1 (Strong solution). A strong solution of the SDE, on a given probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a given B.M. $(W_t, \mathcal{F}_t^W)_{t \geq 0}$, with initial condition ξ is a process $\{X_t, t \geq 0\}$ with continuous sample path and

- (i) $X \in \mathcal{F}_t$ with $(\mathcal{F}_t)_{t \geq 0}$ being the augmented filtration of the B.M. $(\mathcal{F}_t^W)_{t \geq 0}$.
- (ii) $\mathbb{P}(X_0 = \xi) = 1$.
- (iii)

$$\mathbb{P} \left(\int_0^t |b_i(s, X_s)| + \sigma_{ij}^2(s, X_s) ds < \infty \right) = 1$$

for all i, j, t .

- (iv) $X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s$.

Definition 1.9.2 (Strong Uniqueness). Given $b(t, x)$ and $\sigma(t, x)$. Suppose $\forall$ B.M. $\{W_t\}_{t \geq 0}$ on some $(\Omega, \mathcal{F}, \mathbb{P})$ and ξ is the initial condition, and $\{\mathcal{F}_t\}$ is the augmented filtration, if X and $\tilde{X}$ are two strong solutions with respect to the same W , then

$$\mathbb{P}(X_t = \tilde{X}_t, \forall t \geq 0) = 1.$$

Example . (i) $dX_t = dW_t \implies X_t = W_t + \xi_0$.

(ii) $dX_t = b(t, X_t) dt + dW_t$ where $b : [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ is bounded Borel-measurable and $\forall x \leq y$, we have $b(t, x) \geq b(t, y)$, $\forall t \geq 0$. For this SDE, strong uniqueness holds.

Proof. Eg. (ii) $\forall$ B.M. on $(\Omega, \mathcal{F}, \mathbb{P})$, $\forall \xi$. If $X^{(1)}$ & $X^{(2)}$ are strong solutions, then

$$X_t^{(j)} = \xi + \int_0^t b(s, X_s^{(j)}) ds + W_t, \quad \forall j = 1, 2.$$

$$X_t^{(1)} - X_t^{(2)} = \int_0^t [b(s, X_s^{(1)}) - b(s, X_s^{(2)})] ds.$$

$$\implies d\Delta_t = [b(t, X_t^{(1)}) - b(t, X_t^{(2)})] dt.$$

By Itô's formula,

$$\Delta_t^2 = \int_0^t 2\Delta_s d\Delta_s + \frac{1}{2} \int_0^t 2 d[\Delta]_t = 2 \int_0^t (X_s^{(1)} - X_s^{(2)}) [b(s, X_s^{(1)}) - b(s, X_s^{(2)})] ds.$$

If $x \leq y$, then $b(s, x) - b(s, y) \geq 0$.

If $x > y$, then $b(s, x) - b(s, y) \leq 0$.

$\implies (x - y)(b(s, x) - b(s, y)) \leq 0$. So

$$\Delta_t^2 = 2 \int_0^t (X_s^{(1)} - X_s^{(2)}) [b(s, X_s^{(1)}) - b(s, X_s^{(2)})] ds \leq 0, \quad \text{a.s.}$$

$\implies \mathbb{P}(\Delta_t = 0, \forall t \geq 0) = 1$.

$\implies$ strong uniqueness. □

Theorem 1.9.1 (Itô Uniqueness Theorem). Suppose $b(t, x)$ and $\sigma(t, x)$ are locally Lipschitz in the space variable: $\forall n \geq 1, \exists K_n > 0$ s.t. $\forall t \geq 0, \|x\|, \|y\| \leq n$, we have

$$\|b(t, x) - b(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\| \leq K_n \|x - y\|.$$

Then strong uniqueness holds.

Lemma 1.9.1 (Gronwall's Inequality). Suppose a continuous function $g(t)$ satisfies $0 \leq g(t) \leq \alpha(t) + \beta \int_0^t g(s) ds$ for all $t \in [0, T]$ with $\beta \geq 0$ and $\alpha : [0, T] \rightarrow \mathbb{R}$ integrable. Then

$$g(t) \leq \alpha(t) + \beta \int_0^t \alpha(s) e^{\beta(t-s)} ds, \quad \forall t \leq T.$$

Proof. $I(t) := \int_0^t g(s) ds.$

$$\implies I'(t) = g(t) \leq \alpha(t) + \beta I(t).$$

$$\implies I'(t) - \beta I(t) - \alpha(t) \leq 0.$$

Notice that

$$\left[e^{-\beta t} I(t) \right]' = (I'(t) - \beta I(t)) e^{-\beta t} \leq \alpha(t) e^{-\beta t}.$$

$$e^{-\beta t} I(t) - 0 = \int_0^t \left[e^{-\beta s} I(s) \right]' ds \leq \int_0^t \alpha(s) e^{-\beta s} ds.$$

$$\implies I(t) \leq \int_0^t \alpha(s) e^{\beta(t-s)} ds.$$

$$\implies g(t) \leq \alpha(t) + \beta I(t) = \alpha(t) + \beta \int_0^t \alpha(s) e^{\beta(t-s)} ds. \quad \square$$

Proof. of Itô Uniqueness Theorem.

Given W and $(\Omega, \mathcal{F}, \mathbf{P})$ and ξ . If X and $\tilde{X}$ are two strong solutions with respect to the same W . Define

$$\tau_n = \inf\{t \geq 0: \|X_t\| \geq n\}$$

$$\tilde{\tau}_n = \inf\{t \geq 0: \|\tilde{X}_t\| \geq n\}.$$

Set $\tau_n \wedge \tilde{\tau}_n$. Then $\lim_{n \rightarrow \infty} S_n = \infty$, $\mathbf{P}$ -a.s.

Note that

$$X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n} = \int_0^{t \wedge S_n} [b(s, X_s) - b(s, \tilde{X}_s)] ds + \int_0^{t \wedge S_n} [\sigma(s, X_s) - \sigma(s, \tilde{X}_s)] dW_s$$

$$\implies \mathbb{E} \|X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n}\|^2 \leq 4\mathbb{E} \left[\int_0^{t \wedge S_n} (b - \tilde{b})^2 ds \right] + 4\mathbb{E} \left[\int_0^{t \wedge S_n} (\sigma - \tilde{\sigma})^2 ds \right]$$

$$(X_t^1, X_t^2, \dots, X_t^d) = \left(\int_0^t b^1 ds + \int_0^t \sigma^1 dW_s, \dots, \int_0^t b^d ds + \int_0^t \sigma^d dW_s \right).$$

$$\begin{aligned} \implies \|X\|^2 &= \sum_{j=1}^d (X_t^j)^2 \\ &= \sum_{j=1}^d \left(\int_0^t b^j ds + \int_0^t \sigma^j dW_s \right)^2 \\ &\leq 2 \sum_{j=1}^d \left(\int_0^t b^j ds \right)^2 + \left(\int_0^t \sigma^j dW_s \right)^2 \end{aligned}$$

$$\begin{aligned}
\mathbb{E} \left\| X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n} \right\|^2 &\leq 2 \sum_{i=1}^d \mathbb{E} \left(\int_0^{t \wedge S_n} \left(b_i(s, X_s) - b(s, \tilde{X}_s) \right) ds \right)^2 \\
&\quad + 2 \sum_{i=1}^d \mathbb{E} \left(\sum_{j=1}^r \int_0^{t \wedge S_n} \left(\sigma_{ij}(s, X_s) - \sigma_{ij}(s, \tilde{X}_s) \right) dW_s^{(j)} \right)^2 \\
&\leq 2 \sum_{i=1}^d \mathbb{E} \left[t \cdot \int_0^{t \wedge S_n} \left(b_i(s, X_s) - b_i(s, \tilde{X}_s) \right)^2 ds \right] \\
&\quad + 2 \sum_{i=1}^d \sum_{j=1}^r \mathbb{E} \left[\int_0^{t \wedge S_n} (\sigma_{ij} - \sigma_{ij})^2 ds \right]
\end{aligned}$$

And $\mathbb{E} \left[\int_0^{t \wedge S_n} (\sigma_{ij} - \sigma_{ij})^2 ds \right] = \mathbb{E} \left[\int_0^{t \wedge S_n} (\sigma_{ij} - \sigma_{ij})^2 ds \right]$.

We conclude that

$$\begin{aligned}
\mathbb{E} \left\| X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n} \right\|^2 &\leq 2t \mathbb{E} \int_0^{t \wedge S_n} \left\| b(s, X_s) - b(s, \tilde{X}_s) \right\|^2 ds + 2 \mathbb{E} \int_0^{t \wedge S_n} \left\| \sigma(s, X_s) - \sigma(s, \tilde{X}_s) \right\|^2 ds \\
&\leq 2T \mathbb{E} \int_0^{t \wedge S_n} K_n^2 \left\| X_s - \tilde{X}_s \right\|^2 ds + 2 \mathbb{E} \int_0^{t \wedge S_n} K_n^2 \left\| X_s - \tilde{X}_s \right\|^2 ds \\
&\leq 2(T+1) K_n^2 \int_0^t \mathbb{E} \left\| X_{s \wedge S_n} - \tilde{X}_{s \wedge S_n} \right\|^2 ds,
\end{aligned}$$

since

$$\|\sigma\|^2 = \sum_{i=1}^d \sum_{j=1}^r \sigma_{ij}^2.$$

If we let $g(t) = \mathbb{E} \left\| X_{t \wedge S_n} - \tilde{X}_{t \wedge S_n} \right\|^2$, then $g(t) \leq 2(T+1) K_n^2 \int_0^t g(s) ds$.

By Gronwall's inequality, we get $g(t) \leq 0$.

$$\implies X_{t \wedge S_n} = \tilde{X}_{t \wedge S_n} \text{ a.s., } \forall t \geq 0.$$

$$\implies X_{t \wedge S_n} = \tilde{X}_{t \wedge S_n}, \forall t \geq 0, \text{ a.s.}$$

Let $S_n \uparrow \infty$ to conclude $X_t = \tilde{X}_t, \forall t \geq 0$, a.s. □

Theorem 1.9.2 (Existence). Suppose $\exists K > 0$ s.t. $\forall t \geq 0, \forall x, y \in \mathbb{R}^d$,

$$\begin{cases} \|b(t, x) - b(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\| \leq K \|x - y\| \\ \|b(t, x)\| + \|\sigma(t, x)\| \leq K(1 + \|x\|). \end{cases}$$

On $(\Omega, \mathcal{F}, \mathbb{P})$, let ξ be R.V. and r -dimensional $B.M.$ $W = \{W_t\}_{t \geq 0}$ independent of ξ , and $\mathbb{E} \|\xi\|^2 < \infty$. Let $(\mathcal{F}_t)_{t \geq 0}$ be the augmented filtration. Then $\exists$ continuous adapted process $X = \{X_t\}_{t \geq 0}$ which is a strong solution with respect to W and $X_0 = \xi$ a.s.

Moreover, $\forall T > 0, \exists C = C(K, T) > 0$ s.t. $\forall 0 \leq t \leq T$,

$$\mathbb{E} \|X_t\|^2 \leq C(1 + \mathbb{E} \|\xi\|^2) e^{Ct}.$$

Proof. Define $X_t^{(0)} = \xi, \forall t \geq 0$.

For any $k \geq 1$, define

$$X_t^{(k)} \triangleq \xi + \int_0^t b(s, X_s^{(k-1)}) ds + \int_0^t \sigma(s, X_s^{(k-1)}) dW_s.$$

Then $\forall T > 0, \exists C = C(K, T) > 0$ s.t.

$$\mathbb{E} \left\| X_t^{(k)} \right\|^2 \leq C(1 + \mathbb{E} \|\xi\|^2) e^{Ct}, \quad \forall t \leq T, \forall k \geq 0.$$

(Exercise)

For any $k \geq 0$,

$$X_t^{(k+1)} - X_t^{(k)} = \int_0^t \left(b(s, X_s^{(k)}) - b(s, X_s^{(k-1)}) \right) ds + \int_0^t \left(\sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)}) \right) dW_s.$$

$$\begin{aligned} \mathbb{E} \left[\sup_{s \leq t} \|B_s\|^2 \right] &= \mathbb{E} \left\| \sup_{s \leq t} \int_0^s \left(b(u, X_u^{(k)}) - b(u, X_u^{(k-1)}) \right) du \right\|^2 \\ &\leq \mathbb{E} \left[t \cdot \int_0^t \left\| b(u, X_u^{(k)}) - b(u, X_u^{(k-1)}) \right\|^2 du \right] \\ &\leq K^2 + \int_0^t \mathbb{E} \left\| X_u^{(k)} - X_u^{(k-1)} \right\|^2 du. \end{aligned}$$

$$M_t = \int_0^t \sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)}) dW_s.$$

$$[M]_t = \int_0^t \left\| \sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)}) \right\|^2 ds.$$

$$\implies \mathbb{E}[M]_t \leq \int_0^t K^2 \left(1 + \left\| X_s^{(k)} \right\| + \left\| X_s^{(k-1)} \right\| \right)^2 ds < \infty \text{ by } \mathbb{E} \left\| X_t^{(k)} \right\|^2 \leq C e^{Ct}.$$

$$\begin{aligned} \mathbb{E} \left[\sup_{s \leq t} \|M_s\|^2 \right] &\leq C \mathbb{E}[M_t] \\ &= C \mathbb{E} \int_0^t \left\| \sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)}) \right\|^2 ds \\ &\leq C \mathbb{E} \int_0^t K \left\| X_s^{(k)} - X_s^{(k-1)} \right\|^2 ds. \end{aligned}$$

Combine the two inequalities to get

$$\begin{aligned} \mathbb{E} \left[\max_{s \leq t} \left\| X_s^{(k+1)} - X_s^{(k)} \right\|^2 \right] &\leq 2 \mathbb{E} \left[\max_{s \leq t} \|B_s\|^2 \right] + 2 \mathbb{E} \left[\max_{s \leq t} \|M_s\|^2 \right] \\ &\leq C \cdot 4K^2(1 + T) \int_0^t \mathbb{E} \left\| X_s^{(k)} - X_s^{(k-1)} \right\|^2 ds \\ &\leq L \int_0^t \mathbb{E} \left(\left\| X_u^{(k)} - X_u^{(k-1)} \right\|^2 \right) ds. \end{aligned}$$

When $k = 1$, we have

$$\mathbb{E} \left(\left\| X_s^{(1)} - X_s^{(0)} \right\|^2 \right) \leq 2\mathbb{E} \left\| X_t^{(1)} \right\|^2 + 2\mathbb{E} \left\| X_t^{(0)} \right\|^2 \leq C^*$$

for all $0 \leq t \leq T$.

Hence

$$\begin{aligned} \mathbb{E} \left(\max_{0 \leq s \leq t} \left\| X_s^{(2)} - X_s^{(1)} \right\|^2 \right) &\leq L \int_0^t C^* ds = C^*(Lt). \\ \implies \mathbb{E} \left(\max_{0 \leq s \leq t} \left\| X_s^{(3)} - X_s^{(2)} \right\|^2 \right) &\leq L \int_0^t C^*(Ls) ds = C^* \frac{1}{2} (Lt)^2. \end{aligned}$$

By induction, we can prove $\forall k \geq 1$,

$$\mathbb{E} \left(\max_{s \leq t} \left\| X_s^{(k+1)} - X_s^{(k)} \right\|^2 \right) \leq C^* \frac{1}{k!} (Lt)^k.$$

By chebyshev's Inequality,

$$\mathbb{P} \left(\max_{0 \leq t \leq T} \left\| X_t^{(k+1)} - X_t^{(k)} \right\| > \frac{1}{2^{k+1}} \right) \leq 4^{k+1} C^* \frac{1}{k!} (LT)^k.$$

$$\sum_{k=1}^{\infty} \mathbb{P}(\max \|\dots\| > \frac{1}{2^{k+1}}) < \infty.$$

$\implies$ B.C. Lemma gives

$$\mathbb{P} \left(\max_{t \leq T} \left\| X_t^{(k+1)} - X_t^{(k)} \right\| > \frac{1}{2^{k+1}} \text{ i.o.} \right) = 0.$$

$\implies$ P-a.s. $\exists N(\omega) < \infty$ s.t. $\forall k \geq N$,

$$\max_{t \leq T} \left\| X_t^{(k+1)} - X_t^{(k)} \right\| \leq \frac{1}{2^{k+1}}.$$

$\implies \forall k \geq N, \forall m \geq 1$, we have

$$\max_{t \leq T} \left\| X_t^{(k+m)} - X_t^{(k)} \right\| \leq \sum_{j=k}^{k+m-1} \frac{1}{2^{j+1}} \leq \frac{1}{2^k}.$$

Hence $\left\{ X_t^{(k)}(\omega), 0 \leq t \leq T \right\}$ is a Cauchy sequence in $\|\dots\|_{sup}$.

Hence $\exists$ limit $\{X_t(\omega), 0 \leq t \leq T\}$ s.t.

$$\lim_{k \rightarrow \infty} \max_{t \leq T} \left\| X_t^{(k)} - X_t \right\| = 0.$$

Since $T > 0$ is arbitrary, $\exists X = \{X_t, 0 \leq t < \infty\}$ s.t. $\forall T > 0, \sup_{t \leq T} \left\| X_t^{(k)} - X_t \right\| \rightarrow 0$ as $k \rightarrow \infty$.

$$\begin{aligned}\mathbb{E} \|X_t\|^2 &= \mathbb{E} \lim_{k \rightarrow \infty} \|X_t^{(k)}\|^2 \\ &\leq \lim_{k \rightarrow \infty} \mathbb{E} \|X_t^{(k)}\|^2 \leq C(1 + \mathbb{E} \|\xi\|^2) e^{Ct}.\end{aligned}$$

$$\implies \mathbb{P} \left(\int_0^t |b(s, X_s)| ds + \int_0^t \sigma_{ij}^2(s, X_s) ds < \infty \right) = 1.$$

$$X_t^{(k+1)} = \xi + \int_0^t b(s, X_s^{(k)}) ds + \int_0^t \sigma(s, X_s^{(k)}) dW_s.$$

$$X_t = \xi + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s.$$

Need to prove $X_t^{(k+1)} \xrightarrow{k \rightarrow \infty} X_t$ a.s. □

Theorem 1.9.3 (Yamada-Watanabe). Let $dX_t = \sigma(X_t) dW_t + b(X_t) dt$, where

$$\sigma(x) = \begin{pmatrix} \sigma_1(x) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n(x) \end{pmatrix}$$

$$b(x) = (b^1(x), b^2(x), \dots, b^n(x))$$

such that

(i) $\exists$ positive increasing function $\rho(u)$, $u \in (0, \infty)$ s.t.

$$|\sigma_i(\xi) - \sigma_i(\eta)| \leq \rho(|\xi - \eta|)$$

for all $\xi, \eta \in \mathbb{R}^1$, $i = 1, \dots, n$ and

$$\int_{0+} \rho^{-2}(u) du = \infty.$$

i.e. $\forall \varepsilon > 0$, $\int_0^\varepsilon \rho^{-2}(u) du = \infty$.

(ii) $\exists$ positive increasing concave function $k(u)$, $u \in (0, \infty)$ s.t.

$$|b_i(x) - b_j(y)| \leq k(|x - y|)$$

for all $x, y \in \mathbb{R}^n$, $i = 1, \dots, n$ and $\int_{0+} k^{-1}(u) du = \infty$.

Then strong uniqueness holds.

Remark. If $\rho(u) = u^\alpha$, then strong uniqueness holds if $\alpha \geq \frac{1}{2}$; fails if $0 < \alpha < \frac{1}{2}$.

Proposition 1.9.1. (Yamada & Watanabe 1971)

“ $d = 1$ ” Consider

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t.$$

Assume that $\exists$ constant $K > 0$ and $h : [0, \infty) \rightarrow [0, \infty)$ is a strictly increasing function with $h(0) = 0$ such that

$$\begin{cases} |b(t, x) - b(t, y)| \leq K |x - y| \\ |\sigma(t, x) - \sigma(t, y)| \leq h(|x - y|) \end{cases}$$

hold for all $t \geq 0$ and $x, y \in \mathbb{R}$ with

$$\int_0^\varepsilon h(u)^{-2} du = \infty, \quad \forall \varepsilon > 0.$$

Then strong uniqueness holds.

Proof. Idea: Construct a C^2 function $\psi_n(x)$ s.t. $\psi_n(x) \uparrow |x|$. Let $a_0 = 1 > a_1 > \dots > a_n \downarrow 0$ such that

$$\begin{aligned} \int_{a_1}^{a_0} h(u)^{-2} du &= 1, \\ \int_{a_2}^{a_1} h(u)^{-2} du &= 2, \\ &\dots \\ \int_{a_n}^{a_{n-1}} h(u)^{-2} du &= n, \end{aligned}$$

So

$$\int_{a_n}^{a_{n-1}} \frac{2}{n} h(x)^{-2} dx = 2.$$

Then $\exists \rho_n(x)$ s.t.

$$\int_{a_n}^{a_{n-1}} \rho_n(x) dx = 1$$

and $\rho_n(x) = 0, \forall x \notin (a_n, a_{n-1})$, and $x \mapsto \rho_n(x)$ is continuous.

$$\implies \int_0^\infty \rho_n(x) dx = 1.$$

Let $\psi'_n(x) = \int_0^x \rho_n(y) dy, \forall x \geq 0$.

Let $\psi_n(x) = \int_0^x \psi'_n(y) dy$ for $x \geq 0$ and

$$\psi_n(x) = \psi_n(-x), \quad \forall x < 0.$$

One can check that $\psi_n(x) \uparrow |x|, \forall x \in \mathbb{R}$.

Summary:

(i) $\psi_n(x) \rightarrow |x|, \forall x \in \mathbb{R}$,

(ii) $|\psi'_n(x)| \leq 1, \forall x \in \mathbb{R},$

(iii) $|\psi''_n(x)| = |\rho_n(x)| \leq \frac{2}{n}h(x)^{-2}.$

Suppose $X^{(1)}$ & $X^{(2)}$ are two strong solutions with $X_0^{(1)} = X_0^{(2)}$ a.s.

Define

$$\begin{aligned} \tau_N &= \inf \left\{ t \geq 0: \int_0^t \sigma(s, X_s^{(1)})^2 ds \geq N \text{ or } \int_0^t \sigma(s, X_s^{(2)})^2 ds \geq N \right\}. \\ &\implies \mathbb{E} \int_0^{t \wedge \tau_N} \sigma(s, X_s^{(i)})^2 ds \leq N \end{aligned}$$

for $i = 1, 2.$

We may assume that

$$\mathbb{E} \int_0^t \sigma(s, X_s^{(i)})^2 ds < \infty,$$

for $\forall t \geq 0, i = 1, 2,$ due to $\tau_N \uparrow \infty$ a.s.

Otherwise we may prove

$$X_s^{(1)} = X_s^{(2)}, \quad \forall 0 \leq s \leq \tau_N \text{ a.s.} \implies X_s^{(1)} = X_s^{(2)}, \quad \forall s \geq 0.$$

Define $\Delta_t = X_t^{(1)} - X_t^{(2)}.$

$$\begin{aligned} \implies d\Delta_t &= dX_t^{(1)} - dX_t^{(2)} \\ &= [\sigma(t, X_t^{(1)}) - \sigma(t, X_t^{(2)})] dW_t + [b(t, X_t^{(1)}) - b(t, X_t^{(2)})] dt. \end{aligned}$$

Apply Itô's formula to get

$$\begin{aligned} \psi_n(\Delta_t) &= \psi_n(\Delta_0) + \int_0^t \psi'_n(\Delta_s) d\Delta_s + \frac{1}{2} \int_0^t \psi''_n(\Delta_s) d[\Delta]_s \\ &= \psi_n(0) + \int_0^t \psi'_n(\Delta_s) [\sigma - \sigma] dW_s + \int_0^t \psi'_n(\Delta_s) [b - b] ds + \frac{1}{2} \int_0^t \psi''_n(\Delta_s) [\sigma - \sigma]^2 ds. \end{aligned}$$

Take expectation to see that

$$\begin{aligned} \mathbb{E} \psi_n(\Delta_t) &= \mathbb{E} \int_0^t \psi'_n(\Delta_s) [b(t, X_t^{(1)}) - b(t, X_t^{(2)})] ds + \mathbb{E} \int_0^t \frac{1}{2} \psi''_n(\Delta_s) [\sigma - \sigma]^2 ds \\ &\leq \mathbb{E} \int_0^t K |X_s^{(1)} - X_s^{(2)}| ds + \mathbb{E} \int_0^t \frac{1}{2} \frac{2}{n} h(|\Delta_s|)^{-2} \cdot h(|X_s^{(1)} - X_s^{(2)}|)^2 ds \\ &= K \int_0^t \mathbb{E} |X_s^{(1)} - X_s^{(2)}| ds + \frac{1}{n} t. \end{aligned}$$

Let $n \rightarrow \infty$ to get

$$\mathbb{E} |\Delta_t| \leq K \int_0^t \mathbb{E} |\Delta_s| ds.$$

By Gronwall's Inequality,

$$\mathbb{E} |\Delta_t| = 0 \implies X_t^{(1)} = X_t^{(2)} \text{ a.s..}$$

By continuity,

$$X_t^{(1)} = X_t^{(2)}, \quad \forall t \geq 0 \text{ a.s..}$$

□

Proposition 1.9.2. (R & W)

If $Y = X - X'$, and $h: [0, \infty) \rightarrow [0, \infty)$ is increasing s.t.

$$\int_0^\varepsilon h(x)^{-2} dx = \infty, \quad \forall \varepsilon > 0,$$

then $l_t^0(Y) = 0$ if

$$\int_0^t h(Y_s)^{-2} 1_{\{Y_s > 0\}} d[Y]_s < \infty$$

a.s. $\forall t > 0$.

Here $l_t^0(Y)$ is the local time of Y at 0

(Recall $|Y_t| = |Y_0| + \int_0^t \text{sgn}(Y_s) dY_s + l_t^0(Y)$)

Proof.

$$l_t^a \text{ " = " } \int_0^t \delta_a(Y_s) d[Y]_s.$$

Then $\forall$ function $f: \mathbb{R} \rightarrow \mathbb{R}$,

$$\begin{aligned} \int_0^t f(Y_s) d[Y]_s &= \int_0^t d[Y]_s \int_{-\infty}^{\infty} f(a) \delta_a(Y_s) da \\ &= \int_{-\infty}^{\infty} f(a) da \int_0^t \delta_a(Y_s) d[Y]_s \\ &= \int_{-\infty}^{\infty} f(a) l_t^a(Y) da. \end{aligned}$$

So

$$\int_0^t h(Y_s)^{-2} 1_{\{Y_s > 0\}} d[Y]_s = \int_{-\infty}^{\infty} h(a)^{-2} 1_{\{a > 0\}} l_t^a da = \int_0^{\infty} h(a)^{-2} l_t^a da.$$

If $l_t^a(Y) > 0$, then by the right-continuity of $a \mapsto l_t^a(Y)$, we get $l_t^a(Y) \geq \varepsilon_0$ for some $\varepsilon_0 > 0$ and $0 < a < \varepsilon_0$.

$$\implies \int_0^{\infty} h(a)^{-2} l_t^a da \geq \int_0^{\varepsilon_0} h(a)^{-2} \varepsilon_0 da = \infty.$$

Contradiction.

Hence we conclude that $l_t^0(Y) = 0, \forall t \geq 0$, a.s.

□

Assumption: $\begin{cases} |b(t, x) - b(t, y)| \leq K |x - y| \\ |\sigma(t, x) - \sigma(t, y)| \leq h(|x - y|) \end{cases}$ and $\int_0^\varepsilon h(x)^{-2} dx = \infty, \forall \varepsilon > 0$.

Proof. Second proof of (Y-W 1971) (Le Gall)

Since (Recall $dY_s = dX_s^{(1)} - dX_s^{(2)}$)

$$\begin{aligned} \int_0^t h(Y_s)^{-2} \mathbf{1}_{\{Y_s > 0\}} d[Y]_s &= \int_0^t h(Y_s)^{-2} \mathbf{1}_{\{Y_s > 0\}} (\sigma - \sigma)^2 ds \\ &\leq \int_0^t h(Y_s)^{-2} \mathbf{1}_{\{Y_s > 0\}} h(|X_s^{(1)} - X_s^{(2)}|)^2 ds \leq t < \infty \text{ a.s..} \end{aligned}$$

$$\implies l_t^0(Y) = 0 \text{ a.s.}$$

Thus

$$\begin{aligned} |Y_t| &= |Y_0| + \int_0^t \text{sgn}(Y_s) dY_s + l_t^0(Y). \\ \implies |X_t^{(1)} - X_t^{(2)}| &= \int_0^t \text{sgn}(X_s^{(1)} - X_s^{(2)}) (b - b) ds + \int_0^t \text{sgn}(X_s^{(1)} - X_s^{(2)}) (\sigma - \sigma) dW_s. \\ \implies \mathbb{E} |X_t^{(1)} - X_t^{(2)}| &\leq \int_0^t \mathbb{E} K |X_s^{(1)} - X_s^{(2)}| ds. \end{aligned}$$

By Gronwall's, the proof is done. $\square$

Theorem 1.9.4 (Comparison Principle). Given $(\Omega, \mathcal{F}, \mathbb{P})$ and a B.M. $\{W_t, \mathcal{F}_t, t \geq 0\}$. If two continuous adapted process $X^{(1)}, X^{(2)}$ satisfy that

$$\begin{cases} X_t^{(1)} = X_0^{(1)} + \int_0^t b_1(s, X_s^{(1)}) ds + \int_0^t \sigma(s, X_s^{(1)}) dW_s, \\ X_t^{(2)} = X_0^{(2)} + \int_0^t b_2(s, X_s^{(2)}) ds + \int_0^t \sigma(s, X_s^{(2)}) dW_s. \end{cases}$$

Assume

(1) $\sigma(t, x), b_j(t, x)$ are continuous, real-values functions on $[0, \infty) \times \mathbb{R}$.

(2) $|\sigma(t, x) - \sigma(t, y)| \leq h(|x - y|)$ with

$$\int_0^\varepsilon h(x)^{-2} dx = \infty.$$

(3) $X_0^{(1)} \leq X_0^{(2)}$.

(4) $b_1(t, x) \leq b_2(t, x)$.

(5) either $b_1(t, x)$ or $b_2(t, x)$ satisfy $|b(t, x) - b(t, y)| \leq K |x - y|$.

Then $\mathbb{P}(X_t^{(1)} \leq X_t^{(2)}, \forall t \geq 0) = 1$.

Proof. Assume $|b_1(t, x) - b_1(t, y)| \leq K|x - y|$. Again we assume

$$\mathbb{E} \int_0^t \sigma(s, X_s^{(j)})^2 ds < \infty, \quad \forall t \geq 0.$$

Let $\psi_n(x)$ be as before. Define

$$\varphi_n(x) = \psi_n(x) \mathbf{1}_{(0, \infty)}(x).$$

$$\mathbb{E}[\varphi_n(\Delta_t)] \leq \mathbb{E} \int_0^t \varphi'_n(\Delta_s) [b_1(s, X_s^{(1)}) - b_2(s, X_s^{(2)})] ds + \frac{t}{n}$$

where $\Delta_t = X_t^{(1)} - X_t^{(2)}$.

So

$$\begin{aligned} \mathbb{E}\varphi_n(\Delta_t) - \frac{t}{n} &\leq \mathbb{E} \int_0^t \varphi'_n(\Delta_s) [b_1(s, X_s^{(1)}) - b_2(s, X_s^{(2)})] ds \\ &\leq \mathbb{E} \int_0^t K |X_s^{(1)} - X_s^{(2)}| \varphi'_n(\Delta_s) ds \\ &= \mathbb{E} \int_0^t K \Delta_s^+ \varphi'_n(\Delta_s^+) ds \\ &\leq \mathbb{E} \int_0^t K \Delta_s^+ ds, \end{aligned}$$

since $\varphi'_n(\Delta_s) = 0$ if $\Delta_s \leq 0$. $\varphi'_n(\Delta_s) > 0$ if $\Delta_s > 0$.

By $\psi_n(x) \rightarrow |x|$, $\varphi_n(x) = \psi_n(x) \mathbf{1}_{(0, \infty)}(x) \rightarrow x^+$.

Let $n \rightarrow \infty$ to get

$$\mathbb{E}\Delta_t^+ \leq K \int_0^t \mathbb{E}\Delta_s^+ ds.$$

By Gronwall, $\mathbb{E}\Delta_t^+ = 0 \implies \mathbb{P}(X_t^{(1)} \leq X_t^{(2)}) = 1, \forall t > 0$. □

Definition 1.9.3 (Weak Solution). A weak solution of SDE is a triple $(X, W), (\Omega, \mathcal{F}, \mathbb{P}), \{\mathcal{F}_t\}$ where

- (1) $(\Omega, \mathcal{F}, \mathbb{P})$ probability space, $\{\mathcal{F}_t\}$ is filtration.
- (2) $(X_t, \mathcal{F}_t)$ is continuous and adapted $(W_t, \mathcal{F}_t)$ is an r-dim B.M.
- (3) $\mathbb{P}\left(\int_0^t b_j(s, X_s) + \sigma_{ij}(s, X_s)^2 ds < \infty, \quad \forall t \geq 0\right) = 1$.
- (4) $X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s$.

Definition 1.9.4 (Pathwise Uniqueness). Suppose whenever $(X, W), (\Omega, \mathcal{F}, \mathbb{P}), \{\mathcal{F}_t\}$ and $(\tilde{X}, W), (\Omega, \mathcal{F}, \mathbb{P}), \{\tilde{\mathcal{F}}_t\}$ are two weak solutions with $X_0 = \tilde{X}_0$ a.s., if

$$\mathbb{P}(X_t = \tilde{X}_t, \quad \forall t \geq 0) = 1,$$

then pathwise uniqueness holds.

Definition 1.9.5 (Weak Uniqueness (Uniqueness in law)). If for any two weak solutions (X, W) , $(\Omega, \mathcal{F}, \mathbb{P})$, $\{\mathcal{F}_t\}$ and $(\tilde{X}, \tilde{W})$, $(\tilde{\Omega}, \tilde{\mathcal{F}}_t, \tilde{\mathbb{P}})$, $\{\tilde{\mathcal{F}}_t\}$ with same initial distribution $\forall B \in \mathcal{B}(\mathbb{R}^d)$,

$$\mathbb{P}(X_0 \in B) = \tilde{\mathbb{P}}(\tilde{X}_0 \in B).$$

We get X and $\tilde{X}$ have the same law, i.e. $\forall t_0 \leq t_1 \leq \dots \leq t_n$,

$$\mathbb{P}(X_{t_0} \in B_0, \dots, X_{t_n} \in B_n) = \tilde{\mathbb{P}}(\tilde{X}_{t_0} \in B_0, \dots, \tilde{X}_{t_n} \in B_n)$$

for all $B_i \in \mathcal{B}(\mathbb{R}^d)$, then weak uniqueness holds.

Remark. (Tanaka SDE)

$$X_t = \int_0^t \text{sgn}(X_s) dW_s$$

$$\text{with } \text{sgn}(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0 \end{cases}.$$

Weak Uniqueness: If (X, W) , $(\Omega, \mathcal{F}, \mathbb{P})$, $\{\mathcal{F}_t\}$ is a weak solution, then the process (X_t) is a continuous, square-integrable martingale with

$$[X]_t = \int_0^t \text{sgn}(X_s)^2 ds = t.$$

So X is a B.M. $\implies$ Weak Uniqueness holds.

However, $(-X, W)$, $(\Omega, \mathcal{F}, \mathbb{P})$, $\{\mathcal{F}_t\}$ is also a weak solution.

Because

$$X_t = \int_0^t \text{sgn}(X_s) dW_s \implies -X_t = \int_0^t -\text{sgn}(X_s) dW_s = \int_0^t \text{sgn}(-X_s) dW_s.$$

$$\int_0^t (-\text{sgn}(X_s) - \text{sgn}(-X_s)) dW_s \sim \int_0^t (-\text{sgn}(X_s) - \text{sgn}(-X_s))^2 ds.$$

Notice that $-\text{sgn}(x) - \text{sgn}(-x) = 0$ for all $x \neq 0$ and

$$\mathbb{P}(X_s \neq 0, \text{ for a.e. } s \geq 0) = 1.$$

W.P.1

$$\int_0^t (-\text{sgn}(X_s) - \text{sgn}(-X_s))^2 ds = 0 \implies \int_0^t -\text{sgn}(X_s) dW_s = \int_0^t \text{sgn}(-X_s) dW_s \text{ a.s.}$$

$$\implies -X_t = \int_0^t \text{sgn}(-X_s) dW_s.$$

$$\implies (-X, W) \text{ is also a weak solution.}$$

$$\implies \mathbb{P}(X_t \neq -X_t, \text{ for a.e. } t \geq 0) = 1.$$

Hence pathwise uniqueness fails.

Weak Existence: $(X, W), (\Omega, \mathcal{F}, \mathbf{P}), \{\mathcal{F}_t\}$.

Construction: Given $(\Omega, \mathcal{F}, \mathbf{P})$ and a B.M. $X = \{X_t, \mathcal{F}_t^X, t \geq 0\}$. Let $\mathring{\mathcal{F}}_t^X$ = augmentation of $\mathcal{F}_t^X$.

Assume $\mathbf{P}(X_0 = 0) = 1$.

Define $W_t \triangleq \int_0^t \text{sgn}(X_s) dX_s$.

Then W_t is a B.M. with respect to $\mathring{\mathcal{F}}_t^X$.

Also

$$\int_0^t \text{sgn}(X_s) dW_s = \int_0^t \text{sgn}(X_s) \text{sgn}(X_s) dX_s = \int_0^t 1 dX_s = X_t.$$

Hence $(X, W), (\Omega, \mathcal{F}, \mathbf{P}), \mathring{\mathcal{F}}_t^X$ is a weak solution.

No strong solution.

Given $(\Omega, \mathcal{F}, \mathbf{P})$ and a B.M. W , if X is a strong solution, then

$$X_t \in \mathcal{F}_t^W \subseteq \mathring{\mathcal{F}}_t^W \implies \mathcal{F}_t^X \subseteq \mathring{\mathcal{F}}_t^W.$$

By X is a B.M., by Tanaka's formula,

$$|X_t| = |X_0| + \int_0^t \text{sgn}(X_s) dX_s + L_t^X(0).$$

By SDE, we get

$$\int_0^t \text{sgn}(X_s) dX_s = \int_0^t \text{sgn}(X_s)^2 dW_s = W_t.$$

$$\implies W_t = |X_t| - L_t^X(0).$$

$$L_t^X(0) = \int_0^t \delta_0(X_s) ds = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \int_0^t 1_{|X_s| \leq \varepsilon} ds = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} |\{0 \leq s \leq t: |X_s| \leq \varepsilon\}|.$$

Hence $W_t = |X_t| - \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} |\{0 \leq s \leq t: |X_s| \leq \varepsilon\}| \in \mathcal{F}_t^{|X|}$.

$$\implies \mathring{\mathcal{F}}_t^W \subseteq \mathring{\mathcal{F}}_t^{|X|}.$$

$$\implies \mathring{\mathcal{F}}_t^X \subseteq \mathring{\mathcal{F}}_t^{|X|}. \text{ Contradiction.}$$

$$\begin{cases} \text{strong uniqueness} \implies \text{weak uniqueness} \\ \text{weak existence} + \text{pathwise uniqueness} \implies \text{strong existence.} \end{cases}$$

1.9.1 Girsanov Theorem

Given $(\Omega, \mathcal{F}, \mathbf{P})$ and a d-dim B.M. $W = \{W_t, \mathcal{F}_t, t \geq 0\}$ with $\mathbf{P}(W_0 = 0) = 1$. Let $X = \{X_t, \mathcal{F}_t, t \geq 0\}$ be a d-dim measurable, adapted process such that $\forall T \geq 0, \forall 1 \leq i \leq d, \mathbf{P}\left(\int_0^T (X_t^{(i)})^2 dt < \infty\right) = 1$.

Then

$$\int_0^t X_s^{(i)} dW_s^{(i)}$$

is well-defined. Set

$$Z_t(X) = \exp \left(\int_0^t X_s dW_s - \frac{1}{2} \int_0^t \|X_s\|^2 ds \right).$$

Assume $Z(X)$ is a martingale i.e. $\mathbb{E}Z_t(X) = 1, \forall t \geq 0$.

Define a probability measure $\tilde{\mathbb{P}}_T$ for any $T > 0$ by

$$\tilde{\mathbb{P}}_T(A) = \mathbb{E}(Z_T(X) \cdot 1_A)$$

for all $A \in \mathcal{F}_T$, $\frac{d\tilde{\mathbb{P}}_T}{d\mathbb{P}} = Z_T(X)$.

Remark. If $A \in \mathcal{F}_t \subseteq \mathcal{F}_T$ for $0 \leq t \leq T$, we must

$$\tilde{\mathbb{P}}_T(A) = \mathbb{E}(1_A Z_T(X)),$$

$$\tilde{\mathbb{P}}_t(A) = \mathbb{E}(1_A Z_t(X)).$$

Note that

$$\mathbb{E}(1_A Z_T(X)) = \mathbb{E}[\mathbb{E}(1_A Z_T(X) \mid \mathcal{F}_t)] = \mathbb{E}[1_A \mathbb{E}(Z_T(X) \mid \mathcal{F}_t)] = \mathbb{E}[1_A Z_t(X)].$$

Hence $\tilde{\mathbb{P}}_T|_{\mathcal{F}_t} = \tilde{\mathbb{P}}_t$.

Theorem 1.9.5 (Girsanov Theorem). Assume $Z(X)$ is a martingale. Define $\tilde{W} = \{\tilde{W}_t, t \geq 0\}$ by $\tilde{W}_t^{(i)} = W_t^{(i)} - \int_0^t X_s^{(i)} ds$ for $1 \leq i \leq d$, and $t \geq 0$. Then for each fixed $T > 0$, $\{\tilde{W}_t, \mathcal{F}_t, 0 \leq t \leq T\}$ is a d-dim B.M. on $(\Omega, \mathcal{F}_T, \tilde{\mathbb{P}}_T)$.

Lemma 1.9.2. Fix $T > 0$. If $0 \leq s \leq t \leq T$ and $Y \in \mathcal{F}_t$ with $\tilde{E}_T|Y| < \infty$. Then

$$\tilde{E}_T(Y \mid \mathcal{F}_s) = \frac{1}{Z_s} \mathbb{E}(Y Z_t \mid \mathcal{F}_s)$$

a.s. $\tilde{\mathbb{P}}_T$ and $\mathbb{P}$.

$\tilde{\mathbb{P}}_T \ll \mathbb{P}$ and $\mathbb{P} \ll \tilde{\mathbb{P}}_T$. $\tilde{\mathbb{P}}_T(A) = \mathbb{E}(1_A Z_T(X))$.

If $\mathbb{P}(A) = 0 \implies \tilde{\mathbb{P}}_T(A) = 0$.

If $\tilde{\mathbb{P}}_T(A) = 0 \implies \mathbb{P}(A) = 0$. ($Z_T(X) > 0$ a.s.).

Proof. It suffices to prove $\forall A \in \mathcal{F}_s$,

$$\tilde{E}_T(1_A Y) = \tilde{E}_T(1_A \frac{1}{Z_s} \mathbb{E}(Y Z_t \mid \mathcal{F}_s)).$$

$$\begin{aligned}
 RHS &= \tilde{E}_s(1_A \frac{1}{Z_s} \mathbb{E}(Y Z_t \mid \mathcal{F}_s)) \\
 &= \mathbb{E}(1_A \mathbb{E}(Y Z_t \mid \mathcal{F}_s)) \\
 &= \mathbb{E}(1_A Y Z_t) \\
 &= \tilde{E}_t(1_A Y) \\
 &= \tilde{E}_T(1_A Y).
 \end{aligned}$$

□

Proposition 1.9.3. Define $M_T^{c,loc} = \{M : M \text{ is a continuous local martingale on } (\Omega, \mathcal{F}_T, \mathbb{P})\}$.

$\tilde{M}_T^{c,loc} = \{\text{continuous local martingale on } (\Omega, \mathcal{F}_T, \tilde{\mathbb{P}}_T)\}$.

(1) If $M \in M_T^{c,loc}$, then

$$\tilde{M}_t \triangleq M_t - \sum_{i=1}^d \int_0^t X_s^{(i)} d[M, W^{(i)}]_s$$

for $0 \leq t \leq T$ is in $\tilde{M}_T^{c,loc}$.

(2) If $N \in M_T^{c,loc}$ and

$$\tilde{N}_t = N_t - \sum_{i=1}^d \int_0^t X_s^{(i)} d[N, W^{(i)}]_s,$$

then

$$[\tilde{M}, \tilde{N}]_t = [M, N]_t$$

for all $0 \leq t \leq T$, a.s. $\mathbb{P}$ and $\tilde{\mathbb{P}}_T$.

Proof. of Girsanov Theorem:

If $M = W^{(j)}$ with $1 \leq j \leq d$, then

$$\tilde{M}_t = W_t^{(j)} - \int_0^t X_s^{(j)} dW_s^{(j)} \in \tilde{M}_T^{c,loc}$$

$$\implies \tilde{W}_t^{(j)} \in \tilde{M}_T^{c,loc}.$$

Set $N = W^{(k)} \implies \tilde{N}_t = \tilde{W}_t^{(k)} \in \tilde{M}_T^{c,loc}$ and $[\tilde{M}, \tilde{N}]_t = [M, N]_t = \delta_{jk}t$.

$$\implies [\tilde{W}^{(j)}, \tilde{W}^{(k)}]_t = \delta_{jk}t$$

$$\implies \tilde{W} \text{ is a d-dim B.M..}$$

□

Proof. of Prop. Assume M, N are bounded martingale with bounded quadratic variation.

$$[\tau_n^M = \inf \{t \geq 0 : |M_t| \geq n \text{ or } [M]_t \geq n\}]$$

Consider $t \leq \tau_n^M \wedge \tau_n^N$.

Assume also that $Z_t(X)$ and $\sum_{i=1}^d \int_0^t (X_s^{(j)})^2 ds$ are bounded for all $t \geq 0$ a.s.

By Kunita & Watanabe inequality

$$\begin{aligned} \left| \int_0^t X_s^{(j)} d[M, W^{(j)}]_s \right|^2 &\leq \int_0^t (X_s^{(j)})^2 d[W^{(j)}]_s \cdot \int_0^t 1^2 d[M]_s \\ &= \int_0^t (X_s^{(j)})^2 ds \cdot [M]_t \\ &\leq C \quad \text{a.s.} \end{aligned}$$

Hence $\tilde{M}_t$ is also bounded for all $t \geq 0$. To prove $\tilde{M} \in \tilde{M}_T^{c,loc}$, we only need to show that

$$\tilde{E}_T(\tilde{M}_t \mid \mathcal{F}_s) = \tilde{M}_s \quad \text{a.s.}$$

$$LHS = (\text{by Lemma}) = \frac{1}{Z_s(X)} \mathbb{E}(\tilde{M}_t Z_t(X) \mid \mathcal{F}_s).$$

By IBP, we get

$$\tilde{M}_t Z_t(X) = \tilde{M}_0 Z_0(X) + \int_0^t \tilde{M}_s dZ_s(X) + \int_0^t Z_s(X) d\tilde{M}_s + [\tilde{M}, Z]_t.$$

Recall $Z_t(X) = e^{\int_0^t X_s dW_s - \frac{1}{2} \int_0^t \|X_s\|^2 ds}$,

we have

$$dZ_t(X) = Z_t(X) X_t \cdot dW_t = \sum_{i=1}^d Z_t(X) X_t^{(i)} dW_t^{(i)}.$$

Recall $d\tilde{M}_t = dM_t - \sum_{i=1}^d X_t^{(i)} d[M, W^{(i)}]_t$.

So

$$\begin{aligned} &Z_t(X) \tilde{M}_t \\ &= \int_0^t \tilde{M}_s \sum_{i=1}^d Z_s(X) X_s^{(i)} dW_s^{(i)} \\ &\quad + \int_0^t Z_s(X) \left[dM_s - \sum_{i=1}^d X_s^{(i)} d[M, W^{(i)}]_s \right] + \int_0^t Z_s(X) X_s^{(i)} d[M, W^{(i)}]_s \\ &= \int_0^t \tilde{M}_s \sum_{i=1}^d Z_s(X) X_s^{(i)} dW_s^{(i)} + \int_0^t Z_s(X) dM_s. \end{aligned}$$

Hence $\tilde{M}_t Z_t(X)$ is a martingale.

$$\implies LHS = \frac{1}{Z_s(X)} \tilde{M}_s Z_s(X) = RHS.$$

$$(2) \quad d\tilde{N}_t = dN_t = \sum_{i=1}^d X_t^{(i)} d[N, W^{(i)}]_t.$$

$$d\tilde{M}_t = dM_t - \sum_{i=1}^d X_t^{(i)} d[M, W^{(i)}]_t.$$

$$\begin{aligned} \tilde{M}_t \tilde{N}_t &= \tilde{M}_0 \tilde{N}_0 + \int_0^t \tilde{M}_s d\tilde{N}_s + \int_0^t \tilde{N}_s d\tilde{M}_s + [\tilde{M}, \tilde{N}]_t \\ &= 0 + \int_0^t \tilde{M}_s \left(dW_s - \sum_{i=1}^d X_s^{(i)} d[N, W^{(i)}]_s \right) \\ &\quad + \int_0^t \tilde{N}_s \left(dM_s - \sum_{i=1}^d X_s^{(i)} d[M, W^{(i)}]_s \right) + [M, N]_t. \end{aligned}$$

Then

$$\begin{aligned} \tilde{M}_t \tilde{N}_t - [M, N]_t &= \int_0^t \tilde{M}_s dN_s + \int_0^t \tilde{N}_s dM_s - \sum_{i=1}^d \int_0^t X_s^{(i)} \tilde{M}_s d[N, W^{(i)}]_s \\ &\quad - \sum_{i=1}^d \int_0^t X_s^{(i)} \tilde{N}_s d[M, W^{(i)}]_s. \end{aligned}$$

Next

$$\begin{aligned} &Z_t(X) [\tilde{M}_t - \tilde{N}_t - [M, N]_t] \\ &= \int_0^t Z_s(X) \left[\tilde{M}_s dN_s + \tilde{N}_s dM_s - \sum_{i=1}^d X_s^{(i)} \left(\tilde{M}_s d[N, W^{(i)}]_s + \tilde{N}_s d[M, W^{(i)}]_s \right) \right] \\ &\quad + \int_0^t (\tilde{M}_s \tilde{N}_s - [M, N]_s) \sum_{i=1}^d Z_s(X) X_s^{(i)} dW_s^{(i)} \\ &\quad + \int_0^t \sum_{i=1}^d Z_s(X) X_s^{(i)} \left(\tilde{M}_s d[N, W^{(i)}]_s + \tilde{N}_s d[M, W^{(i)}]_s \right). \end{aligned}$$

Therefore

$$\tilde{E}_T(\tilde{M}_t \tilde{N}_t - [M, N]_t \mid \mathcal{F}_s) = \frac{1}{Z_s(X)} \mathbb{E}(Z_t(X)(\tilde{M}_t \tilde{N}_t - [M, N]_t) \mid \mathcal{F}_s) = (\tilde{M}_s \tilde{N}_s - [M, N]_s).$$

This proves $[\tilde{M}, \tilde{N}]_t = [M, N]_t$. □

Weak solutions constructed by means of the Girsanov Theorem.

Proposition 1.9.4. Consider the SDE

$$dX_t = dW_t + b(t, X_t) dt$$

for $0 \leq t \leq T$, W is B.M. and $b(t, x)$ is Borel-measurable with

$$\|b(t, x)\| \leq K(1 + \|x\|)$$

for all $0 \leq t \leq T$, $x \in \mathbb{R}^d$.

Then for any probability measure μ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ the SDE has a weak solution (X, W) , $(\Omega, \mathcal{F}, \mathbb{P})$, $\{\mathcal{F}_t\}$ such that

$$\mathbb{P}(X_0 \in \cdot) = \mu.$$

Proof. Let $X = \{X_t, \mathcal{F}_t, t \geq 0\}$ be a B.M. on $(\Omega, \mathcal{F})$ with law $(\mathbb{P}^X)_{X \in \mathbb{R}^d}$ s.t.

$$\mathbb{P}^X(X_0 = x) = 1.$$

By corollary 3.5.16, by assuming

$$\|b(t, x)\| \leq K(1 + \|x\|)$$

we get

$$Z_t \triangleq \exp \left(\sum_{i=1}^d \int_0^t b_j(s, X_s) dX_s^{(j)} - \frac{1}{2} \int_0^t \|b(s, X_s)\|^2 ds \right)$$

is a martingale.

By Girsanov Theorem, if we define

$$Q^X(A) = \mathbb{P}^X(A \cdot Z_T)$$

for all $A \in \mathcal{F}_T$, then

$$W_t \triangleq X_t - X_0 - \int_0^t b(s, X_s) ds$$

is a B.M. with $Q^X(W_0 = 0) = 1$.

Then $X_t = X_0 + W_t + \int_0^t b(s, X_s) ds$ for all $0 \leq s \leq T$.

Note that $\mathbb{P}^X(X_0 = x) = 1$.

Then $Q^X(X_0 = x) = 1$.

Define $Q^\mu(A) = \int_{\mathbb{R}^d} Q^X(A) \mu(dx)$.

Then $Q^\mu(W_0 = 0) = 1$ and

$$\begin{aligned} Q^\mu(X_0 \in A) &= \int_{\mathbb{R}^d} Q^X(X_0 \in A) \mu(dx) \\ &= \int_{\mathbb{R}^d} 1_{x \in A} \mu(dx) = \mu(A). \end{aligned}$$

So under Q^μ we get $X_0 \sim \mu$ and (X, W) , $(\Omega, \mathcal{F}, Q^\mu)$, $\{\mathcal{F}_t\}$.

□

(1) pathwise uniqueness $\implies$ weak uniqueness.

Proof. If

$$\begin{cases} (X^{(1)}, W^{(1)}) \\ (\Omega_1, \mathcal{F}_1, v_1) \\ (\mathcal{F}_t^{(1)})_{t \geq 0}, \end{cases} \quad \begin{cases} (X^{(2)}, W^{(2)}) \\ (\Omega_2, \mathcal{F}_2, v_2) \\ (\mathcal{F}_t^{(2)})_{t \geq 0}, \end{cases}$$

are two weak solutions.

If $v_1(X_0^{(1)} \in \cdot) = v_2(X_0^{(2)} \in \cdot)$, then we will show that

$$v_1(X^{(1)} \in \cdot) = v_2(X^{(2)} \in \cdot).$$

In fact, we will prove that if

$$Y^{(j)} = X^{(j)} - X_0^{(j)},$$

then

$$v_1((X_0^{(1)}, W^{(1)}, Y^{(1)}) \in A) = v_2((X_0^{(2)}, W^{(2)}, Y^{(2)}) \in A)$$

for all $A \in \mathcal{B}(\Theta)$ with $\Theta = \mathbb{R}^d \times C[0, \infty)^r \times C[0, \infty)^d$ and $\mathcal{B}(\Theta) = \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{B}(C[0, \infty)^r) \otimes \mathcal{B}(C[0, \infty)^d)$.

Let $\mu(\cdot) = v_1(X_0^{(1)} \in \cdot) = v_2(X_0^{(2)} \in \cdot)$.

Let P_* denote the wiener measure on $C[0, \infty)^r$.

Define $(\Omega, \mathcal{F})$ by $\Omega = \Theta \times C[0, \infty)^d$, $\mathcal{F} = \mathcal{B}(\Theta) \otimes \mathcal{B}(C[0, \infty)^d)$.

For each $\omega \in \Omega$, we write $\omega = (x, w, y_1, y_2)$.

Define

$$P(d\omega) = \mu(dx)P_*(dw)Q_1(x, w, dy_1)Q_2(x, w, dy_2),$$

where $Q_j(x, w, y_i)$ is the regular conditional probability s.t.

- (1) $\forall x \in \mathbb{R}^d, w \in C[0, \infty)^r, Q_j(x, w, \cdot)$ is a probability measure on $(C[0, \infty)^d, \mathcal{B}(C[0, \infty)^d))$.
- (2) $\forall F \in \mathcal{B}(C[0, \infty)^d)$, the map $(x, w) \rightarrow Q_j(x, w; F)$ is measurable w.r.t $\mathcal{B}(\mathbb{R}^d) \otimes \mathcal{B}(C[0, \infty)^r)$.
- (3) $\forall G \in \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{B}(C[0, \infty)^r), \forall F \in \mathcal{B}(C[0, \infty)^d)$,

$$v_j(X_0^{(j)}, W^{(j)} \in G, Y^{(j)} \in F) = \int_G \mu(dx)P_*(dw)Q_j(x, w; F)$$

i.e. $Q_j(x, w; F) = v_j(Y^{(j)} \in F \mid X_0^{(j)} = x, W^{(j)} = w)$.

Let $G_t \triangleq \sigma((x, w(s), y_1(s), y_2(s)): x \in \mathbb{R}^d, 0 \leq s \leq t)$.

Let $F_t \triangleq$ augmentation of σ_t .

Now we conclude that

$$\begin{aligned} P(\{\omega \in \Omega: (x, w, y_1) \in A\}) &= \int P(d\omega)1_{(x, w, y_1) \in A} \\ &= \int_A \mu(dx)P_*(dw)Q_1(x, w; dy_1) \\ &= v_1((X_0^{(1)}, W^{(1)}, Y^{(1)}) \in A). \end{aligned}$$

So

$$(x + y_1, w) \stackrel{d}{=} (X^{(1)}, X^{(1)})$$

and

$$(x + y_2, w) \stackrel{d}{=} (X^{(2)}, W^{(2)}).$$

pathwise uniqueness implies

$$\mathbb{P}(x + y_1(t) = x + y_2(t), \forall t \geq 0) = 1.$$

$$\begin{aligned} v_1((X_0^{(1)}, W^{(1)}, Y^{(1)}) \in A) &= \mathbb{P}(\{\omega \in \Omega : (x, w, y_1) \in A\}) \\ &= \mathbb{P}(\{\omega \in \Omega : (x, w, y_2) \in A\}) \\ &= v_2((X_0^{(2)}, W^{(2)}, Y^{(2)}) \in A). \end{aligned}$$

□

1.10 Harmonic Function

$\forall$ smooth function $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\frac{\Delta}{2} f(x) = \lim_{t \downarrow 0} \frac{E^x[f(W_t) - f(x)]}{t}.$$

We call $\frac{\Delta}{2}$ the generator of Brownian motion.

$$\Delta u \triangleq \sum_{i=1}^d \frac{\partial^2 u}{\partial x_i^2}.$$

Proof.

$$f(W_t) = f(W_0) + \int_0^t \sum_{i=1}^d \frac{\partial f}{\partial x_i}(W_s) dW_s^{(i)} + \frac{1}{2} \int_0^t \sum_{i=1}^d \frac{\partial^2 f}{\partial x_i^2}(W_s) ds.$$

So

$$\frac{1}{t} E^x[f(W_t) - f(W_0)] = \frac{1}{2} \frac{1}{t} \int_0^t E[\Delta f(W_s)] ds.$$

Let $t \downarrow 0$, we get the limit is $\frac{1}{2} \Delta f(x)$. □

A function u defined on some open subset $D \subseteq \mathbb{R}^d$ is called harmonic if $u \in C^2(D)$ and $\Delta u = 0$. Let $\{W_t, \mathcal{F}_t, 0 \leq t < \infty\}$, $(\Omega, \mathcal{F})$ and $\{\mathbb{P}^x\}_{x \in \mathbb{R}^d}$ is a d-dim B.M. such that

$$\mathbb{P}^x(W_0 = x) = 1.$$

For any $D \subseteq \mathbb{R}^d$ open, we set $\tau_D = \inf\{t \geq 0 : W_t \notin D\}$.

If D is bounded, then

$$\mathbb{P}^x(\tau_D < \infty) = 1, \quad \forall x \in D,$$

and $B_{\tau_D} \in \partial D$ with ∂D = topological boundary, $\overline{D} = D \cup \partial D$ is the closure of D .

Let $B_r = \{x \in \mathbb{R}^d : \|x\| < r\}$, $B_r(x) = \{y \in \mathbb{R}^d : \|y - x\| < r\}$ be the open ball centered at $0/x$ with radius $r > 0$. Volume

$$V_r = \frac{2r^d \pi^{\frac{d}{2}}}{d\Gamma(\frac{d}{2})}.$$

Surface area

$$S_r \triangleq \frac{2r^{d-1} \pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})}.$$

Define $\mu_r(dx) = \mathbb{P}^0(W_{\tau_{B_r}} \in dx)$. Then μ_r is a measure on ∂B_r .

By rotational invariance μ_r is a uniform measure on ∂B_r . In particular, we have

$$\mu_r = \frac{\sigma_r}{S_r}$$

where $\sigma_r(dx)$ = surface measure on ∂B_r .

Recall that for any function f ,

$$\int_{B_r} f(x) dx = \int_0^t d\rho \int_{\partial B_\rho} f(x) \sigma_\rho(dx) = \int_0^r d\rho \int_{\partial B_\rho} f(x) S_\rho \mu_\rho(dx).$$

Definition 1.10.1 (Mean Value Property (MVP)). A function $u : D \rightarrow \mathbb{R}$ has MVP if $\forall a \in D$, $\forall r > 0$ with $B_r(a) \subseteq D$, we have

$$u(a) = \int_{\partial B_r} \mu(a+x) u_r(dx) = \int_{\partial B_r(a)} u(x) \mu_r(dx).$$

Assume MVP, we have

$$\int_{B_r(a)} u(x) dx = \int_0^t d\rho \int_{\partial B_\rho(a)} u(x) S_\rho \mu_\rho(dx) = \int_0^r d\rho \cdot u(a) S_\rho = V_r \cdot u(a).$$

Then

$$u(a) = \frac{1}{V(r)} \int_{B_r(a)} u(x) dx.$$

Proposition 1.10.1. If u is harmonic on D , then u has MVP. And if u has MVP on D , then u is harmonic & $u \in C^\infty$.

Proof. (1). Assume $\Delta u = 0$, $\forall x \in D$. Then $\forall r > 0$ with $\overline{B_r(a)} \subseteq D$, $\forall a \in D$.

By Itô's formula

$$u(W_{t \wedge \tau_{B_r(a)}}) = u(W_0) + \int_0^{t \wedge \tau_{B_r(a)}} \nabla u(W_s) dW_s + \frac{1}{2} \int_0^{t \wedge \tau_{B_r(a)}} \Delta u(W_s) ds, \quad \forall t \geq 0.$$

By $\nabla u(x)$ is bounded on $\overline{B_r(a)}$, we get

$$E \left[\int_0^{t \wedge \tau_{B_r(a)}} \nabla u(W_0) dW_s \right].$$

Hence

$$E^a \left[u(W_{t \wedge \tau_{B_r(a)}}) \right] = u(a).$$

Let $t \rightarrow \infty$ to get

$$\begin{aligned} u(a) &= \lim_{t \rightarrow \infty} E^a \left[u(W_{t \wedge \tau_{B_r(a)}}) \right] = E^a \left[u(W_{\tau_{B_r(a)}}) \right] \\ &= E^0 \left[u(W_{\tau_{B_r}} + a) \right] \\ &= \int_{\partial B_r} u(x + a) \mu_r(dx). \end{aligned}$$

(2) If u has MVP, then $\forall a \in D, \forall r > 0$ with $\overline{B_r(a)} \subseteq D$,

$$u(a) = \int_{\partial B_r(a)} u(x) \mu_r(dx).$$

For any $\varepsilon > 0$, define $g_\varepsilon : \mathbb{R}^d \rightarrow \mathbb{R}$ by

$$g_\varepsilon(x) = \begin{cases} c(\varepsilon) e^{-\frac{1}{\varepsilon^2 - \|x\|^2}}, & \|x\| < \varepsilon \\ 0, & \|x\| \geq \varepsilon. \end{cases}$$

where $c(\varepsilon) > 0$ is a constant s.t.

$$\int_{\mathbb{R}^d} g_\varepsilon(x) dx = \int_{B_\varepsilon} g_\varepsilon(x) dx = 1.$$

Mollifier. For any $a \in D$ and $\varepsilon > 0$ with $\overline{B_\varepsilon(a)} \subseteq D$, define

$$u_\varepsilon(a) \triangleq \int_{B_\varepsilon} u(a+x) g_\varepsilon(x) dx = \int_{\mathbb{R}^d} u(y) g_\varepsilon(y-a) dy.$$

Then $u_\varepsilon(a) \in C^\infty$.

$$\begin{aligned} \lim_{\delta \downarrow 0} \frac{u_\varepsilon(a+\delta) - u_\varepsilon(a)}{\delta} &= \lim_{\delta \downarrow 0} \int_{\mathbb{R}^d} u(y) \frac{g_\varepsilon(y-a-\delta) - g_\varepsilon(y-a)}{\delta} dy \\ &= - \int_{\mathbb{R}^d} u(y) \left(\frac{d}{dx} g_\varepsilon(x) \Big|_{x=y-a} \right) dy. \end{aligned}$$

And

$$\begin{aligned}
u_\varepsilon(a) &= c(\varepsilon) \int_{B_\varepsilon} u(a+x) e^{-\frac{1}{\varepsilon^2 - \|x\|^2}} dx \\
&= c(\varepsilon) \int_0^\varepsilon S_\rho d\rho \int_{\partial B_\rho} u(a+x) e^{-\frac{1}{\varepsilon^2 - \rho^2}} \mu_\rho(dx) \\
&= c(\varepsilon) \int_0^\varepsilon S_\rho d\rho \cdot e^{-\frac{1}{\varepsilon^2 - \rho^2}} u(a) \\
&= u(a) \int_0^\varepsilon S_\rho c(\varepsilon) e^{-\frac{1}{\varepsilon^2 - \rho^2}} d\rho \\
&= u(a) \int_{B_\varepsilon} c(\varepsilon) e^{-\frac{1}{\varepsilon^2 - \|x\|^2}} dx = u(0).
\end{aligned}$$

So $u(a) = u_\varepsilon(0)$ for all $a \in D$ with $\overline{B_\varepsilon(a)} \subseteq D$.

Then $u \in C^\infty(D)$.

Next, we will prove $\Delta u = 0$. $\forall a \in D, \exists \varepsilon > 0$ s.t. $\overline{B_\varepsilon(a)} \subseteq D$. By Taylor's expansion at a ,

$$u(a+y) = u(a) + \sum_{i=1}^d y_i \frac{\partial u}{\partial x_i}(a) + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d y_i y_j \frac{\partial^2 u}{\partial x_i \partial x_j}(a) + o(\|y\|^2), \quad y \in \overline{B_\varepsilon}.$$

By symmetry,

$$\int_{\partial B_\varepsilon} y_i \mu_\varepsilon(dy) = 0, \quad \int_{\partial B_\varepsilon} y_i y_j \mu_\varepsilon(dy) = 0, \quad \forall i \neq j.$$

Now integrate $u(a+y)$ over ∂B_ε to obtain

$$\int_{\partial B_\varepsilon} u(a+y) \mu_\varepsilon(dy) = u(a) + 0 + \int_{\partial B_\varepsilon} \frac{1}{2} \Delta u(a) \sum_{i=1}^d y_i^2 \mu_\varepsilon(dy) + o(\varepsilon^2).$$

For

$$\frac{1}{2} \sum_{i=1}^d \int_{\partial B_\varepsilon} \frac{\partial^2 u}{\partial x_i^2}(a) \cdot y_i^2 \mu_\varepsilon(dy).$$

Notice that

$$\int_{\partial B_\varepsilon} y_i^2 \mu_\varepsilon(dy) = \frac{1}{d} \sum_{i=1}^d \int_{\partial B_\varepsilon} y_i^2 \mu_\varepsilon(dy) = \frac{1}{d} \int_{\partial B_\varepsilon} \|y\|^2 \mu_\varepsilon(dy) = \frac{1}{d} \varepsilon^2 \cdot 1 = \frac{1}{d} \varepsilon^2.$$

Hence

$$\frac{1}{2} \sum_{i=1}^d \int_{\partial B_\varepsilon} \frac{\partial^2 u}{\partial x_i^2}(a) \cdot y_i^2 \mu_\varepsilon(dy) = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 u}{\partial x_i^2}(a) \cdot \frac{\varepsilon^2}{d} = \frac{1}{2} \Delta u(a) \cdot \frac{\varepsilon^2}{d}.$$

Now we conclude

$$u(a) = \int_{\partial B_\varepsilon} u(a+y) \mu_\varepsilon(dy) = u(a) + \frac{\Delta}{2} u(a) \cdot \frac{\varepsilon^2}{d} + o(\varepsilon^2).$$

So $0 = \frac{\Delta}{2} u(a) \cdot \frac{1}{d} + o(1)$, be letting $\varepsilon \downarrow 0$ to get $\Delta u(a) = 0$.

□

1.10.1 Dirichlet Problem

Given an open set $D \subseteq \mathbb{R}^d$, and a continuous function f on ∂D , the Dirichlet probability (D.f) is to find $u \in C(\overline{D})$ s.t.

$$\begin{cases} \Delta u = 0, & \text{in } D \\ u = f, & \text{on } \partial D. \end{cases}$$

$$\frac{\partial u(t, x)}{\partial t} = \frac{\Delta}{2} u(t, x).$$

Heat Equation $u(t, x)$ = temperature at $x \in D$ at time t .

Existence.

Let $u(x) = E^x[f(W_{\tau_D})]$ for all $x \in \overline{D}$, provided that $E^x |f(W_{\tau_D})| < \infty$, $\forall x \in D$. Then for $x \in \partial D$, we get

$$P^x(\tau_D = 0) = 1$$

by $\tau_D = \inf\{t \geq 0 : W_t \notin D\}$. Then

$$P^x(W_{\tau_D} = x) = 1,$$

which implies $E^x[f(W_{\tau_D})] = f(x)$.

So $u(x) = f(x)$, $\forall x \in \partial D$.

Next, we will prove $\Delta u = 0$, $\forall x \in D \Leftrightarrow u$ has MVP.

For all $a \in D$ and $r > 0$ s.t. $\overline{B_r(a)} \subseteq D$, we have

$$\begin{aligned} u(a) &= E^a[f(W_{\tau_D})] = E^a \left[E^a \left[f(W_{\tau_D}) \mid \mathcal{F}_{\tau_{B_r(a)}} \right] \right] \\ (\tau_{B_r(a)} \leq \tau_D) &= E^a \left[E^{W_{\tau_{B_r(a)}}} [f(W_{\tau_D})] \right] \\ &= E^a \left[u(W_{\tau_{B_r(a)}}) \right] \\ &= \int_{\partial B_r} u(a + y) \mu_r(dy). \end{aligned}$$

Therefore u has MVP. It remains to prove $u \in C(\overline{D})$ i.e. $\forall a \in \partial D$

$$\lim_{\substack{x \rightarrow a \\ x \in D}} u(x) = u(a) = f(a).$$

$\forall a \in \partial D$, we need that

$$\lim_{\substack{x \rightarrow a \\ x \in D}} E^x[f(W_{\tau_D})] = f(a).$$

Uniqueness.

If f is bdd and $P^a(\tau_D < \infty) = 1$, $\forall a \in D$. Then any bounded solution u to Dirichlet Problem (D,f) is given by $u(a) = E^a[f(W_{\tau_D})]$.

Proof. Let $u \in C(\overline{D})$ be bounded and

$$\begin{cases} \Delta u = 0, & \text{in } D \\ u = f, & \text{on } \partial D. \end{cases}$$

For each $n \geq 1$, we let

$$\begin{aligned} D_n &= \{x \in D : \inf_{y \in \partial D} \|x - y\| > \frac{1}{n}\} \\ &= \{x \in D : d(x, \partial D) > \frac{1}{n}\}. \end{aligned}$$

By Itô's formula

$$u(W_{t \wedge \tau_{D_n} \wedge \tau_{B_n}}) = u(W_0) + \sum_{i=1}^d \int_0^{t \wedge \tau_{D_n} \wedge \tau_{B_n}} \frac{\partial u}{\partial x_i}(W_s) dW_s^{(i)}.$$

Since $\frac{\partial u}{\partial x_i}(W_s)$ is bounded for all $0 \leq s \leq t \wedge \tau_{B_n} \wedge \tau_{D_n}$, we get

$$E \left[\int_0^{t \wedge \tau_{D_n} \wedge \tau_{B_n}} \frac{\partial u}{\partial x_i}(W_s) dW_s^{(i)} \right] = 0.$$

Then

$$u(a) = E^a [u(W_{t \wedge \tau_{B_n} \wedge \tau_{D_n}})].$$

By $P^a(\tau_D < \infty) = 1$, we have $(t \rightarrow \infty, n \rightarrow \infty)$

$$u(W_{t \wedge \tau_{B_n} \wedge \tau_{D_n}}) \xrightarrow{a.s.} u(W_{\tau_D}) = f(W_{\tau_D}).$$

So by bounded convergence theorem,

$$u(a) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow \infty} E^a [u(W_{t \wedge \tau_{B_n} \wedge \tau_{D_n}})] = E^a(f(W_{\tau_D})).$$

□

To prove $\forall a \in \partial D$,

$$\lim_{\substack{x \rightarrow a \\ x \in D}} E^x [f(W_{\tau_D})] = f(a.)$$

Define $\sigma_D = \inf\{t > 0 : W_t \notin D\}$. It is obvious that $\tau_D \leq \sigma_D$.

Definition 1.10.2 (regular point). We say a point $a \in \partial D$ is regular for D if $P^a(\sigma_D = 0) = 1$.

$$\sigma_D = \frac{1}{2} \Leftrightarrow \forall 0 < t < \frac{1}{2}, W_t \in D.$$

$$\sigma_D = 0 \Leftrightarrow \exists t_n \downarrow 0 \text{ s.t. } W_{t_n} \notin D.$$

When $d = 1$, $D = (a, b)$. Then $\partial D = \{a, b\}$. By LIL, a and b are both regular points.

$$\limsup_{t \downarrow 0} \frac{B_t - B_0^n}{\sqrt{2t \log \log \frac{1}{t}}} = 1 \quad \text{a.s.}$$

$\implies \exists t_n \downarrow 0$ s.t.

$$B_t - b \geq (\sqrt{2} - \varepsilon) \sqrt{t_n \log \log \frac{1}{t_n}} > 0.$$

Definition 1.10.3 (irregular point). A point $a \in \partial D$ is irregular if $\mathbf{P}^a(\sigma_D = 0) < 1 \Leftrightarrow \mathbf{P}^a(\sigma_D > 0) > 0$.

By Blumenthal 0-1 law, $\mathbf{P}^a(\sigma_D > 0) \in \{0, 1\}$. So if $a \in \partial D$ is irregular, then $\mathbf{P}^a(\sigma_D > 0) = 1$.

Theorem 1.10.1. ($d \geq 2$) and fix $a \in \partial D$. The following are equivalent

(i)

$$\lim_{\substack{x \rightarrow a \\ x \in D}} \frac{E^x [f(W_{\tau_D})]}{u(x)} = f(a)$$

holds for every bounded, measurable function f on ∂D s.t. f is continuous at a .

(ii) a is regular.

(iii) $\forall \varepsilon > 0$,

$$\lim_{\substack{x \rightarrow a \\ x \in D}} \mathbf{P}^x(\tau_D > \varepsilon) = 0.$$

$$(\tau_D \xrightarrow{\mathbf{P}^x} 0 \text{ as } x \rightarrow a)$$

Proof. (iii) $\implies$ (i) By continuity of Brownian path,

$$\mathbf{P}^x \left(\max_{0 \leq t \leq \varepsilon} \|W_t - W_0\| < r \right) \rightarrow 1$$

as $\varepsilon \downarrow 0$, and

$$\mathbf{P}^0 \left(\max_{0 \leq t \leq \varepsilon} \|W_t\| < r \right) \rightarrow 1.$$

Then

$$\begin{aligned} \mathbf{P}^x (\|W_{\tau_D} - W_0\| < r) &\geq \mathbf{P}^x \left(\max_{0 \leq t \leq \varepsilon} \|W_t - W_0\| < r, \tau_D \leq \varepsilon \right) \\ &\geq \mathbf{P}^x \left(\max_{0 \leq t \leq \varepsilon} \|W_t - W_0\| < r \right) - \mathbf{P}^x(\tau_D > \varepsilon). \end{aligned}$$

Let $x \in D \rightarrow a$ to get

$$\lim_{\substack{x \rightarrow a \\ x \in D}} \mathbf{P}^x (\|W_{\tau_D} - W_0\| < r) \geq \mathbf{P}^0 \left(\max_{0 \leq t \leq \varepsilon} \|W_t\| < r \right) - 0.$$

$\forall \varepsilon > 0, \exists r > 0$ s.t. $|f(x) - f(a)| < \varepsilon$ for all $|x - a| < r$,

$$E^x |f(W_{\tau_D}) - f(a)| \leq \|f\|_\infty \cdot E^x \left(\mathbf{1}_{\|W_{\tau_D} - x\| > \frac{r}{2}} \right) + E^x \left(|f(W_{\tau_D}) - f(a)| \cdot \mathbf{1}_{\|W_{\tau_D} - x\| < \frac{r}{2}} \right).$$

If we let $|x - a| < \frac{r}{2}$, then

$$\|W_{\tau_D} - a\| \leq \|W_{\tau_D} - x\| + \|x - a\| < r.$$

$$\implies |f(W_{\tau_D}) - f(a)| < \varepsilon.$$

□

Definition 1.10.4 (defn example). Example

Theorem 1.10.2 (thm example). Example

Proof. proof □

Lemma 1.10.1 (lemma example). Example

Proof for Lemma.

■. proof for lemma example □

Proposition 1.10.2. pro example

Proof for Proposition.

■. Example □

Claim 1.10.1. claim example Example

Proof for Claim.

■. proof for claim example □

Corollary 1.10.1 (cor example). example

Proof for Corollary.

■. proof for corollary example □

Fact 1.10.1. Example fact

Remark. Example remark

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